ECEN 662: Estimation & Detection Theory

Lecture 0: Course Overview

Tie Liu

Texas A&M University

Statistical inference

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  - We will use f_θ to denote probability density function (pdf) and p_θ for probability mass function (pmf), and f_θ for *generic* descriptions
  - Both X and θ can be *vectors* of *finite* dimensions
- In *statistical inference*, we observe a *realization* x of X and need to answer some questions about θ, such as:
  - *Estimation*: What is a good *guess* of θ?
  - *Detection*: Is $\theta \in \Theta_1$, or is $\theta \in \Theta_2$?

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- A probabilistic model f_θ is said to be *identifiable* if

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- Model identifiability is *not* important if our goal is to infer about the likelihood function f_θ, i.e., *generative modeling* in machine learning, rather than about the unknown parameter θ

Multiple independent measurements

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$$X_i \overset{iid}{\sim} f_\theta(x_i)$$

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- Note that for *iid* measurements, when the sample size $n \to \infty$, our ability to infer about the unknown parameter θ boils down to the *identifiability* of the *marginal* distribution of X
- In practice, n is always *finite* and inference is always subject to the *randomness* of observation

Topics

- Part I: Point Estimation
  - Principles:
    - Minimum-variance unbiased estimation (MVUE)
    - Maximum-likelihood estimation (MLE)
    - Minimum-risk equivariant estimation (MREE)
    - Bayes estimators
    - Empirical Bayes estimation
    - Minimax estimators
  - Algorithms:
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- Part II: Testing Statistical Hypotheses
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- Part III: Sampling
  - Transform and rejection sampling
  - Monte Carlo integration
  - Markov chain Monte Carlo (MCMC) sampling
  - Diffusion Monte Carlo sampling (if time permits)

Textbooks, Suggested Papers, and Lecture Slides

- (LC) E. L. Lehmann and George Casella: *Theory of Point Estimation, Second Edition*. Springer 1998
- (LR) E. L. Lehmann and Joseph P. Romano: *Testing Statistical Hypotheses, Third Edition*. Springer 2008
- (RC) Christian P. Robert and George Casella: *Monte Carlo Statistical Methods, Second Edition*. Springer 2004
- *Suggested papers and lecture slides* will be posted on Canvas as we move along

Course Requirement

- Weekly homework (75%):
  - Theoretical in nature
  - Occasionally *very* challenging
  - Discussions among peers are allowed. However, each student must submit his/her own *independent* work
- Term project (25%):
  - The project can be either a specific application, or some preliminary research on methodology/algorithm
  - The topic of the project needs to be approved by the instructor
- Grading scale: A: 85-100; B: 70-84; C: 60-69; D: 50-59; F: 0-49

Next lecture

Principle of data reduction and sufficiency

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Lecture 1: Principle of Data Reduction and Sufficiency

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- In other words, does there exist $T = T(X)$ of dimension $m < n$ such that T carries *all* the useful information about θ?

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- In other words, does there exist $T = T(X)$ of dimension $m < n$ such that T carries *all* the useful information about θ?
- If so, for the purpose of studying θ, we could discard the raw observation x and retain only the compressed statistic $t = T(x)$

Example: Bernoulli trials

- More specifically, let us consider the example of *Bernoulli trials*, in which $X = [X_1, \ldots, X_n]^t$,

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and the parameter $\theta \in (0, 1)$ is unknown

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- Intuitively, to infer about θ, it *suffices* to know how many 1's are in the realization of X - the positions of 1's appear to be rather irrelevant
- In another word, our intuition suggests that

$$T(X) = \sum_{i=1}^{n} X_i$$

is *sufficient* for inferring about θ. But how do we justify this *probabilistically*?

A formula for $p_\theta(x|t)$

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- We thus have

$$p_\theta(x) = p_\theta(x,T(x)) = p_\theta(T(x))p_\theta(x|T(x))$$

and hence

$$p_\theta(x|T(x)) = \frac{p_\theta(x)}{p_\theta(T(x))}$$

Back to Bernoulli trials

- For Bernoulli trials, we have Since $x_i \in \{0, 1\}$. We may thus write

$$p_\theta(x_i) = \theta^{x_i}(1 - \theta)^{1-x_i}$$

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- It follows that

$$\begin{aligned} p_\theta(x) &= \prod_{i=1}^{n} p_\theta(x_i) \\ &= \prod_{i=1}^{n} \theta^{x_i}(1-\theta)^{1-x_i} \\ &= \theta^{T(x)}(1-\theta)^{n-T(x)} \end{aligned}$$

where

$$T(x) = \sum_{i=1}^{n} x_i$$

- Further note that

$$T \sim Binomial(n, \theta)$$

so

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$$p_\theta(t) = \left(\begin{array}{c} n \\ t \end{array} \right) \theta^t (1-\theta)^{n-t}$$

- By the previous formula, for any (x, t) such that $T(x) = t$ we have
$$p_\theta(x|t) = \frac{p_\theta(x)}{p_\theta(t)} = \frac{\theta^{T(x)}(1-\theta)^{n-T(x)}}{\left(\begin{array}{c} n \\ t \end{array} \right) \theta^t (1-\theta)^{n-t}} = \frac{1}{\left(\begin{array}{c} n \\ t \end{array} \right)}$$

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- Note that the above conditional distribution is *independent* of θ, i.e, given $T = T(X)$, full knowledge of X brings *no* additional information about θ. This is the *statistical* notion of *sufficiency* that we look for

Sufficient statistic

- Formally, let $X \sim f_\theta(x)$. A statistic $T = T(X)$ is said to be a *sufficient statistic* for θ if the conditional distribution of X given T is *independent* of θ, i.e., the functional form of $f_\theta(x|t)$ does *not* involve θ

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- If the conditional distribution $p_\theta(x|t)$ does *not* depend on θ, the dependence of $p_\theta(x)$ on θ is manifested *entirely* in $p_\theta(t)$, and in this case, any inference strategy based on $p_\theta(x)$ can be replaced by a strategy based on $p_\theta(t)$

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- (*Fisher-Neyman factorization theorem*) Let $f_\theta(x)$ be a pdf or pmf of X. A statistic $T = T(X)$ is sufficient for θ *if and only if* there exist functions $g_\theta(t)$ and $h(x)$ such that:

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- In particular, if the likelihood $f_\theta(x)$ depends x *only* through $T(x)$, then $T = T(X)$ is a sufficient statistic for θ

Bernoulli trials revisited

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- Note that $p_\theta(x)$ depends on x *only* through

$$T(x) = \sum_{i=1}^{n} x_i,$$

so by the *Fisher-Neyman factorization theorem*,

$$T = \sum_{i=1}^{n} X_i$$

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- For the other direction, assume that
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- Note that
$$p_\theta(x|t) = \frac{p_\theta(x)}{p_\theta(t)}$$
for any (x, t) such that $T(x) = t$

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p_\theta(x|t) = \frac{p_\theta(x)}{p_\theta(t)} &= \frac{p_\theta(x)}{\sum_{y:T(y)=t} p_\theta(y)} \\
&= \frac{g_\theta(T(x))h(x)}{\sum_{y:T(y)=t} g_\theta(T(y))h(y)} \\
&= \frac{g_\theta(t)h(x)}{g_\theta(t)\sum_{y:T(y)=t} h(y)} \\
&= \frac{h(x)}{\sum_{y:T(y)=t} h(y)}
\end{aligned}$$

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- Since $p_\theta(x|t)$ does *not* depend on θ, by definition $T = T(x)$ is a sufficient statistic for θ

Another formula for $p_\theta(x|t)$

- If
$$p_\theta(x) = h(x) g_\theta(T(x))$$
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for any (x, t) such that $T(x) = t$

- For Bernoulli trials, we have $h(x) = 1$ for all $x \in \{0, 1\}^n$ so
$$p_\theta(x|t) = \frac{1}{\sum_{y:\sum_{i=1}^n y_i = t} 1} = \frac{1}{\binom{n}{t}}$$
for any (x, t) such that $\sum_{i=1}^n x_i = t$

Example: Gaussian with unknown mean

- Suppose we observe $X = [X_1, \ldots, X_n]^t$, where

$$X_i \overset{iid}{\sim} \mathcal{N}(\mu, \sigma^2)$$

σ^2 is a *known* constant, and μ is the *unknown* parameter

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- We have

$$f_\mu(x_i) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$

and hence

$$\begin{aligned}
f_\mu(x) &= \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2\right) \\
&= \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(\frac{\mu}{\sigma^2}\sum_{i=1}^{n} x_i - \frac{n\mu^2}{2\sigma^2}\right) \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n} x_i^2\right) \\
&= g_\mu(T(x))h(x)
\end{aligned}$$

where

$$T(x) = \sum_{i=1}^{n} x_i$$

$$g_\mu(t) = \exp\left(\frac{\mu t}{\sigma^2} - \frac{n\mu^2}{2\sigma^2}\right)$$

$$h(x) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n} x_i^2\right)$$

$$T(x) = \sum_{i=1}^{n} x_i$$

$$g_\mu(t) = \exp\left(\frac{\mu t}{\sigma^2} - \frac{n\mu^2}{2\sigma^2}\right)$$

$$h(x) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n} x_i^2\right)$$

- By the *Fisher-Neyman factorization theorem*,

$$T = \sum_{i=1}^{n} X_i$$

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$$f_\theta(x_i) = \frac{1}{\theta} 1_{\{x_i \leq \theta\}}$$

for any $x_i \geq 0$ and hence

$$f_\theta(x) = \prod_{i=1}^{n} \frac{1}{\theta} 1_{\{x_i \leq \theta\}} = \frac{1}{\theta^n} \prod_{i=1}^{n} 1_{\{x_i \leq \theta\}} = \frac{1}{\theta^n} 1_{\{\max_{i \in [n]} x_i \leq \theta\}}$$

for any $x \geq 0$, where $[n] := \{1, \ldots, n\}$

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- Note that $f_\theta(x)$ depends on x only through $T(x) = \max_{i \in [n]} x_i$. By the *Fisher-Neyman factorization theorem*,

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Next lecture

Minimal and complete sufficient statistics

ECEN 662: Estimation & Detection Theory

Lecture 2: Minimal and Complete Sufficient Statistics

Tie Liu

Texas A&M University

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A sufficient condition for minimality given sufficiency

- The minimality of a sufficient statistic can be verified based on the following proposition: A sufficient statistic T is *minimal* if for *any* $x, y \in \mathcal{X}$,

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- By the above property, we must have $T(x) = T(y)$, which completes the proof of the minimality of T

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- Combined with the previous proposition, a statistic $T = T(X)$ is a *minimal sufficient* statistic if for *any* $x, y \in \mathcal{X}$,

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- The advantage of the above result is that it allows us to check sufficiency and minimality in one shot

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- We thus conclude that for Bernoulli trials, $T = \sum_{i=1}^{n} X_i$ is a *minimal sufficient* statistic

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- Let

$$\begin{aligned} \psi(t') &:= \mathbb{E}_\theta[T|T' = t'] \\ \rho(t) &:= \mathbb{E}_\theta[\psi(T')|T = t] \end{aligned}$$

  Note that both T' and T are *sufficient* statistics, so neither ψ nor ρ depends depend on θ

- Next, by the *law of total expectation*,

$$\begin{aligned}\mathbb{E}_\theta[T] &= \mathbb{E}_\theta[\mathbb{E}_\theta[T|T']] \\ &= \mathbb{E}_\theta[\psi(T')] \\ &= \mathbb{E}_\theta[\mathbb{E}_\theta[\psi(T')|T]] \\ &= \mathbb{E}_\theta[\rho(T)]\end{aligned}$$

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- Finally, recall the *law of total variance*:

$$\mathrm{Var}[Y] = \mathbb{E}[\mathrm{Var}[Y|Z]] + \mathrm{Var}[\mathbb{E}[Y|Z]]$$

  for any *real-valued* random variable Y

- For any $i \in [m]$, we have

$$
\begin{aligned}
&\mathrm{Var}_\theta[\rho_i(T)] \\
&\quad = \mathbb{E}_\theta[\mathrm{Var}_\theta[\rho_i(T)|T']] + \mathrm{Var}_\theta[\mathbb{E}_\theta[\rho_i(T)|T']] \\
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$$

- Since variances are always *non-negative*, we must have

$$
\begin{aligned}
\mathrm{Var}_\theta[\rho_i(T)|T'] = 0 &\qquad \text{with probability 1} \\
\mathrm{Var}_\theta[\psi_i(T')|T] = 0 &\qquad \text{with probability 1}
\end{aligned}
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for all $i \in [m]$. That is, $\rho(T)$ must be a *deterministic* function of T'

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- The "mild conditions" under which the result is valid are that the various expectations and variances used throughout the proof are well defined and finite

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\mathbb{E}_\theta[\phi(T)] &= \sum_{t=0}^{n} \phi(t) \binom{n}{t} \theta^t (1-\theta)^{n-t} \\
&= (1-\theta)^n \sum_{t=0}^{n} \phi(t) \binom{n}{t} \left(\frac{\theta}{1-\theta}\right)^t \\
&= 0
\end{aligned}$$

we must have

$$\sum_{t=0}^{n} \phi(t) \binom{n}{t} \left(\frac{\theta}{1-\theta}\right)^t = 0$$

for all $\theta \in (0, 1)$

- Let $\eta = \frac{\theta}{1-\theta}$. We have

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$$\mathbb{E}_\theta[\phi(T)] = n\theta^{-n} \int_0^\theta \phi(t)t^{n-1}dt = 0$$
for all $\theta > 0$, we must have
$$\int_0^\theta \phi(t)t^{n-1}dt = 0, \quad \forall \theta > 0$$

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$$
\begin{aligned}
f_\theta(x) &= \prod_{i=1}^{n} \left(\frac{1}{\sqrt{2\pi\theta^2}} \exp\left(-\frac{(x_i - \theta)^2}{2\theta^2} \right) \right) \\
&= \left(\frac{1}{\sqrt{2\pi\theta^2}} \right)^n \exp\left(-\frac{\sum_{i=1}^{n} x_i^2}{2\theta^2} + \frac{\sum_{i=1}^{n} x_i}{\theta} - \frac{n}{2} \right)
\end{aligned}
$$

A minimal sufficient statistic is not necessarily complete

- Let $(X_i : i \in [n])$ be n independent Gaussian random variables with mean θ and variance θ^2, where $\theta > 0$ is an unknown parameter
- The likelihood function for $X = (X_1, \dots, X_n)$ is given by

$$\begin{aligned} f_\theta(x) &= \prod_{i=1}^{n} \left(\frac{1}{\sqrt{2\pi\theta^2}} \exp\left(-\frac{(x_i - \theta)^2}{2\theta^2} \right) \right) \\ &= \left(\frac{1}{\sqrt{2\pi\theta^2}} \right)^n \exp\left(-\frac{\sum_{i=1}^{n} x_i^2}{2\theta^2} + \frac{\sum_{i=1}^{n} x_i}{\theta} - \frac{n}{2} \right) \end{aligned}$$

- By the Fisher-Neyman factorization theorem,

$$T = \left(\sum_{i=1}^{n} X_i, \sum_{i=1}^{n} X_i^2 \right)$$

is a two-dimensional sufficient statistic for θ

- The likelihood ratio

$$\frac{f_\theta(x)}{f_\theta(y)} = \exp\left(-\frac{\sum_{i=1}^n x_i^2 - \sum_{i=1}^n y_i^2}{2\theta^2} + \frac{\sum_{i=1}^n x_i - \sum_{i=1}^n y_i}{\theta}\right)$$

for which the exponent is a *quadratic* polynomial of $1/\theta$

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- Note that for a quadratic polynomial to be constant over all positive reals, the polynomial must be a *constant* polynomial
- Therefore, the likelihood ratio $f_\theta(x)/f_\theta(y)$ is constant over $\theta > 0$ only if

$$\sum_{i=1}^n x_i^2 = \sum_{i=1}^n y_i^2 \quad \text{and} \quad \sum_{i=1}^n x_i = \sum_{i=1}^n y_i$$

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- We may thus conclude that

$$T = \left(\sum_{i=1}^n X_i, \sum_{i=1}^n X_i^2\right)$$

is in fact a *minimal* sufficient statistic for θ

- In your homework assignment, you are asked to find a real-valued function $\phi(t)$ such that $\mathbb{E}_\theta[\phi(T)] = 0$ for all $\theta > 0$ but $\mathbb{P}_\theta(\phi(T) = 0) < 1$ for some $\theta > 0$, i.e., T is *not* complete

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- Consequently, if there is a minimal sufficient statistic that is *not* complete, then a complete sufficient does *not* exist
- As we will see in later lectures, while both minimal and complete sufficient statistics are "optimal" in terms of data reduction, only complete sufficient statistics guarantee optimality when it comes to *risk reduction*

Next lecture

Exponential family of distributions

ECEN 662: Estimation & Detection Theory

Lecture 3: Exponential Family of Distributions

Tie Liu

Texas A&M University

Exponential family of distributions

- A model $f_\theta(x)$ is called an *exponential family of distributions* if it can be written as:

$$f_\theta(x) = h(x) \exp\left(\eta^t T - A(\eta)\right)$$

  where:

  - $\eta = \eta(\theta)$ is an s-dimensional function of the model parameter θ
  - $T = T(x)$ is an s-dimensional function of the observation x
  - $h(x)$ is a one-dimensional function of the observation x known as the *base measure*
  - $A(\eta)$ is the *normalization* factor known as the *log-partition function* and can be computed as:

$$A(\eta) = \log\left(\int_{\mathcal{X}} h(x) \exp\left(\eta^t T(x)\right) dx\right)$$

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$$A(\eta) = \log\left(\int_{\mathcal{X}} h(x) \exp\left(\eta^t T(x)\right) dx\right)$$

- In addition, it is required that the *support set* $\{x : f_\theta(x) > 0\}$ does *not* depend on θ

Natural parameter and canonical representation

- Note that the likelihood function f_θ depends on the model parameter θ *only* through $\eta = \eta(\theta)$. Therefore, η is known as a *natural parameter* of the model, and any representation of the aforementioned form is known as a *canonical* representation of the exponential family

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- Note that for any given exponential family of distributions, canonical presentations are *not* unique. For example, if $(\eta, T, h, A(\eta))$ is a canonical representation, then

$$(\eta' = B\eta, T' = B^{-1}T, h' = h, A'(\eta') = A(B^{-1}\eta'))$$

  is also a canonical representation for any *invertible* matrix B

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  is also a canonical representation for any *invertible* matrix B

- A canonical representation is called *irreducible* if neither $(\eta_i(\theta) : i \in [s])$ nor $(T_i(x) : i \in [s])$ satisfies a *linear-equality* constraint; otherwise, we can find an alternative canonical representation with a natural parameter of a *reduced* dimension

Full-ranked and curved families

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- We say that an (irreducible) s-parameter exponential family is *full-ranked* if the space of the natural parameter $\eta(\Theta)$ contains an *s-dimensional* rectangle; otherwise, it is called a *curved* family

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- Let Θ be the space of the model parameter θ and let $\eta(\Theta) := \{\eta(\theta) : \theta \in \Theta\}$ be the space of the natural parameter η induced by $\eta = \eta(\theta)$
- We say that an (irreducible) s-parameter exponential family is *full-ranked* if the space of the natural parameter $\eta(\Theta)$ contains an *s-dimensional* rectangle; otherwise, it is called a *curved* family
- Note that by the Fisher-Neyman factorization theorem, $T(X)$ is a *sufficient statistic* for θ for any exponential family. In addition, if the exponential family is *full-ranked*, then $T(X)$ is also *complete* for θ

Multiple independent measurements

- Let $X = [X_1, \ldots, X_n]^t$, where

$$X_i \overset{iid}{\sim} f_\theta(x_i)$$

and

$$f_\theta(x_i) = h(x_i) \exp\left(\eta^t T(x_i) - A(\eta)\right)$$

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$$f_\theta(x_i) = h(x_i) \exp\left(\eta^t T(x_i) - A(\eta)\right)$$

- We thus have

$$f_\theta(x) = \prod_{i=1}^{n} f_\theta(x_i) = \left(\prod_{i=1}^{n} h(x_i)\right) \exp\left(\eta^t \sum_{i=1}^{n} T(x_i) - nA(\eta)\right),$$

which is (again) an exponential family of distribution with natural parameter η, sufficient statistic $\sum_{i=1}^{n} T(x_i)$, base measure $\prod_{i=1}^{n} h(x_i)$, and log-partition function $nA(\eta)$

Example: Bernoulli trials

- For Bernoulli trials, we have

$$
\begin{aligned}
p_\theta(x_i) &= \theta^{x_i}(1-\theta)^{1-x_i} \\
&= \exp\left(x_i \log\theta + (1-x_i)\log(1-\theta)\right) \\
&= \exp\left(x_i \log\frac{\theta}{1-\theta} - \log\frac{1}{1-\theta}\right)
\end{aligned}
$$

which is a one-parameter exponential family of distributions with

$$
\eta(\theta) = \log\frac{\theta}{1-\theta}, \quad T(x_i) = x_i, \quad h(x_i) = 1,
$$

$$
\text{and} \quad A(\eta) = \log\frac{1}{1-\theta} = \log\left(1+e^\eta\right)
$$

- Thus,

$$p_\theta(x) = \prod_{i=1}^{n} p_\theta(x_i)$$

is also a one-parameter exponential family of distributions with

$$\eta(\theta) = \log \frac{\theta}{1-\theta}, \quad T(x) = \sum_{i=1}^{n} x_i, \quad h(x) = 1,$$

$$\text{and} \quad A(\eta) = n \log\left(1 + e^{\eta}\right)$$

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$$\text{and} \quad A(\eta) = n \log\left(1 + e^{\eta}\right)$$

- Note that the space of the model parameter θ is $(0,1)$, and the space of the natural parameter

$$\eta = \log \frac{\theta}{1-\theta}$$

is $\mathcal{R}$

- Therefore, this exponential family is *full-ranked*, and the sufficient statistic
$$T(X) = \sum_{i=1}^{n} X_i$$
is also *complete*

Example: Gaussian with unknown mean

- For Gaussian with unknown mean, we have

$$f_\mu(x_i) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{x_i^2}{2\sigma^2}\right) \exp\left(\frac{\mu x_i}{\sigma^2} - \frac{\mu^2}{2\sigma^2}\right)$$

which is a one-parameter exponential family of distributions with

$$\eta(\mu) = \frac{\mu}{\sigma^2}, \quad T(x_i) = x_i, \quad h(x_i) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{x_i^2}{2\sigma^2}\right)$$

$$\text{and} \quad A(\eta) = \frac{\mu^2}{2\sigma^2} = \frac{\sigma^2\eta^2}{2}$$

- Thus,

$$f_\mu(x) = \prod_{i=1}^{n} f_\mu(x_i)$$

is also a one-parameter exponential family of distributions with

$$\eta(\mu) = \frac{\mu}{\sigma^2}, \quad T(x) = \sum_{i=1}^{n} x_i,$$

$$h(x) = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n \exp\left(-\frac{\sum_{i=1}^{n} x_i^2}{2\sigma^2}\right), \quad \text{and} \quad A(\eta) = \frac{n\sigma^2\eta^2}{2}$$

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- Note that the space of the model parameter μ is $\mathcal{R}$, and the space of the natural parameter

$$\eta = \frac{\mu}{\sigma^2}$$

is also $\mathcal{R}$

- Therefore, this exponential family is *full-ranked*. As a result, the sufficient statistic
$$T(X) = \sum_{i=1}^{n} X_i$$
is also *complete*

Example: Gaussian with unknown variance

- For Gaussian with unknown variance, we have

$$f_{\sigma^2}(x_i) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i-\mu)^2}{2\sigma^2}\right)$$

which is a one-parameter exponential family of distributions with

$$\eta(\sigma^2) = -\frac{1}{2\sigma^2}, \quad T(x_i) = (x_i-\mu)^2, \quad h(x_i) = \frac{1}{\sqrt{\pi}}$$

$$\text{and} \quad A(\eta) = \frac{1}{2}\log(2\sigma^2) = -\frac{1}{2}\log(-\eta)$$

- Thus,

$$f_{\sigma^2}(x) = \prod_{i=1}^{n} f_{\sigma^2}(x_i)$$

is also a one-parameter exponential family of distributions with

$$\eta(\sigma^2) = -\frac{1}{2\sigma^2}, \quad T(x) = \sum_{i=1}^{n}(x_i - \mu)^2,$$
$$h(x) = \left(\frac{1}{\sqrt{\pi}}\right)^n, \quad \text{and} \quad A(\eta) = -\frac{n}{2}\log(-\eta)$$

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- Note that the space of the model parameter σ^2 is $\mathcal{R}_{++}$, and the space of the natural parameter

$$\eta = \frac{1}{2\sigma^2}$$

is $\mathcal{R}_{--}$

- Therefore, this exponential family is *full-ranked*. As a result, the sufficient statistic

$$T(X) = \sum_{i=1}^{n} (X_i - \mu)^2$$

is also *complete*

Example: Gaussian with unknown mean and variance

- For Gaussian with unknown mean and variance, we have

$$\begin{aligned} f_\theta(x_i) &= \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(\frac{\mu x_i}{\sigma^2} - \frac{x_i^2}{2\sigma^2} - \frac{\mu^2}{2\sigma^2} \right) \\ &= \frac{1}{\sqrt{\pi}} \exp\left(\frac{\mu x_i}{\sigma^2} - \frac{x_i^2}{2\sigma^2} - \frac{\mu^2}{2\sigma^2} - \frac{1}{2}\log(2\sigma^2) \right) \end{aligned}$$

  which is a two-parameter exponential family of distributions with

$$\eta(\theta) = \left(\frac{\mu}{\sigma^2}, -\frac{1}{2\sigma^2} \right), \quad T(x_i) = (x_i, x_i^2), \quad h(x_i) = \frac{1}{\sqrt{\pi}},$$

  and log-partition function

$$A(\eta) = \frac{\mu^2}{2\sigma^2} + \frac{1}{2}\log(2\sigma^2) = -\frac{\eta_1^2}{4\eta_2} - \frac{1}{2}\log(-\eta_2)$$

- Thus,

$$f_\theta(x) = \prod_{i=1}^{n} f_\theta(x_i)$$

is also a two-parameter exponential family of distributions with

$$\eta(\theta) = \left(\frac{\mu}{\sigma^2}, -\frac{1}{2\sigma^2}\right),\ T(x) = \left(\sum_{i=1}^{n} x_i, \sum_{i=1}^{n} x_i^2\right),\ h(x) = \left(\frac{1}{\sqrt{\pi}}\right)^n,$$

and log-partition function

$$A(\eta) = -\frac{n\eta_1^2}{4\eta_2} - \frac{n}{2}\log(-\eta_2)$$

- Thus,
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and log-partition function
$$A(\eta) = -\frac{n\eta_1^2}{4\eta_2} - \frac{n}{2}\log(-\eta_2)$$
- Note that the space of model parameter $\theta = (\mu, \sigma^2)$ is $\mathcal{R} \times \mathcal{R}_{++}$ and the space of the natural parameter $\eta(\theta) = \left(\frac{\mu}{\sigma^2}, -\frac{1}{2\sigma^2}\right)$ is $\mathcal{R} \times \mathcal{R}_{--}$

- Therefore, this exponential family is *full-ranked*, and the two-dimensional sufficient statistic

$$T(X) = \left(\sum_{i=1}^{n} X_i, \sum_{i=1}^{n} X_i^2 \right)$$

is also *complete*

Example: $X_i \sim \mathcal{N}(\theta, \theta^2)$ for some unknown $\theta > 0$

- In this case,

$$f_\theta(x_i) = \frac{1}{\sqrt{2\pi\theta^2}} \exp\left(-\frac{(x_i-\theta)^2}{2\theta^2}\right)$$
$$= \frac{1}{\sqrt{2\pi\theta^2}} \exp\left(-\frac{x_i^2}{2\theta^2} + \frac{x_i}{\theta} - \frac{1}{2}\right)$$

which is a two-parameter exponential family of distributions with

$$\eta(\theta) = \left(\frac{1}{\theta}, -\frac{1}{2\theta^2}\right), \quad T(x) = \left(x_i, x_i^2\right), \quad h(x_i) = \frac{1}{\sqrt{\pi}},$$

and log-partition function

$$A(\eta) = \frac{1}{2} + \frac{1}{2}\log(2\theta^2) = \frac{1}{2} - \frac{1}{2}\log(-\eta_2)$$

- Thus,

$$f_\theta(x) = \prod_{i=1}^{n} f_\theta(x_i)$$

is also a two-parameter exponential family of distributions with

$$\eta(\theta) = \left(\frac{1}{\theta}, -\frac{1}{2\theta^2}\right),\ T(x) = \left(\sum_{i=1}^{n} x_i, \sum_{i=1}^{n} x_i^2\right),\ h(x) = \left(\frac{1}{\sqrt{2\pi}}\right)^n,$$

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$$A(\eta) = \frac{n}{2} - \frac{n}{2}\log(-\eta_2)$$

- Note that the space of model parameter θ is $\mathcal{R}_{++}$ and the space of the natural parameter $\eta(\theta) = \left(\frac{1}{\theta}, -\frac{1}{2\theta^2}\right)$ is a *one-dimensional curve* in $\mathcal{R}^2$. Therefore, this exponential family is *curved*

- As discussed in the previous lecture, in this case, the two-dimension sufficient statistic

$$T(X) = \left(\sum_{i=1}^{n} X_i, \sum_{i=1}^{n} X_i^2 \right)$$

is *not* complete for any $n \geq 1$

Uniform with unknown support

- Recall that for each $i \in [n]$, the likelihood function

$$f_\theta(x_i) = \frac{1}{\theta} 1_{\{x_i \leq \theta\}}$$

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Uniform with unknown support

- Recall that for each $i \in [n]$, the likelihood function

$$f_\theta(x_i) = \frac{1}{\theta} 1_{\{x_i \leq \theta\}}$$

- Because the support of $f_\theta(x_i)$ depends on θ, this is *not* an exponential family of distributions
- As we have shown previously, for this example, it is

$$T(X) = \max_{i \in [n]} X_i$$

rather than

$$T(X) = \sum_{i=1}^{n} X_i$$

that is a sufficient statistic for θ

Distribution and moments of $T(X)$

- Let $f_\theta(x)$ be an exponential family such that

$$f_\theta(x) = h(x) \exp\left(\eta^t T - A(\eta)\right)$$

The distribution of T is given by

$$f_\theta(t) = k(t) \exp\left(\eta^t t - A(\eta)\right)$$

for some new base measure $k(t)$, i.e, it is again an *exponential* family

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  for some new base measure $k(t)$, i.e, it is again an *exponential* family

- For example, when $X = (X_1, \ldots, X_n)$ where each X_i is Bernoulli with an unknown parameter θ, we can write

$$p_\theta(x) = \exp\left(\eta T(x) - A(\eta)\right),$$

  where $\eta = \log \frac{\theta}{1-\theta}$, $h(x) = 1$, $T(x) = \sum_{i=1}^n x_i$, and $A(\eta) = n \log\left(1 + e^\eta\right)$

- The sufficient statistic $T(X) = \sum_{i=1}^{n} X_i$ is binomial whose pmf can be written as

$$\begin{aligned} p_\theta(t) &= \begin{pmatrix} n \\ t \end{pmatrix} \theta^t (1-\theta)^{n-t} \\ &= \begin{pmatrix} n \\ t \end{pmatrix} \exp\left(\left(\log \frac{\theta}{1-\theta}\right) t - n \log \frac{1}{1-\theta}\right) \\ &= k(t) \exp(\eta t - A(\eta)), \end{aligned}$$

where

$$k(t) = \begin{pmatrix} n \\ t \end{pmatrix}$$

- The moment and cumulant generating functions of T can be computed from the log-partition function $A(\eta)$ as:

$$\begin{aligned} M_T(u;\theta) &:= \mathbb{E}_\eta\left[e^{u^t T}\right] \\ &= \int_{\mathcal{T}} k(t) \exp\big((u+\eta)^t t - A(\eta)\big)\, dt \\ &= \exp(A(u+\eta) - A(\eta)) \end{aligned}$$

and hence

$$K_T(u;\theta) = \log M_T(u;\theta) = A(u+\eta) - A(\eta)$$

- The moment and cumulant generating functions of T can be computed from the log-partition function $A(\eta)$ as:

$$\begin{aligned} M_T(u;\theta) &:= \mathbb{E}_\eta\left[e^{u^t T}\right] \\ &= \int_{\mathcal{T}} k(t) \exp\left((u+\eta)^t t - A(\eta)\right) dt \\ &= \exp(A(u+\eta) - A(\eta)) \end{aligned}$$

and hence

$$K_T(u;\theta) = \log M_T(u;\theta) = A(u+\eta) - A(\eta)$$

- It follows immediately that

$$\mathbb{E}_\theta[T] = \nabla A(\eta) \quad \text{and} \quad \mathrm{Cov}_\theta[T] = \nabla^2 A(\eta)$$

- Going back to the example of Bernoulli trials, we have $\eta = \log \frac{\theta}{1-\theta}$, $T(x) = \sum_{i=1}^{n} x_i$, and $A(\eta) = n \log\left(1 + e^{\eta}\right)$

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- We thus have

$$\mathbb{E}_{\theta}[T] = \nabla A(\eta) = \frac{ne^{\eta}}{1 + e^{\eta}} = n\theta$$

and

$$\mathrm{Var}_{\theta}[T] = \nabla^2 A(\eta) = \frac{ne^{\eta}}{(1 + e^{\eta})^2} = n\theta(1 - \theta)$$

Next lecture

Uniformly Minimum-Variance Unbiased Estimation

ECEN 662: Estimation & Detection Theory

Lecture 5: Uniformly Minimum-Variance Unbiased Estimation (Part II)

Tie Liu

Texas A&M University

UMVUE without a complete sufficient statistic

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- What difference does this additional knowledge make?

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$$T = \max_{i \in [n]} X_i$$

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- As a result, the unbiased estimator

$$\hat{\theta} = \frac{n+1}{n} T$$

is an UMVUE of θ with the minimum variance given by

$$\mathrm{Var}_\theta[\hat{\theta}] = \frac{\theta^2}{n(n+2)}$$

Example: Uniform with unknown support ($\theta > 1$)

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- For any $\theta > 1$, the expected value of $\hat{\theta}'$ can be calculated as

$$\begin{aligned} \mathbb{E}_\theta[\hat{\theta}'] &= \int_0^1 f_\theta(t)dt + \int_1^\theta \frac{n+1}{n} t f_\theta(t)dt \\ &= n\theta^{-n} \int_0^1 t^{n-1}dt + (n+1)\theta^{-n} \int_1^\theta t^n dt \\ &= \theta^{-n} + \theta^{-n}(\theta^{n+1} - 1) \\ &= \theta \end{aligned}$$

- That is,

$$\hat{\theta}' = \begin{cases} 1, & \text{if } T \leq 1 \\ \frac{n+1}{n} T, & \text{if } T > 1 \end{cases}$$

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- Let $\phi(T)$ be a function of T such that

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- As before, it can be shown that we *must* have

$$\phi(t) = 0, \quad \forall t > 1$$

- However, for $t \leq 1$, $\phi(t)$ does *not* have to be a zero function to satisfy

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- For example, if we let

$$\phi(t) = \begin{cases} t - \frac{n}{n+1}, & \text{if } t \le 1 \\ 0, & \text{if } t > 1, \end{cases}$$

we have for any $\theta > 1$,

$$\begin{aligned}
\mathbb{E}_\theta[\phi(T)] &= \int_0^1 \left(t - \frac{n}{n+1} \right) f_\theta(t) dt \\
&= n\theta^{-n} \int_0^1 t^{n-1} dt - (n+1)\theta^{-n} \int_0^1 t^n dt \\
&= \theta^{-n} - \theta^{-n} \\
&= 0
\end{aligned}$$

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- Is the (very sensible) $\hat{\theta}'$ an UMVUE in this case?
- Next, we shall provide an alternative characterization of UMVUE based on the notion of *unbiased estimators of* 0 and use this characterization to show that $\hat{\theta}'$ is indeed an UMVUE of θ, knowing that $\theta > 1$

Variance reduction via unbiased estimators of 0

- Mathematically, for a given likelihood family $f_\theta(x)$, a statistic $U = U(X)$ is called an *unbiased estimator of* 0 if $\mathbb{E}_\theta[U] = 0$ for *all* $\theta \in \Theta$

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- Let $\delta_0 = \delta_0(X)$ be an unbiased estimator of $g(\theta)$ and let $U = U(X)$ be an unbiased estimator of 0
- Note that for any real α,

$$\delta_\alpha = \delta_0 + \alpha U$$

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- Further note that

$$\begin{aligned}\mathrm{Var}_\theta[\delta_\alpha] &= \mathrm{Var}_\theta[\delta_0 + \alpha U] \\ &= \mathrm{Var}_\theta[\delta_0] + \alpha^2 \mathrm{Var}_\theta[U] + 2\alpha \mathrm{Cov}_\theta[\delta_0, U],\end{aligned}$$

  which is a *quadratic* function of α

- Let

$$g(\alpha) := \mathrm{Var}_\theta[\delta_\alpha] = \mathrm{Var}_\theta[\delta_0] + \alpha^2 \mathrm{Var}_\theta[U] + 2\alpha \mathrm{Cov}_\theta[\delta_0, U]$$

and note that

$$g(0) = \mathrm{Var}_\theta[\delta_0] \quad \text{and} \quad g'(0) = 2\mathrm{Cov}_\theta[\delta_0, U]$$

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we can always find a nonzero α such that

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- Therefore, for any unbiased estimator δ_0 of $g(\theta)$, if there exists an unbiased estimator U of 0 such that

$$\mathrm{Cov}_\theta[\delta_0, U] \neq 0,$$

then δ_0 is *not* an UMVUE of $g(\theta)$

UMVUE and unbiased estimators of 0

- That is,

$$\text{Cov}_\theta[\delta_0, U] = 0$$

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- Let δ be any unbiased estimator $g(\theta)$. Then $\delta - \delta_0$ is an unbiased estimator of 0 and hence by our assumption,
$$\mathrm{Cov}_\theta[\delta_0, \delta - \delta_0] = 0, \quad \forall \theta \in \Theta$$

- It follows that

$$\begin{aligned}
\mathrm{Var}_\theta[\delta] &= \mathrm{Var}_\theta[(\delta_0 + (\delta - \delta_0))] \\
&= \mathrm{Var}_\theta[\delta_0] + \mathrm{Var}_\theta[\delta - \delta_0] + 2\mathrm{Cov}_\theta[\delta_0, \delta - \delta_0] \\
&= \mathrm{Var}_\theta[\delta_0] + \mathrm{Var}_\theta[\delta - \delta_0] \\
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- (*Theorem*) That is,

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- In light of the *Rao-Blackwell* theorem, to find an UMVUE, it suffices to consider estimators that are functions of a *sufficient* statistic. We have the following corollary to the previous theorem

- (Corollary) Let T be a sufficient statistic, and let $\delta_0 = \delta_0(T)$ be an unbiased estimator of $g(\theta)$. Then δ_0 is an UMVUE of $g(\theta)$ *if and only if*

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- Note that if T is a *complete* sufficient statistic, the *only* unbiased estimator U of 0 that is a function of T is $U = 0$ with probability 1

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- (Proof of the corollary) The "only if" part of the corollary follows directly from the "only if" part of the theorem

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Then, U'' is an unbiased estimator of 0 that is a function of T
- We thus have
$$\begin{aligned} \mathrm{Cov}_\theta[\delta_0, U'] &= \mathbb{E}_\theta[(\delta_0(T) - g(\theta))U'] \\ &= \mathbb{E}_\theta\left[\mathbb{E}_\theta[(\delta_0(T) - g(\theta))U'|T]\right] \\ &= \mathbb{E}_\theta\left[(\delta_0(T) - g(\theta)U''\right] \\ &= \mathrm{Cov}_\theta[\delta_0, U''] \\ &= 0 \end{aligned}$$
by the assumption that δ_0 is uncorrelated with any unbiased estimator of 0 that is also a function of T

- The fact that δ_0 is uncorrelated with any unbiased estimator of 0 (which may or may not be a function of T) proves that it is an UMVUE

Example: Uniform with unknown support ($\theta > 1$)

- Recall that for any $\theta > 1$, we have shown that

$$\hat{\theta}' = \begin{cases} 1, & \text{if } T \leq 1 \\ \frac{n+1}{n}T, & \text{if } T > 1 \end{cases}$$

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- To show that is is an UMVUE, it suffices to show that it is uncorrelated with any unbiased estimator of 0 that is a function of T

- Let $U = U(T)$ be an unbiased estimator of 0. For any $\theta > 1$, we have

$$0 = \mathbb{E}_\theta[U(T)] = \int_0^\theta U(t) f_\theta(t) dt = n\theta^{-n} \int_0^\theta U(t) t^{n-1} dt$$

  and hence

$$\int_0^\theta U(t) t^{n-1} dt = 0$$

- Taking derivative over θ gives

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- Combining the facts that $U(t) = 0$ for any $t > 1$ and $\int_0^\theta U(t)t^{n-1}dt = 0$ for any $\theta > 1$ gives

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- That is, for *any* unbiased estimator U of 0 that is a function of T, we must have 1) $U(t) = 0$ for any $t > 1$; and 2)

$$\int_0^1 U(t)t^{n-1}dt = 0$$

- For any $\theta > 1$, the covariance

$$\begin{aligned}
\mathrm{Cov}_\theta[\hat{\theta}', U] &= \mathbb{E}_\theta[(\hat{\theta}' - \theta)U] \\
&= \mathbb{E}_\theta[\hat{\theta}' U] \\
&= \int_0^\theta \hat{\theta}'(t) U(t) f_\theta(t) dt \\
&= \int_0^1 \hat{\theta}'(t) U(t) f_\theta(t) dt \\
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- This completes the proof that

$$\hat{\theta}' = \begin{cases} 1, & \text{if } T \leq 1 \\ \frac{n+1}{n}T, & \text{if } T > 1 \end{cases}$$

is indeed an UMVUE of θ, knowing that $\theta > 1$

The uniqueness of UMVUE

- We conclude this lecture by proving that UMVUE is *unique* in the sense that if both $\delta(X)$ and $\delta'(X)$ are an UMVUE of $g(\theta)$, then we have

$$\mathbb{P}_\theta(\delta(X) = \delta'(X)) = 1, \quad \forall \theta \in \Theta$$

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- (Proof) Let $\phi = \delta(X) - \delta'(X)$. Since both $\delta(X)$ and $\delta'(X)$ are unbiased estimators of $g(\theta)$, ϕ is an unbiased estimator of 0

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- (Proof) Let $\phi = \delta(X) - \delta'(X)$. Since both $\delta(X)$ and $\delta'(X)$ are unbiased estimators of $g(\theta)$, ϕ is an unbiased estimator of 0
- Since both $\delta(X)$ and $\delta'(X)$ are an UMVUE of $g(\theta)$, by the previous theorem, we must have

$$\mathrm{Cov}_\theta[\delta(X), \phi] = 0 \quad \text{and} \quad \mathrm{Cov}_\theta[\delta'(X), \phi] = 0$$

for all $\theta \in \Theta$

The uniqueness of UMVUE

- We conclude this lecture by proving that UMVUE is *unique* in the sense that if both $\delta(X)$ and $\delta'(X)$ are an UMVUE of $g(\theta)$, then we have

$$\mathbb{P}_\theta(\delta(X) = \delta'(X)) = 1, \quad \forall \theta \in \Theta$$

- (Proof) Let $\phi = \delta(X) - \delta'(X)$. Since both $\delta(X)$ and $\delta'(X)$ are unbiased estimators of $g(\theta)$, ϕ is an unbiased estimator of 0
- Since both $\delta(X)$ and $\delta'(X)$ are an UMVUE of $g(\theta)$, by the previous theorem, we must have

$$\mathrm{Cov}_\theta[\delta(X), \phi] = 0 \quad \text{and} \quad \mathrm{Cov}_\theta[\delta'(X), \phi] = 0$$

  for all $\theta \in \Theta$
- It follows that

$$\begin{aligned}
\mathrm{Var}_\theta[\phi] &= \mathbb{E}_\theta[\phi^2] = \mathbb{E}_\theta[(\delta(X) - \delta'(X))\phi] \\
&= \mathbb{E}_\theta[((\delta(X) - g(\theta)) - (\delta'(X) - g(\theta)))\phi] \\
&= \mathrm{Cov}_\theta[\delta(X), \phi] - \mathrm{Cov}_\theta[\delta'(X), \phi] = 0, \quad \forall \theta \in \Theta
\end{aligned}$$

- We thus conclude that

$$\mathbb{P}_\theta[\phi = 0] = \mathbb{P}_\theta[\delta(X) = \delta'(X)] = 1$$

for all $\theta \in \Theta$

ECEN 662: Estimation & Detection Theory

Lecture 6: Uniformly Minimum-Variance Unbiased Estimation (Part III)

Tie Liu

Texas A&M University

Finding the UMVUE

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- In this lecture, we shall introduce a *universal* lower bound on the variance of *any* unbiased estimator

- Therefore, any unbiased estimator whose variance *matches* this universal lower bound is the UMVUE

Fisher information

- Assume that our likelihood function $f_\theta(x)$ satisfies the following two *regularity* assumptions:
  - The *support* $\{x : f_\theta(x) > 0\}$ does *not* depend on θ
  - For *any* scalar statistic $T(X)$ such that $\mathbb{E}_\theta[|T(X)|] < \infty$ for all $\theta \in \Theta$,
  $$\nabla_\theta\left(\mathbb{E}_\theta\left[T(X)\right]\right) = \int_{\mathcal{X}} T(x)\nabla_\theta\left(f_\theta(x)\right)dx$$

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- (*The fundamental identity of score function*)

$$\mathbb{E}_\theta[s_\theta(X)T(X)] = \nabla_\theta\left(\mathbb{E}_\theta[T(X)]\right)$$

for any scalar statistic $T(X)$ such that $\mathbb{E}_\theta[|T(X)|] < \infty$ for all $\theta \in \Theta$

Proof of the score identity

$$
\begin{aligned}
\mathbb{E}_\theta[s_\theta(X)T(X)] &= \int_{\mathcal{X}} s_\theta(x)T(x)f_\theta(x)dx \\
&= \int_{\mathcal{X}} \nabla_\theta \left(\log f_\theta(x)\right) T(x)f_\theta(x)dx \\
&= \int_{\mathcal{X}} \nabla_\theta \left(f_\theta(x)\right) T(x)dx \\
&= \nabla_\theta \left(\int_{\mathcal{X}} f_\theta(x)T(x)dx\right) \\
&= \nabla_\theta \left(\mathbb{E}_\theta[T(X)]\right)
\end{aligned}
$$

Fisher information and average curvature

- Applying the score identity with $T(x) = 1$ gives

$$\mathbb{E}_\theta[s_\theta(X)] = \nabla_\theta \left(\mathbb{E}_\theta[1]\right) = 0,$$

i.e., the score function $s_\theta(X)$ always has a *zero* mean

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- The *covariance matrix* of the score function

$$\mathrm{Var}_\theta[s_\theta(X)] = \mathbb{E}_\theta[s_\theta(X) s_\theta^t(X)]$$

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- If the log-likelihood $\log f_\theta(x)$ is *twice* differentiable with respect to θ and under the additional regularity assumption that for any scalar statistic $T(x)$ such that $\mathbb{E}_\theta[|T(x)|] < \infty$ for all $\theta \in \Theta$,

$$\nabla_\theta^2\left(\mathbb{E}_\theta\left[T(x)\right]\right) = \int_{\mathcal{X}} T(x)\nabla_\theta^2\left(f_\theta(x)\right)dx$$

- We have

$$\nabla^2_\theta \left(\log f_\theta(x)\right) = \frac{\nabla^2_\theta \left(f_\theta(x)\right)}{f_\theta(x)} - s_\theta(x) s^t_\theta(x)$$

and hence

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  and hence

$$I(\theta) = -\mathbb{E}_\theta \left[\nabla_\theta^2 \left(\log f_\theta(X)\right)\right]$$

- Note that $-\frac{\partial^2}{\partial\theta^2} \log f_\theta(x)$ is the *curvature* of the log-likelihood $\log f_\theta(x)$ with respect to θ for a given x, so Fisher information measures the *average* curvature of the log-likelihood $\log f_\theta(x)$ at a given value of θ

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- More specifically, let $s_\eta(x)$ and $I(\eta)$ be Fisher score and the Fisher information with respect to the parameter η, respectively
- If $\eta = \eta(\theta)$ is a *continuous and differentiable* mapping from θ to η, we have

$$s_\theta(x) = \nabla_\theta \left(\log \tilde{f}_\eta(x)\right) = J_\eta^t \nabla_\eta \left(\log \tilde{f}_\eta(x)\right) = J_\eta^t s_\eta(x)$$

where $J_\eta = \left[\frac{\partial \eta_i}{\partial \theta_j}\right]$ is the *Jacobian* of $\eta = \eta(\theta)$

- We thus have the following *re-parameterization* formula for Fisher information:

$$\begin{aligned} I(\theta) &= \mathbb{E}_\theta[s_\theta(X)s_\theta^t(X)] \\ &= J_\eta^t \mathbb{E}_\theta[s_\eta(X)s_\eta^t(X)]J_\eta \\ &= J_\eta^t \mathbb{E}_\eta[s_\eta(X)s_\eta^t(X)]J_\eta \\ &= J_\eta^t I(\eta) J_\eta \end{aligned}$$

Exponential family of distributions

- In particular, an *exponential* family of distributions can be re-parameterized in terms of the natural parameter η as:

$$\tilde{f}_\eta(x) = h(x) \exp(\eta^t T(x) - A(\eta))$$

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- By the *re-parameterization* formula, the Fisher information with respect to θ is given by

$$I(\theta) = J_\eta^t I(\eta) J_\eta = J_\eta^t \nabla_\eta^2 A(\eta) J_\eta$$

Additivity over independent measurements

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- By the *independence* of the measurements, the Fisher information

$$I_n(\theta) = \mathrm{Var}_\theta\left[s_\theta(X)\right] = \sum_{i=1}^{n} \mathrm{Var}_\theta[s_\theta(X_i)] = nI_1(\theta)$$

  i.e., Fisher information is *additive* over independent measurements

Example: Gaussian with unknown mean and variance

- Recall that in this case ($\theta_1 = \mu$ and $\theta_2 = \sigma^2$), the marginal distributions are exponential families of distributions with

$$\eta(\theta) = \left(\frac{\theta_1}{\theta_2}, -\frac{1}{2\theta_2}\right), \quad T(x_i) = (x_i, x_i^2)$$

  and log-partition function

$$A(\eta) = -\frac{\eta_1^2}{4\eta_2} - \frac{1}{2}\log(-\eta_2)$$

- The Fisher information

$$
\begin{aligned}
I_n(\theta) &= nI_1(\theta) \\
&= nJ_\eta^t \nabla_\eta^2 A(\eta) J_\eta \\
&= n \begin{bmatrix} \frac{1}{\theta_2} & -\frac{\theta_1}{\theta_2^2} \\ 0 & \frac{1}{2\theta_2^2} \end{bmatrix}^t \begin{bmatrix} -\frac{1}{2\eta_2} & \frac{\eta_1}{2\eta_2^2} \\ \frac{\eta_1}{2\eta_2^2} & -\frac{\eta_1^2}{2\eta_2^3} + \frac{1}{2\eta_2^2} \end{bmatrix} \begin{bmatrix} \frac{1}{\theta_2} & -\frac{\theta_1}{\theta_2^2} \\ 0 & \frac{1}{2\theta_2^2} \end{bmatrix} \\
&= n \begin{bmatrix} \frac{1}{\theta_2} & -\frac{\theta_1}{\theta_2^2} \\ 0 & \frac{1}{2\theta_2^2} \end{bmatrix}^t \begin{bmatrix} \theta_2 & 2\theta_1\theta_2 \\ 2\theta_1\theta_2 & 4\theta_1^2\theta_2 + 2\theta_2^2 \end{bmatrix} \begin{bmatrix} \frac{1}{\theta_2} & -\frac{\theta_1}{\theta_2^2} \\ 0 & \frac{1}{2\theta_2^2} \end{bmatrix} \\
&= \begin{bmatrix} \frac{n}{\theta_2} & 0 \\ 0 & \frac{n}{2\theta_2^2} \end{bmatrix} \\
&= \begin{bmatrix} \frac{n}{\sigma^2} & 0 \\ 0 & \frac{n}{2\sigma^4} \end{bmatrix}
\end{aligned}
$$

Cramér-Rao lower bound (CRLB)

- Assume that the likelihood function $f_\theta(x)$ satisfies the aforementioned *regularity* assumptions. We have

$$\mathrm{Var}_\theta[\hat{g}(X)] \geq J_\phi I^{-1}(\theta) J_\phi^t, \quad \forall \theta \in \Theta$$

for any estimator $\hat{g}(X)$ of $g(\theta)$, where

$$\phi(\theta) := \mathbb{E}_\theta[\hat{g}(X)],$$

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- Therefore, any *unbiased* estimator whose variance *equals*

$$J_g I^{-1}(\theta) J_g^t$$

is the *UMVUE* for $g(\theta)$

Proof

- For simplicity, let us assume that θ is a *scalar* parameter. The vector case follows essentially the same idea, but requires a bit more machinery

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- By the *Cauchy-Schwarz inequality*, we have

$$\mathrm{Var}_\theta[s_\theta(X)]\mathrm{Var}_\theta[\hat{g}(X)] \geq \left(\mathrm{Cov}_\theta[s_\theta(X), \hat{g}(X)]\right)^2$$

so

$$\mathrm{Var}_\theta[\hat{g}(X)] \geq \frac{\left(\mathrm{Cov}_\theta[s_\theta(X), \hat{g}(X)]\right)^2}{\mathrm{Var}_\theta[s_\theta(X)]}$$

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$$\mathrm{Var}_\theta[\hat{g}(X)] \geq \frac{(\mathrm{Cov}_\theta[s_\theta(X), \hat{g}(X)])^2}{I(\theta)}$$

- By the score identity, the covariance

$$\begin{aligned}\mathrm{Cov}_\theta[s_\theta(X), \hat{g}(X)] &= \mathbb{E}_\theta[s_\theta(X)\hat{g}(X)] - \mathbb{E}_\theta[s_\theta(X)]\mathbb{E}_\theta[\hat{g}(X)] \\ &= \mathbb{E}_\theta[s_\theta(X)\hat{g}(X)] \\ &= \frac{d}{d\theta}\mathbb{E}_\theta[\hat{g}(X)] \\ &= \frac{d}{d\theta}\phi(\theta)\end{aligned}$$

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  for all $\theta \in \Theta$

Example: Gaussian with unknown mean and variance

- Recall that $\theta = (\mu, \sigma^2)$, and the Fisher information matrix is given by:
$$I_n(\theta) = \begin{bmatrix} \frac{n}{\sigma^2} & 0 \\ 0 & \frac{n}{2\sigma^4} \end{bmatrix}$$

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- Thus, the CRLB for $g(\theta) = \theta_1 = \mu$ is

$$\begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} \frac{n}{\sigma^2} & 0 \\ 0 & \frac{n}{2\sigma^4} \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{\sigma^2}{n},$$

and the CRLB for $g(\theta) = \theta_2 = \sigma^2$ is

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- Recall that the *empirical mean*

$$\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

is unbiased for μ and its variance σ^2/n *matches* the CRLB. We can thus conclude that it is the UMVUE for μ

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- Also recall that for $n \geq 2$,

$$\widehat{\sigma^2} = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \hat{\mu})^2$$

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- Therefore, for σ^2, we *cannot* conclude from the CRLB that $\widehat{\sigma^2}$ is the UMVUE (even though it, in fact, is)

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- However, as shown in the previous example, the variance of the UMVUE may be strictly greater than the CRLB. In this case, an efficient estimator does *not* exist
- Also keep in mind that Fisher information and, by extension, CRLB are *only* defined for likelihood families that satisfy the *regularity* conditions

Efficient estimators

- In statistics, we say $\hat{g}(X)$ is an *efficient* estimator of $g(\theta)$ if it is *unbiased* and its variance *equals* the CRLB
- By definition, an *efficient* estimator, if exists, must be the UMVUE (and hence unique)
- However, as shown in the previous example, the variance of the UMVUE may be strictly greater than the CRLB. In this case, an efficient estimator does *not* exist
- Also keep in mind that Fisher information and, by extension, CRLB are *only* defined for likelihood families that satisfy the *regularity* conditions
- Next, we shall briefly mention the *Hammersley-Chapman-Robbins lower bound (HCRLB)*, which can be applied *without* the regularity conditions

Hammersley-Chapman-Robbins lower bound (HCRLB)

- We have

$$\mathrm{Var}_\theta[\hat{g}(X)] \geq \sup_{\theta' \in \Theta} \frac{\left(\phi(\theta') - \phi(\theta)\right)^2}{\mathbb{E}_\theta\left[\left(\frac{f_{\theta'}(X)}{f_\theta(X)} - 1\right)^2\right]}, \quad \forall \theta \in \Theta$$

for any estimator $\hat{g}(X)$ of $g(\theta)$, where

$$\phi(\theta) := \mathbb{E}_\theta[\hat{g}(X)]$$

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for any estimator $\hat{g}(X)$ of $g(\theta)$, where

$$\phi(\theta) := \mathbb{E}_\theta[\hat{g}(X)]$$

- In particular, if $\hat{g}(X)$ is *unbiased*, we have $\phi(\theta) = g(\theta)$ and hence

$$\mathrm{Var}_\theta[\hat{g}(X)] \geq \sup_{\theta' \in \Theta} \frac{\left(g(\theta') - g(\theta)\right)^2}{\mathbb{E}_\theta\left[\left(\frac{f_{\theta'}(X)}{f_\theta(X)} - 1\right)^2\right]}, \quad \forall \theta \in \Theta$$

- Let $\theta' = \theta + \epsilon$. Under the *regularity* conditions, it can be shown that

$$\lim_{\|\epsilon\| \to 0} \frac{\left(\phi(\theta') - \phi(\theta)\right)^2}{\mathbb{E}_\theta\left[\left(\frac{f_{\theta'}(X)}{f_\theta(X)} - 1\right)^2\right]} = J_\phi I^{-1}(\theta) J_\phi^t, \quad \forall \theta \in \Theta$$

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- Therefore, the HCRLB is at least as tight as the CRLB under the *regularity* conditions

- The real power of the HCRLB, however, lies in the fact that it can be applied even when the regularity conditions are *not* met

Example: Uniform with unknown support ($\theta > 0$)

- Note that for this example, the support of $f_\theta(x)$ depends on θ. Therefore, the regularity conditions are *not* met for this likelihood family and the CRLB does not apply

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- This gives

$$\mathrm{Var}_\theta[\hat{g}(X)] \geq \sup_{\theta' > \theta} \frac{(\theta' - \theta)^2}{\left(\frac{\theta'}{\theta}\right)^n - 1} = \sup_{\epsilon > 0} \frac{\theta^2 \epsilon^2}{(1+\epsilon)^n - 1}$$

ECEN 662: Estimation & Detection Theory

Lecture 7: Maximum-Likelihood Estimation

Tie Liu

Texas A&M University

Problems with UMVUE

- Finding an UMVUE can be mathematically challenging, and in some cases, it might *not* even exist

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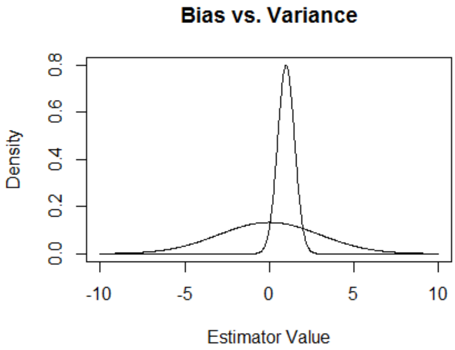
- Finding an UMVUE can be mathematically challenging, and in some cases, it might *not* even exist
- More importantly, while an UMVUE minimizes variance among unbiased estimators, strictly adhering to the *unbiased* constraint can sometimes lead to a *significantly* higher variance than a slightly biased estimator



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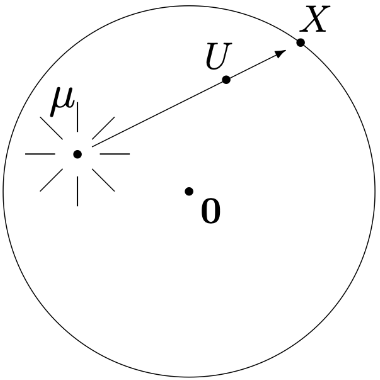
- Another illuminating example is to estimate the location of a light source based on its projection on the boundary of a dart board over a randomly placed dart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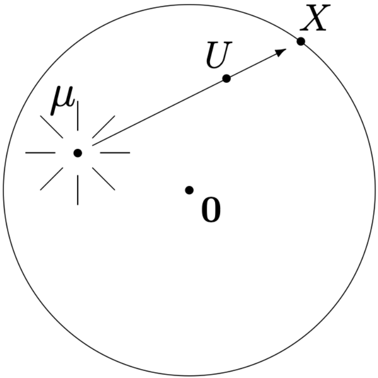

- The details can be found in Michael Hardy's short exposition

Maximum-likelihood estimator (MLE)

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Maximum-likelihood estimator (MLE)

- An alternative *frequentist* estimator is *maximum-likelihood estimator (MLE)*
- Let us first focus on estimating the model parameter θ itself (and defer the discussions on estimating a function of θ to the later part of the lecture)
- Given a likelihood family $f_\theta(x)$, an estimator $\hat{\theta} = \hat{\theta}(X)$ is an *MLE* of θ if

$$\hat{\theta}(x) \in \arg\max_{\theta \in \Theta} f_\theta(x), \quad \forall x \in \mathcal{X}$$

or, equivalently,

$$\hat{\theta}(x) \in \arg\max_{\theta \in \Theta} \log f_\theta(x), \quad \forall x \in \mathcal{X}$$

Maximizing likelihood and minimizing K-L divergence

- While mainly motivated by *intuition*, MLE has an interesting interpretation when the observation $X = (X_1, \ldots, X_n)$, where

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$$\hat{p}_x(z) = \frac{1}{n} \sum_{i=1}^{n} 1_{\{x_i = z\}}, \quad \forall z \in \mathcal{X}$$

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- For two pmfs p and q over the same alphabet $\mathcal{X}$, the *K-L divergence* between p and q is defined as:

$$D_{KL}(p \| q) := \mathbb{E}_p \left[\log \frac{p(Z)}{q(Z)} \right] = \sum_{z \in \mathcal{X}} p(z) \log \frac{p(z)}{q(z)}$$

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- (*Claim*) Maximizing the likelihood

$$\sum_{i=1}^{n} \log p_\theta(x_i)$$

is *equivalent* to minimizing the K-L divergence

$$D_{KL}(\hat{p}_x\|p_\theta)$$

Proof

- First note that

$$\begin{aligned}D_{KL}(\hat{p}_x \| p_\theta) &= \mathbb{E}_{\hat{p}_x}\left[\log \frac{\hat{p}_x(Z)}{p_\theta(Z)}\right] \\ &= \mathbb{E}_{\hat{p}_x}\left[\log \hat{p}_x(Z)\right] - \mathbb{E}_{\hat{p}_x}\left[\log p_\theta(Z)\right]\end{aligned}$$

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\end{aligned}
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- Further note that

$$
\begin{aligned}
\mathbb{E}_{\hat{p}_x}\left[\log p_\theta(Z)\right] &= \sum_{z \in \mathcal{X}} \hat{p}_x(z) \log p_\theta(z) \\
&= \frac{1}{n} \sum_{z \in \mathcal{X}} \sum_{i=1}^{n} 1_{\{x_i = z\}} \log p_\theta(z) \\
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&= \frac{1}{n} \sum_{i=1}^{n} \log p_\theta(x_i)
\end{aligned}
$$

- Combining the above two results gives

$$D_{KL}(\hat{p}_x \| p_\theta) = \mathbb{E}_{\hat{p}_x}\left[\log \hat{p}_x(Z)\right] - \frac{1}{n}\sum_{i=1}^{n} \log p_\theta(x_i)$$

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- We may thus conclude that minimizing the K-L divergence $D_{KL}(\hat{p}_x \| p_\theta)$ is *equivalent* to maximizing the likelihood $\sum_{i=1}^{n} \log p_\theta(x_i)$
- The above result also suggests that if our model is *identifiable*, MLE will recover θ in the limit as $n \to \infty$. We will have more to say about this result a bit later

Re-parameterization

- Assume that the likelihood function $f_\theta(x)$ can be *re-parameterized* in terms of a more *natural* parameter

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- In this case, for any given observation x, we have

$$\arg\max_{\theta \in \Theta} f_\theta(x) = \left\{ \theta \in \Theta : \eta(\theta) \in \arg\max_{\eta \in \eta(\Theta)} \tilde{f}_\eta(x) \right\},$$

where $\tilde{f}_\eta(x)$ is the *re-parameterized* model

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- That is, to find an MLE of θ, we can first find an MLE of η and then apply the "inverse mapping" of $\eta(\theta)$

Exponential family of distributions

- Consider an *exponential* family of distribution re-parameterized by the *natural* parameter η:

$$\tilde{f}_\eta(x) = h(x) \exp(\eta^t T(x) - A(\eta))$$

where the space of the natural parameter is an *open and connected* subset of $\mathbb{R}^s$

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- To find an MLE for η, note that

$$\nabla_\eta \log \tilde{f}_\eta(x) = T(x) - \nabla_\eta A(\eta) = T(x) - \mathbb{E}_\eta[T(X)]$$
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- A covariance matrix is always positive semi-definite, so for any $x \in \mathcal{X}$, the log-likelihood $\log \tilde{f}_\eta(x)$ is a *concave* function of η, for which any *stationary* point is a global maxima

- Therefore, any solution for η to the equation

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- Furthermore, if the covariance matrix of $T(X)$ is *positive definite* for all η, for any $x \in \mathcal{X}$ the log-likelihood $\log \tilde{f}_\eta(x)$ is a *strictly concave* function of η. In this case, the MLE is *unique* if it exists

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- Finally, by the *re-parameterization* property, *any* solution for θ to the equation

$$\mathbb{E}_\theta[T(X)] = T(x)$$

is an MLE for θ

Gaussian with unknown mean and variance

- Recall that in this case, a sufficient statistic is given by

$$T_1 = \sum_{i=1}^{n} X_i \quad \text{and} \quad T_2 = \sum_{i=1}^{n} X_i^2$$

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- By our previous discussion, any solution of (μ, σ^2) to the equations

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- The above equations have a *unique* solution given by

$$\mu = \frac{T_1(x)}{n} \quad \text{and} \quad \sigma^2 = \frac{T_2(x)}{n} - \left(\frac{T_1(x)}{n}\right)^2$$

- We may thus conclude that the *empirical mean*

$$\hat{\mu} = \frac{T_1(x)}{n} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

is the MLE for μ and the *empirical variance*

$$\widehat{\sigma^2} = \frac{T_2(x)}{n} - \left(\frac{T_1(x)}{n}\right)^2 = \frac{1}{n}\sum_{i=1}^{n} X_i^2 - \hat{\mu}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \hat{\mu})^2$$

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- By *Cochran's theorem*,

$$\mathbb{E}_\theta\left[\sum_{i=1}^{n}(X_i - \hat{\mu})^2\right] = (n-1)\sigma^2$$

so

$$\mathbb{E}_\theta\left[\widehat{\sigma^2}\right] = \frac{n-1}{n}\sigma^2$$

i.e., the empirical variance is a *biased* estimator for the true variance σ^2

- However, compared with the UMVUE

$$\frac{1}{n-1}\sum_{i=1}^{n}(X_i - \hat{\mu})^2$$

whose variance is given by $2\sigma^4/(n-1)$, the MLE

$$\frac{1}{n}\sum_{i=1}^{n}(X_i - \hat{\mu})^2$$

has a *smaller* variance $2\sigma^4/n$ (at the expense of being slightly biased)

Consistency and asymptotic normality

- Note that in the previous example, while the MLE for σ^2 is biased, the bias *vanishes asymptotically* in the limit as the sample size $n \to \infty$

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- Under certain regularity conditions, it can be shown that the above observations hold for a *general* likelihood function

- More specifically, let $X = (X_1, \ldots, X_n)$, where $X_i \overset{iid}{\sim} f_\theta(x_i)$. For any θ that is in the *interior* of Θ, let $\hat{\theta}$ be the MLE for θ given by the *unique* solution to the equation

$$\nabla_\theta \left(\sum_{i=1}^{n} \log f_\theta(x_i) \right) = 0$$

  We have $\sqrt{n}(\hat{\theta} - \theta)$ converges in distribution to $\mathcal{N}(0, I_1^{-1}(\theta))$ in the limit as the sample size $n \to \infty$

- That is, when the sample size n is large, the MLE $\hat{\theta}$ is *approximately* distributed as

$$\mathcal{N}\left(\theta, \frac{1}{n} I_1^{-1}(\theta)\right)$$

or, equivalently,

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- The proof of the *asymptotic normality* of the MLE is based on the well-known *central limit theorem*. The details are omitted from this note

Multinomial with unknown probabilities

- Let $X \sim p_\theta(x)$, where the observation

$$X = (N_1, \ldots, N_K)$$

is the *empirical counts* over K categories and n total counts, the unknown parameter

$$\theta = (p_1, \ldots, p_K)$$

is a *probability* vector over K categories, and the likelihood function is *multinomial*:

$$p_\theta(x) = \frac{n!}{n_1! \cdots n_K!} \prod_{k=1}^{K} p_k^{n_k}$$

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  $$p_\theta(x) = \frac{n!}{n_1! \cdots n_K!} \prod_{k=1}^{K} p_k^{n_k}$$

- The total counts n and the number of categories K are assumed to be *known*

Multinomial with unknown probabilities

- Let $X \sim p_\theta(x)$, where the observation

$$X = (N_1, \dots, N_K)$$

is the *empirical counts* over K categories and n total counts, the unknown parameter

$$\theta = (p_1, \dots, p_K)$$

is a *probability* vector over K categories, and the likelihood function is *multinomial*:

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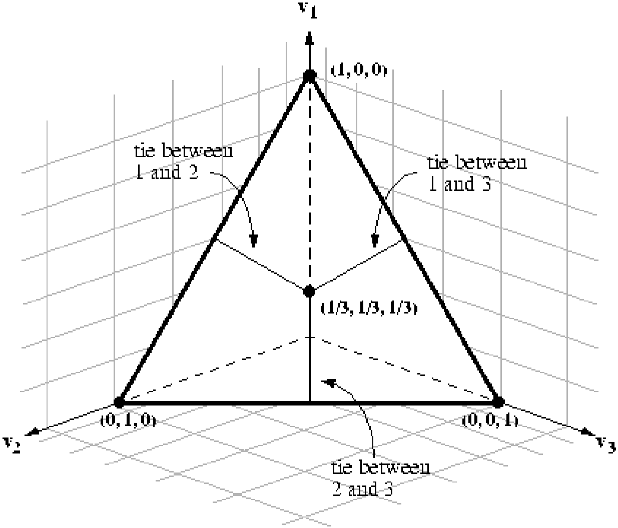
- Note that for this example, the parameter space Θ is the $(K-1)$-*simplex* and has *no* interior in $\mathcal{R}^K$



v_1
(1, 0, 0)
tie between
1 and 2
tie between
1 and 3
(1/3, 1/3, 1/3)
(0, 1, 0)
(0, 0, 1)
v_2
v_3
tie between
2 and 3

Finding MLE via the Lagrangian method

- For this example, the problem of finding the MLE is given by the following *constrained* optimization problem:

$$\begin{aligned} \text{Maximize} \quad & \log p_\theta(x) \\ \text{Subject to} \quad & \sum_{k=1}^{K} p_k = 1 \\ & p_k \geq 0, \quad \forall k \in [K] \end{aligned}$$

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- In optimization theory, the standard approach for solving constrained optimization problems are the *Lagrangian method*
- More specifically, let

$$L_\lambda(\theta) := \log p_\theta(x) - \lambda \sum_{k=1}^{K} p_k$$

where λ is the *Lagrangian multiplier*

- The MLE can then be found by solving the equation

$$\nabla_\theta L_\lambda(\theta) = 0$$

and then finding a Lagrangian multiplier $\lambda \geq 0$ to satisfy the constraint

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$$\frac{\partial}{\partial p_k} L_\lambda(\theta) = \frac{n_k}{p_k} - \lambda = 0$$

gives

$$p_k = \frac{n_k}{\lambda}, \quad \forall k \in [K]$$

- Finally, to satisfy the constraint

$$\sum_{k=1}^{K} p_k = 1$$

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and hence

$$\lambda = \sum_{k=1}^{K} n_k = n$$

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- We thus conclude that the MLE for θ is given by the *empirical frequency* estimator

$$\hat{\theta} = \left(\frac{N_1}{n}, \ldots, \frac{N_K}{n} \right)$$

- Note that for any $k \in [K]$

$$\mathbb{E}_\theta[N_k] = np_k$$

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- Thus by the *Lehmann–Scheffé theorem*, for this example, the MLE is also the (unique) UMVUE

Finding MLE via iterative numerical techniques

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- Gradient descent methods for minimization such as *Newton-Raphson* and *Fisher's scoring* methods are commonly used
- In the next lecture, we will discuss a method known as *expectation-maximization (EM)* algorithm, which is specifically designed for computing MLE for models with *latent* variables

Estimating $g(\theta)$

- Let $\eta = g(\theta)$. Note that if the likelihood function $f_\theta(x)$ can be *re-parameterized* in terms of η, then it is *natural* to use an *MLE* of η as an "*MLE*" estimate of $g(\theta)$

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- In statistics, it is *customary* to define the MLE of $g(\theta)$ as $g(\theta^*)$, where θ^* is an MLE of θ, whether the likelihood function $f_\theta(x)$ can be re-parameterized in terms of $g(\theta)$ or not. This is known as the *invariance* property of MLE

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- Many other estimators such as UMVUE do *not* enjoy the invariance property

ECEN 662: Estimation & Detection Theory

Lecture 8: The Expectation-Maximization Algorithm

Tie Liu

Texas A&M University

Models with latent variables

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- The above optimization problem can be difficult to solve analytically due to the *marginalization* of the latent variables, which occurs *inside* the logarithm of the objective function
- Next, let us use the *mixture Gaussian model* as a concrete example to see what exactly the aforementioned challenge is and how the problem may be solved with *iterative numerical techniques*

Mixture Gaussian model

- Let $X = (X_1, \ldots, X_n)$ and $U = (U_1, \ldots, U_n)$, where (X_i, U_i) are i.i.d. according to:

$$U_i \sim Cat(u_i; \pi_1, \ldots, \pi_K) \quad \text{and} \quad X_i | U_i \sim \mathcal{N}(x_i; \mu_{u_i}, \sigma^2_{u_i})$$

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- *Marginalizing* over U_i gives

$$f_\theta(x_i) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x_i; \mu_k, \sigma_k^2)$$

which is a *mixture Gaussian* distribution, and

$$\theta = (\pi_1, \ldots, \pi_K, \mu_1, \ldots, \mu_K, \sigma_1^2, \ldots, \sigma_K^2)$$

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- The log-likelihood for all measurements is given by

$$\log f_\theta(x) = \sum_{i=1}^{n} \log f_\theta(x_i) = \sum_{i=1}^{n} \log \left(\sum_{k=1}^{K} \pi_k \mathcal{N}(x_i; \mu_k, \sigma_k^2) \right)$$

Finding MLE for $(\mu_1, \ldots, \mu_K)$

- Assume for the moment that $(\pi_1, \ldots, \pi_K)$ and $(\sigma_1^2, \ldots, \sigma_K^2)$ are known. To find the MLE for $(\mu_1, \ldots, \mu_K)$, note that for any $j \in [K]$ we have

$$\frac{\partial}{\partial \mu_j} \log f_\theta(x) = \sum_{i=1}^{n} \frac{\tilde{\pi}_j(x_i)(x_i - \mu_j)}{\sigma_j^2}$$

where

$$\tilde{\pi}_j(x_i) = \frac{\pi_j \mathcal{N}(x_i; \mu_j, \sigma_j^2)}{\sum_{k=1}^{K} \pi_k \mathcal{N}(x_i; \mu_k, \sigma_k^2)} = \mathbb{P}_\theta(U_i = j | X_i = x_i)$$

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- Setting the partial derivative equal to zero for all j gives the following set of equations:

$$\sum_{i=1}^{n} \frac{\tilde{\pi}_j(x_i)(x_i - \mu_j)}{\sigma_j^2} = 0, \quad \forall j \in [K]$$

  for which the solution for $(\mu_1, \ldots, \mu_K)$ is the MLE

A vicious circle

- Note that if $\tilde{\pi}_j(x_i)$'s are *known*, the above equations are *separate* for each μ_j and hence are very easy to solve:

$$\mu_j = \frac{\sum_{i=1}^n \tilde{\pi}_j(x_i) x_i}{\sum_{i=1}^n \tilde{\pi}_j(x_i)}, \quad \forall j \in [K]$$

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- We are now in the following situation:
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- One man's vicious circle is another man's *successive approximation* procedure

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- Next, we shall *generalize* the above procedure to general likelihood models with *latent* variables, and the resulting procedure is known as the *expectation-maximization (EM)* algorithm

A variational characterization of the log-likelihood

- The main idea behind the *EM* algorithm is the following *variational* characterization of the log-likelihood:

$$\log f_\theta(x) = \max_q F(q, \theta)$$

where

$$\begin{aligned} F(q, \theta) &= \log f_\theta(x) - D_{KL}(q(u) \| f_\theta(u|x)) && (1) \\ &= \mathbb{E}_q \left[\log f_\theta(x, U)\right] - \mathbb{E}_q \left[\log q(U)\right], && (2) \end{aligned}$$

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- The main advantage of this *variational* characterization is that it allows us to rewrite the problem of finding the MLE as a *double-maximization* problem:

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$$(\hat{\theta}, q^*) \in \arg\max_{(\theta, q)} F(q, \theta)$$

- q^* can be disregarded if our sole objective is to find the MLE for θ, but is usually quite useful under specific settings

Proof

- First note that by the *information* inequality, for any density function q we have

$$D_{KL}(q(u)\|f_\theta(u|x)) \geq 0$$

where the equality holds if and only if

$$q(u) = f_\theta(u|x), \quad \forall u \in \mathcal{U}$$

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- We thus have

$$\log f_\theta(x) \geq \log f_\theta(x) - D_{KL}(q(u) \| f_\theta(u|x))$$

where the equality holds if and only if

$$q(u) = f_\theta(u|x), \quad \forall u \in \mathcal{U}$$

and hence

$$\log f_\theta(x) = \max_q \left(\log f_\theta(x) - D_{KL}(q(u) \| f_\theta(u|x))\right)$$

- Further note that

$$
\begin{aligned}
&\log f_\theta(x) - D_{KL}(q(u) \| f_\theta(u|x)) \\
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&\quad = \mathbb{E}_q \left[\log \left(f_\theta(x) f_\theta(U|x)\right)\right] - \mathbb{E}_q \left[\log q(U)\right] \\
&\quad = \mathbb{E}_q \left[\log f_\theta(x, U)\right] - \mathbb{E}_q \left[\log q(U)\right]
\end{aligned}
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\end{aligned}
$$

- We have thus proved that expressions (1) and (2) are in fact *equivalent* to each other

Finding MLE via alternating maximization

- Recall that the MLE can be found by solving the *double-maximization* problem:

$$\max_{(\theta,q)} F(q,\theta)$$

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- Consider the following *alternating-maximization* strategy for solving the above *double-maximization* problem:
  - Initialize $\theta = \theta^{(0)}$ and $t = 1$
  - Repeat the following steps until the values of (θ, q) converge:
    - Fix $\theta = \theta^{(t-1)}$ and find

    $$q^{(t)} \in \arg\max_q F\left(q, \theta^{(t-1)}\right)$$

    - Fix $q = q^{(t)}$ and find

    $$\theta^{(t)} = \arg\max_\theta F\left(q^{(t)}, \theta\right)$$

    - $t \leftarrow t + 1$

- The problem of fixing $\theta = \theta^{(t-1)}$ and finding

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$$q^{(t)}(u) = f_{\theta^{(t-1)}}(u|x), \quad \forall u \in \mathcal{U}$$

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- By comparison, the problem of fixing $q = q^{(t)}$ and finding

$$\theta^{(t)} \in \arg\max_\theta F\left(q^{(t)}, \theta\right)$$

is much more involved

- To solve this optimization problem, we shall use expression (2) and write

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- The above two equations allow us to continuously update the value of (θ, q), leading to the famous *EM* algorithm

The EM algorithm

- The algorithm:
  - Initialize $\theta = \theta^{(0)}$ and $t = 1$
  - Do the following steps until the values of (θ, q) converge:
    - (E-step) Compute the *expectation*

      $$\mathbb{E}_{q^{(t)}}\left[\log f_\theta(x, U)\right]$$

      where

      $$q^{(t)}(u) = f_{\theta^{(t-1)}}(u|x), \quad \forall u \in \mathcal{U}$$
    - (M-step) Solve the *expectation-maximization*:

      $$\theta^{(t)} = \arg\max_\theta \mathbb{E}_{q^{(t)}}\left[\log f_\theta(x, U)\right]$$
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- Note that

$$\begin{aligned}\log f_{\theta^{(t-1)}}(x) &= F\left(q^{(t)}, \theta^{(t-1)}\right) \\ &\leq F\left(q^{(t)}, \theta^{(t)}\right) \leq F\left(q^{(t+1)}, \theta^{(t)}\right) = \log f_{\theta^{(t)}}(x)\end{aligned}$$

  so the algorithm increases *monotonically* the log-likelihood $\log f_\theta(x)$

Back to the mixture Gaussian model

- Recall that the M-step in each iteration is to solve

$$\max_{\theta \in \Theta} \mathbb{E}_{q^{(t)}} \left[\log f_\theta(x, U)\right]$$

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- For the *mixture Gaussian* model, we have

$$\begin{aligned}
&\mathbb{E}_{q^{(t)}} [\log f_\theta(x, U)] \\
&\quad = \sum_{i=1}^n \mathbb{E}_{q^{(t)}} [\log f_\theta(x_i, U_i)] \\
&\quad = \sum_{i=1}^n \mathbb{E}_{q^{(t)}} [\log f_\theta(x_i | U_i)] + \sum_{i=1}^n \mathbb{E}_{q^{(t)}} [\log f_\theta(U_i)] \\
&\quad = \sum_{i=1}^n \sum_{k=1}^K \pi_k^{(t)}(x_i) \log \mathcal{N}(x_i; \mu_k, \sigma_k^2) + \sum_{i=1}^n \sum_{k=1}^K \pi_k^{(t)}(x_i) \log \pi_k
\end{aligned}$$

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- By the standard *Lagrangian* method, we can find the optimal solution to the above optimization problem as:

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\mu_k^{(t)} &= \frac{\sum_{i=1}^n \pi_k^{(t)}(x_i) x_i}{\sum_{i=1}^n \pi_k^{(t)}(x_i)} \\
\sigma_k^{2\,(t)} &= \frac{\sum_{i=1}^n \pi_k^{(t)}(x_i) \left(x_i - \mu_k^{(t)}\right)^2}{\sum_{i=1}^n \pi_k^{(t)}(x_i)} \\
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- These estimates reduce to the standard empirical mean, variance, and frequency estimates when the posterior probability vectors $(\pi_1^{(t)}(x_i), \ldots, \pi_K^{(t)}(x_i))$ are *0-1* vectors

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- *Expectation conditional maximization (ECM)*: Performs the M-step by updating parameters *one at a time*
- *Generalized EM (GEM)*: Instead of performing the M-step exactly, it only requires a *non-decreasing* update

ECEN 662: Estimation & Detection Theory

Lecture 9: Generative Adversarial Network

Tie Liu

Texas A&M University

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- In this supplemental lecture, we shall see, yet again, that the idea of *variational characterization* leads to a highly successful learning algorithm, known as *generative adversarial network (GAN)*, for learning a *generative model*
- Together, these two examples highlight the power of *variational characterization* across both *inference* and *learning* problems
- These two examples will also help us to understand some of key *differences* between an *inference* problem and a *learning* problem

Learn to generate

- For the problem of *learn-to-generate*, the learner has access to m training data examples $(x_i : i \in [m])$ such that each x_i is drawn i.i.d. from an *unknown* data-generating distribution $p(x)$

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- The goal is to learn a *generator*, i.e. a conditional distribution $p_\theta(x|z)$, such that:
  - it is *easy* to draw samples of X from $p_\theta(x|z)$ for a given sample of Z
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- The generator $p_\theta(x|z)$ is commonly known as the *generative model* in the machine learning literature

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- First, GAN uses a *deterministic* generator $g_\theta(z)$ to produce new data samples
- Second, GAN uses the *Jensen-Shannon (JS) divergence* to measure the distance between the *target* data-generating distribution $p_\theta(x)$ and the *true* data-generating distribution $p(x)$:

$$D_{JS}(p_\theta \| p) = \frac{D_{KL}\left(p_\theta \,\middle\|\, \frac{p_\theta+p}{2}\right) + D_{KL}\left(p \,\middle\|\, \frac{p_\theta+p}{2}\right)}{2}$$

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- Note that unlike the *K-L divergence*, the *J-S divergence* is *symmetric* between its two input distributions

Inference vs. learning

- One may be tempted to use the J-S divergence $D_{JS}(p_\theta \| p)$ as the learning objective, but would quickly come to realization that this learning objective is *not* directly available because the true data-generating distribution $p(x)$ is *unknown*

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- This is the key difference between an inference problem such as finding the MLE and a learning problem such as learning a generative model: While for inference problems the challenge is entirely *computational*, learning problems face additional challenge of *estimating* the learning objectives from the training data examples

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- This is the key difference between an inference problem such as finding the MLE and a learning problem such as learning a generative model: While for inference problems the challenge is entirely *computational*, learning problems face additional challenge of *estimating* the learning objectives from the training data examples
- It is also important to note that learning algorithms have to take into account that the *estimated* learning objectives are subject to the *sampling noise*, because the number of training data examples is always *finite*

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- One way to address this challenge is through the following *variational characterization* of the J-S divergence:

$$D_{JS}(p_\theta \| p) = \max_d \frac{\mathbb{E}_{p_\theta}[\log d(X)] + \mathbb{E}_p[\log(1 - d(X))] + 2\log 2}{2}$$

  where the maximization is over a *discriminator* $d : \mathcal{X} \to [0, 1]$

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  where the maximization is over a *discriminator* $d : \mathcal{X} \to [0, 1]$
- An immediate consequence of the above *variational characterization* is that the new objective

$$L(\theta, d) := \mathbb{E}_{p_\theta}[\log d(X)] + \mathbb{E}_p[\log(1 - d(X))]$$

  can now be easily estimated from the training data examples:

$$L(\theta, d) \approx \frac{1}{m'} \sum_{i=1}^{m'} \log d(g_\theta(z_i)) + \frac{1}{m} \sum_{i=1}^{n} \log(1 - d(x_i))$$

  where z_i's are i.i.d. samples of the random source Z

- The problem of learning a generator can thus be posed as a *min-max* optimization problem:

$$\min_{\theta} \max_{d} \left(\frac{1}{m'} \sum_{i=1}^{m'} \log d(g_\theta(z_i)) + \frac{1}{m} \sum_{i=1}^{n} \log(1 - d(x_i)) \right)$$

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- In practice, one often *parametrizes* the discriminator d as well and optimizes it within the parameter family:

$$\min_{\theta} \max_{\phi} \left(\frac{1}{m'} \sum_{i=1}^{m'} \log d_\phi(g_\theta(z_i)) + \frac{1}{m} \sum_{i=1}^{n} \log(1 - d_\phi(x_i)) \right)$$

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- In each iterative step, both *partial* optimization problems can be solved efficiently using *gradient descent (GD)* or *stochastic gradient descent (SGD)* algorithms

Proof of the variational characterization

- First note that

$$
\begin{aligned}
&\mathbb{E}_{p_\theta}[\log d(X)] + \mathbb{E}_p[\log(1 - d(X))] \\
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- To maximize the above objective over d, we can choose $d(x) \in [0, 1]$ to maximize

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- Setting $g'(t) = 0$ gives:

$$\frac{p_\theta(x)}{t} - \frac{p(x)}{1-t} = 0 \quad \Rightarrow \quad t = \frac{p_\theta(x)}{p_\theta(x)+p(x)}$$

- We thus have

$$
\begin{aligned}
\max_{d(x)\in[0,1]} & \left(p_\theta(x)\log d(x) + p(x)\log(1-d(x))\right) \\
&= p_\theta(x)\log\frac{p_\theta(x)}{p_\theta(x)+p(x)} + p(x)\log\left(1-\frac{p_\theta(x)}{p_\theta(x)+p(x)}\right) \\
&= p_\theta(x)\log\frac{p_\theta(x)}{p_\theta(x)+p(x)} + p(x)\log\frac{p(x)}{p_\theta(x)+p(x)}
\end{aligned}
$$

and hence

$$
\begin{aligned}
\max_d & \left(\mathbb{E}_{p_\theta}[\log d(X)] + \mathbb{E}_p[\log(1-d(X))]\right) \\
&= \int_{\mathcal{X}} \left(p_\theta(x)\log\frac{p_\theta(x)}{p_\theta(x)+p(x)} + p(x)\log\frac{p(x)}{p_\theta(x)+p(x)}\right) dx \\
&= D_{KL}\left(p_\theta \,\middle\|\, \frac{p_\theta+p}{2}\right) + D_{KL}\left(p \,\middle\|\, \frac{p_\theta+p}{2}\right) - 2\log 2
\end{aligned}
$$

ECEN 662: Estimation & Detection Theory

Lecture 10: Decision-Theoretic Framework

Tie Liu

Texas A&M University

Decision-theoretic framework

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- For *parameter estimation*, $\mathcal{D} = \Theta$, and for *detection*, $\mathcal{D} = \{0, 1\}$. More generally, however, $\mathcal{D}$ can be the collection of any viable decisions
- A *loss function* is any function $L : \Theta \times \mathcal{D} \to [0, \infty)$, which is supposed to evaluate the *penalty* (or *error*) associated with the decision d when the parameter takes the value θ

Decision-theoretic framework

- In a *decision-theoretic* framework, the performance of an estimator is evaluated through a *loss function*
- Formally, a *decision rule* $\delta : \mathcal{X} \to \mathcal{D}$ is a mapping from the observation space $\mathcal{X}$ to a *decision space* $\mathcal{D}$
- For *parameter estimation*, $\mathcal{D} = \Theta$, and for *detection*, $\mathcal{D} = \{0, 1\}$. More generally, however, $\mathcal{D}$ can be the collection of any viable decisions
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$$R_\delta(\theta) := \mathbb{E}_\theta[L(\theta, \delta(X))]$$

where the expectation is over $X \sim f_\theta(x)$

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  for parameter estimation problems and

$$L(\theta, d) = 1_{\{\theta \notin \Theta_d\}}$$

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Mean squared-error and bias-variance tradeoff

- Under the *squared-error* loss, the frequentist risk is given by the *mean squared-error (MSE)*, which can be further decomposed as the sum of a *bias* term and a *variance* term:

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\begin{aligned}
R_\delta(\theta) &= \mathbb{E}_\theta[(\theta - \delta(X))^2] \\
&= \mathbb{E}_\theta[(\theta - \mathbb{E}_\theta[\delta(X)] + \mathbb{E}_\theta[\delta(X)] - \delta(X))^2] \\
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- Potentially, there is a *tradeoff* between the *bias* term and the *variance* terms in achieving a smaller value of the MSE

Gaussian with unknown mean and variance

- Consider the following family of estimators

$$\delta_b(X) = bS^2$$

for the variance parameter σ^2, where

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \hat{\mu})^2$$

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- By *Cochran's theorem*, we have

$$Bias^2 = (b-1)^2\sigma^4$$
$$Variance = \frac{2b^2\sigma^4}{n-1}$$

and hence

$$MSE = \left((b-1)^2 + \frac{2b^2}{n-1}\right)\sigma^4$$

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$$\mathbb{E}_\theta[(\theta - \delta_{\frac{n-1}{n}}(X))^2] = \frac{(2n-1)\sigma^4}{n^2} < \frac{2\sigma^4}{n-1} = \mathbb{E}_\theta[(\theta - \delta_1(X))^2]$$

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- However, *neither* achieves the minimum MSE, which is achieved at $b = \frac{n-1}{n+1}$ with the minimum MSE given by

$$\mathbb{E}_\theta[(\theta - \delta_{\frac{n-1}{n+1}}(X))^2] = \frac{2\sigma^4}{n+1}$$

Frequentist risk as an evaluation criterion

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- One may hope for a decision rule δ that *uniformly* minimizes the risk $R_\delta(\theta)$, but such cases *rarely* occur unless the space of decision rules is *restricted*. We shall come back to the restriction idea a bit later in this lecture
- More generally, however, we can use the frequentist risk to weed out some of the decision rules from being considered, at least from the risk-reduction point of view

Admissibility

- Formally, a decision rule δ is *inadmissible* if it is *dominated* by another decision rule δ', i.e., for *every* $\theta \in \Theta$ we have

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- For example, for Gaussian with unknown mean and variance, both the UMUVE $\delta_1 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2$ and the MLE $\delta_{\frac{n-1}{n}} = \frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})^2$ are *inadmissible* with respect to the squared-error loss, as both are *dominated* by $\delta_{\frac{n-1}{n+1}} = \frac{1}{n+1}\sum_{i=1}^{n}(X_i - \overline{X})^2$

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- (*Generalized Rao-Blackwell theorem*) Let δ be a decision rule and

  $$\delta'(T) = \mathbb{E}_\theta[\delta(X)|T]$$

  where T is a *sufficient* statistic of θ. Then for any *convex* loss function $L(\theta, d)$, we have

  $$R_{\delta'}(\theta) \leq R_\delta(\theta), \quad \forall \theta \in \Theta$$

  where

  $$R_\delta(\theta) = \mathbb{E}_\theta[L(\theta, \delta(X))] \quad \text{and} \quad R_{\delta'}(\theta) = \mathbb{E}_\theta[L(\theta, \delta'(T))]$$

Proof

- First note that by *Jensen's inequality*, we have

$$L(\theta, \delta'(T)) = L(\theta, \mathbb{E}_\theta[\delta(X)|T]) \leq \mathbb{E}_\theta[L(\theta, \delta(X))|T]$$

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- This immediately gives us

$$R_{\delta'}(\theta) \leq R_\delta(\theta)$$

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- Formally, let Z be a random source over $\mathcal{Z}$ that is *independent* of the observation X. A *randomized* decision rule $\delta^* : \mathcal{X} \times \mathcal{Z} \to \mathcal{D}$ is a mapping that maps an observation x to a decision d based on the realization of the random source Z

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- The *risk function* that corresponds to a *randomized* decision rule δ^* is defined as

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where the expectation is over both $X \sim f_\theta(x)$ and Z

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- Now let $\mathcal{R}$ be the set of *all* possible risk vectors that can be achieved by any *deterministic* decision rule. Then $\mathcal{R}$, as a subset of $\mathbb{R}^K$, can take some very *arbitrary* shapes, and in general there are *no* meaningful ways to characterize such sets

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- On the other hand, let $\mathcal{R}^*$ be the set of *all* possible risk vectors that can be achieved by any *randomized* decision rule. Then, it is easy to see that $\mathcal{R}^*$ is the *convex hull* of $\mathcal{R}$ and hence must be *convex* set, which can be much easier to characterize

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- Two most often collapsing strategies are *minimax* and *Bayes* estimators

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- By the *generalized Rao-Blackwell* theorem, the above result holds not just for the squared-error loss, but for any *convex* loss functions

ECEN 662: Estimation & Detection Theory

Lecture 11: Uniformly Minimum-Risk Equivariant Estimation

Tie Liu

Texas A&M University

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  where the function f is fixed and known, and θ is the unknown *location parameter*

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  where the function f is fixed and known, and θ is the unknown *location parameter*
- Note that by the *change-of-variable* formula, we can write

  $$X_i = U_i + \theta, \quad \forall i \in [n]$$

  such that

  $$(U_1, \ldots, U_n) \sim f(u_1, \ldots, u_2),$$

  which is *independent* of θ

Location invariance

- Note that for the aforementioned location family, we have

$$f_{\theta+c}(x_1 + c, \ldots, x_n + c) = f_\theta(x_1, \ldots, x_n)$$

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- A decision problem is *location invariant* if both the family of distributions and the loss function are *location invariant*
- When our decision problem displays this sort of *invariance*, it is reasonable to constrain our estimator to *respect* these symmetries as well

Location equivariant estimator

- An estimator δ is *location equivariant* if

$$\delta(x_1 + c, \ldots, x_n + c) = \delta(x_1, \ldots, x_n) + c$$

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- (Lemma) If δ is a *location-equivariant* estimator for a *location-invariant* decision problem, then its bias, variance, and risk do *not* depend on θ

- To see this, note that

$$\mathbb{E}_\theta[\delta(X_1, \ldots, X_n) - \theta] = \mathbb{E}_\theta[\delta(X_1 - \theta, \ldots, X_n - \theta)] = \mathbb{E}[\delta(U_1, \ldots, U_n)]$$

where $U_i = X_i - \theta$ and

$$(U_1, \ldots, U_n) \sim f(u_1, \ldots, u_n),$$

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- Next, we shall first develop a characterization of *all* location-equivariant estimators

A characterization of location-equivariant estimators

- First note that if δ_0 is a *location-equivariant* estimator, then any other estimator δ is location equivariant if and only if

$$\delta = \delta_0 - U$$

where U is *location invariant*, i.e.,

$$U(x_1 + c, \ldots, x_n + c) = U(x_1, \ldots, x_n), \quad \forall c \in \mathcal{R}$$

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- Next, a statistic U is *location invariant* if and only if it is a function of $(Y_1, \ldots, Y_{i-1})$, where $Y_i = X_i - X_n$ for $i \in [n-1]$. Note that the difference above is taken between i and n simply for convenience

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- Combining the above two observations, we may conclude that *every* location-equivariant estimator has the form

$$\delta(X_1, \ldots, X_n) = \delta_0(X_1, \ldots, X_n) - v(Y_1, \ldots, Y_{i-1})$$

  where δ_0 is a *reference* location-equivariant estimator, v is an arbitrary function, and $Y_i = X_i - X_n$

Finding the UMREE

- Given the aforementioned characterization and the fact that the risk is *independent* of θ, to find an UMREE, it suffices to find a function $v(Y)$ where $Y = (Y_1, \ldots, Y_{i-1})$ to minimize the risk

$$R_\delta(0) = \mathbb{E}_0[\rho(\delta_0(X) - v(Y))]$$

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- For the *squared-error* loss $\rho(t) = t^2$, it is straightforward to see that

$$v^*(Y) = \mathbb{E}_0[\delta_0(X)|Y]$$

  so the unique MREE is given by

$$\delta^*(X) = \delta_0(X) - \mathbb{E}_0[\delta_0(X)|Y]$$

The Pitman estimator of location

- The following more explicit form of δ^* is known as the *Pitman estimator of location*:

$$\delta^*(X) = \frac{\int_{-\infty}^{\infty} u f(X_1 - u, \ldots, X_n - u) du}{\int_{-\infty}^{\infty} f(X_1 - u, \ldots, X_n - u) du}$$

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- To this end, consider $Z = (Y_1, \ldots, Y_{n-1}, X_n)$ and denote its density by g_θ. We have

$$\mathbb{E}_0[X_n | Y] = \frac{\int_{-\infty}^{\infty} t g_0(Y_1, \ldots, Y_{n-1}, t) dt}{\int_{-\infty}^{\infty} g_0(Y_1, \ldots, Y_{n-1}, t) dt}$$

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- Replacing t by $X_n - u$ gives

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&\mathbb{E}_0[X_n|Y] \\
&= \frac{\int_{-\infty}^{\infty} (X_n - u) f(Y_1 + X_n - u, \ldots, Y_{n-1} + X_n - u, X_n - u) du}{\int_{-\infty}^{\infty} f(Y_1 + X_n - u, \ldots, Y_{n-1} + X_n - u, X_n - u) du} \\
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\end{aligned}$$

- The UMREE is thus given by

$$\delta^*(X) = X_n - \mathbb{E}_0[X_n|Y] = \frac{\int_{-\infty}^{\infty} u f(X_1 - u, \ldots, X_n - u)du}{\int_{-\infty}^{\infty} f(X_1 - u, \ldots, X_n - u)du}$$

Example: Uniform with unknown center

- Suppose that $X_i \overset{iid}{\sim} \mathcal{U}\left(\left[\theta - \frac{b}{2}, \theta + \frac{b}{2}\right]\right)$, where b is known and θ is the unknown parameter

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- The likelihood

$$f_\theta(x_1, \ldots, x_n) = \frac{1}{b^n} 1_{\left\{\theta - \frac{b}{2} \leq X_{(1)} \leq X_{(n)} \leq \theta + \frac{b}{2}\right\}}$$

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- Thus, any

$$\delta(X) \in \left[X_{(n)} - \frac{b}{2}, X_{(1)} + \frac{b}{2}\right]$$

  is an *MLE* of θ

- Under the squared-error loss, the *UMREE* is unique and is given by

$$\begin{aligned}
\delta^*(X) &= \frac{\int_{-\infty}^{\infty} u \frac{1}{b^n} 1_{\{X_{(n)}-b/2 \le u \le X_{(1)}+b/2\}} du}{\int_{-\infty}^{\infty} \frac{1}{b^n} 1_{\{X_{(n)}-b/2 \le u \le X_{(1)}+b/2\}} du} \\
&= \frac{\frac{1}{2}(X_{(1)} + b/2)^2 - \frac{1}{2}(X_{(n)} - b/2)^2}{(X_{(1)} + b/2) - (X_{(n)} - b/2)} \\
&= \frac{X_{(1)} + X_{(n)}}{2}
\end{aligned}$$

The unbiasedness of UMREE

- Note that in the previous example, the UMREE

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- (Lemma) If $\delta(X)$ is location equivariant with constant bias b, then $\delta'(X) = \delta(X) - b$ is unbiased, location equivariant, and has a *smaller* risk than $\delta(X)$

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- To see that $\delta'(X)$ has a *smaller* risk than $\delta(X)$, note that

$$\begin{aligned} R_{\delta'}(\theta) &= \mathbb{E}_0[(\delta'(X))^2] \\ &= \mathbb{E}_0[(\delta(X) - b)^2] \\ &= \mathbb{E}_0[(\delta(X))^2] + b^2 - 2b\mathbb{E}_0[\delta(X)] \\ &= R_\delta(\theta) + b^2 - 2b^2 = R_\delta(\theta) - b^2 \end{aligned}$$

ECEN 662: Estimation & Detection Theory

Lecture 12: Bayes Estimators and Average-Risk Optimality

Tie Liu

Texas A&M University

Lecture Roadmap: The Bayesian Journey

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- **Act III: The Bridge to Classical Statistics**
  - *Improper priors* and total ignorance
  - Recovering the frequentist MLE and UMREE (Pitman)

Setting the Stage: The Bayesian Framework

- **The Goal: Average-Risk Optimality**
  In this lecture, we are moving beyond standard estimators to find *average-risk optimality* using the *Bayesian* decision framework.
  - As always, we start with a data model $f_\theta(x)$ and a loss function $L(\theta, d)$.

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- **The Bayesian Shift (The "New World")**
  - We assign a probability measure over the parameter space Ω, called the *prior distribution*, $\pi(\theta)$.
  - We now treat the parameter as a *random variable*, $\Theta \sim \pi(\theta)$.

- **The Conditional Risk**
  - Under this new generative model, the data is drawn conditionally: $X \sim f(x|\theta) = f_\theta(x)$.
  - Consequently, our familiar frequentist risk is now formally defined as a *conditional expectation*:

$$R_\delta(\theta) = \mathbb{E}_{X\,|\,\Theta}[L(\Theta, \delta(X)) \mid \Theta = \theta]$$

Average Risk Optimality & The Bayes Estimator

- **Averaging Out the Unknown**
  To solve the problem of the frequentist risk $R_\delta(\theta)$ depending on the unknown true parameter, we simply average it out over our prior distribution $\pi(\theta)$.

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- **Deriving the Bayes Risk ($R_{\delta,\pi}$)**
  By applying the Law of Total Expectation, we collapse the conditional frequentist risk into a single, globally defined scalar called the *Bayesian risk*:

$$
\begin{aligned}
R_{\delta,\pi} :=\ & \underbrace{\mathbb{E}_\Theta[R_\delta(\Theta)]}_{\text{Average of Frequentist Risk}} \\
&= \underbrace{\mathbb{E}_\Theta\left[\mathbb{E}_{X|\Theta}[L(\Theta,\delta(X)) \mid \Theta]\right]}_{\text{Law of Total Expectation}} \\
&= \underbrace{\mathbb{E}_{\Theta,X}[L(\Theta,\delta(X))]}_{\text{Joint Expectation}}
\end{aligned}
$$

- **The Bayes Estimator (δ_π)**
  Because the Bayesian risk $R_{\delta,\pi}$ is a single, deterministic number (no longer a function of θ), we can globally minimize it.
  - Any estimator $\delta(X)$ that achieves this absolute minimum is called a *Bayes estimator*, denoted as δ_π.

Why Bayes? The Guarantee of Admissibility

- **The Flaw in Classical Methods**
  Previous frameworks like the Maximum Likelihood Estimate (MLE) and Uniformly Minimum Variance Unbiased Estimator (UMVUE) frequently produce *inadmissible* estimators.
  - Reminder: An estimator is inadmissible if another estimator exists that has lower risk for some θ, and never has worse risk for any θ.

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  - Reminder: An estimator is inadmissible if another estimator exists that has lower risk for some θ, and never has worse risk for any θ.
- **The Bayesian Guarantee**
  By design, Bayes estimators are highly robust against this flaw. We will prove shortly that if a Bayes estimator is *unique*, it is mathematically guaranteed to be *admissible*.

Why Bayes? The Guarantee of Admissibility

- **The Flaw in Classical Methods**
  Previous frameworks like the Maximum Likelihood Estimate (MLE) and Uniformly Minimum Variance Unbiased Estimator (UMVUE) frequently produce *inadmissible* estimators.
  - Reminder: An estimator is inadmissible if another estimator exists that has lower risk for some θ, and never has worse risk for any θ.
- **The Bayesian Guarantee**
  By design, Bayes estimators are highly robust against this flaw. We will prove shortly that if a Bayes estimator is *unique*, it is mathematically guaranteed to be *admissible*.
- **The Complete Class Theorem**
  This relationship is surprisingly profound. Under very weak mathematical conditions on the decision problem:
  - *Every* admissible estimator is either a Bayes estimator or a limit of Bayes estimators.

- **The Strategic Takeaway**
  If your engineering goal is to deploy an optimal, admissible estimator, you can safely restrict your search exclusively to the Bayesian framework.
  - Roadmap: As we will see later, finding a highly conservative *minimax* estimator almost always requires finding a Bayes estimator first!

Theorem: Unique Bayes Estimators are Admissible

- **The Premise**
  Suppose we find an estimator δ_π that minimizes the Bayes risk for a given prior π, and we can prove it is the *unique* minimum.

Theorem: Unique Bayes Estimators are Admissible

- **The Premise**
  Suppose we find an estimator δ_π that minimizes the Bayes risk for a given prior π, and we can prove it is the *unique* minimum.
- **The Proof (By Contradiction)**
  Assume the opposite: suppose our unique Bayes estimator δ_π is actually *inadmissible*.
  - Step 1 (The Dominating Estimator): By definition of inadmissibility, there must exist some other estimator δ' that dominates it. For all possible parameters $\theta \in \Omega$:
  $$R_{\delta'}(\theta) \leq R_{\delta_\pi}(\theta)$$
  - Step 2 (Averaging over the Prior): If we take the expected value of both sides over our prior distribution $\Theta \sim \pi(\theta)$, the inequality must hold:
  $$R_{\delta',\pi} \leq R_{\delta_\pi,\pi}$$
  - Step 3 (The Contradiction): Because δ_π is a Bayes estimator, no estimator can have a strictly lower Bayes risk. Thus, the risks must be equal ($R_{\delta',\pi} = R_{\delta_\pi,\pi}$). This means δ' is also a Bayes estimator, which completely violates our premise that δ_π was unique.

- **A Critical Caveat**
  Do not assume the converse! A Bayes estimator can still be admissible even if it is *not* unique.
  - Example: If the parameter space Ω is strictly *discrete*, any Bayes estimator is guaranteed to be admissible, as long as the prior assigns a non-zero probability to every possible state.

How to Compute the Bayes Estimator

- **Reversing the Condition: The Posterior Distribution**
  Instead of averaging over the data first, let's look at the problem after the data $X = x$ has been observed. We use Bayes' Rule to find the posterior distribution:

$$\pi(\theta|x) = \frac{\pi(\theta)f(x|\theta)}{\int_{\Omega} \pi(\theta)f(x|\theta)d\theta}$$

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- **The Posterior Risk**
  We can now define the expected loss strictly with respect to this new posterior distribution. This is the *posterior risk*:

$$R_{\delta,\pi}(x) := \mathbb{E}_{\Theta|X}[L(\Theta, \delta(x)) \mid X = x]$$

- **The Pointwise Optimization Trick**
  By the Law of Total Expectation, the total Bayesian risk $R_{\delta,\pi}$ is simply the average of the posterior risk over the marginal distribution of the data (the *evidence*, $m(x)$):

$$R_{\delta,\pi} = \mathbb{E}_X[R_{\delta,\pi}(X)]$$

  Because the evidence $m(x)$ is always non-negative, to minimize the massive total integral, we just need to minimize the posterior risk individually for each *specific data point* x.

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- **The Practical Computation**
  To find the Bayes estimator $\delta_\pi(x)$, you do not need to solve a functional optimization problem. For any observed x, simply pick the decision d that minimizes the posterior risk:

$$\delta_\pi(x) \in \arg\min_d \mathbb{E}_{\Theta|X}[L(\Theta, d) \mid X = x]$$

Bayes Estimators & Sufficient Statistics

- **The Fundamental Link**
  If $T(X)$ is a sufficient statistic for the parameter θ, then any Bayes estimator $\delta_\pi(x)$ will automatically depend on the data x only through $T(x)$.
  - The Bayesian framework naturally compresses the data, discarding irrelevant noise and keeping only the core information.

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- **The Proof via Neyman-Fisher**
  Let's look at the posterior risk we derived on the last slide. By expanding the posterior distribution and applying the *Neyman-Fisher Factorization Theorem* ($f_\theta(x) = h(x)g_\theta(T(x))$), we get:

$$\begin{aligned}
\mathbb{E}_\pi[L(\Theta, d) \mid X = x] &= \int_\Omega L(\theta, d)\pi(\theta|x)d\theta \\
&= \frac{1}{m(x)} \int_\Omega L(\theta, d)\pi(\theta)f_\theta(x)d\theta \\
&= \frac{h(x)}{m(x)} \int_\Omega L(\theta, d)\pi(\theta)g_\theta(T(x))d\theta
\end{aligned}$$

- **The Optimization Constraint**
  Notice that the scaling factor $\frac{h(x)}{m(x)}$ does not depend on our decision variable d. Therefore, to minimize the posterior risk with respect to d, we only need to minimize the core integral:

$$\arg\min_d \int_\Omega L(\theta, d)\pi(\theta)g_\theta(T(x))d\theta$$

  Because the data x only appears inside the sufficient statistic $T(x)$ within this integral, the optimal decision d *must strictly be a function of* $T(x)$.

Squared-error loss

- **The Posterior Risk is Quadratic**
  Under squared-error loss, the risk with respect to our estimate d forms a *parabola*:

  $$\mathbb{E}_{\Theta|X}[(\Theta - d)^2|X = x] = d^2 - 2\mathbb{E}_{\Theta|X}[\Theta|X = x]d + \mathbb{E}_{\Theta|X}[\Theta^2|X = x]$$

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- **The Unique Bayes Estimator**
  Because the risk is strictly quadratic, the minimum is *unique*. This gives us the *posterior mean* (often called the *MMSE estimator* in engineering):

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- **The Computational Bottleneck**
  Calculating the posterior mean requires the full posterior distribution $\pi(\theta|x)$. This is often *analytically impossible* due to the evidence integral:

$$m(x) = \int_\Omega \pi(\theta) f(x|\theta) d\theta$$

- **The Workaround: Conjugate Priors**
  - Problem: High-dimensional integrals are *intractable*.
  - Solution: Choosing a *conjugate prior* $\pi(\theta)$ allows us to bypass the integral and find the posterior mean in *closed-form*.

The Impact of the Loss Function

- The optimal Bayes estimator $\delta_\pi(x)$ depends entirely on our choice of the loss function $L(\theta, d)$.

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  - More robust to extreme outliers.
  - $\delta_\pi(x) = \text{Median}(\Theta|X = x)$ (*Posterior Median*)
- **Zero-One (Hit-or-Miss) Loss:** $L(\theta, d) = \mathbb{I}_{\{\theta \neq d\}}$
  - Penalizes all errors equally, regardless of distance.
  - $\delta_\pi(x) = \arg\max_\theta \pi(\theta|x)$ (*Maximum a Posteriori / MAP*)

Conjugate priors

- **The Core Definition**
  A family of prior distributions is *conjugate* to a specific likelihood if the resulting posterior belongs to the same family.

$$\text{Prior} \times \text{Likelihood} \propto \text{Posterior (Same Family)}$$

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  - The Useless Case: Defining the family as "all possible probability distributions."
  - The Useful Case: Restricting the prior to a specific parameterized family.
- **Why it matters**
  Bayesian updating reduces to simply updating the *parameters* of the distribution, completely *avoiding* the intractable evidence integral.

- **Key Examples**
  We will examine two highly practical conjugacies:
  - *Multinomial-Dirichlet* (common in machine learning and text processing)
  - *Gaussian-Gaussian* (the foundation of signal processing and control)

Multinomial-Dirichlet Conjugacy (The Setup)

- **The Data Model: Categorical Observations**
  Suppose we have n independent observations $X = (X_1, \ldots, X_n)$ falling into K known categories:

  $$X_i \overset{iid}{\sim} \mathrm{Cat}(K, \theta_1, \ldots, \theta_K),$$

  where our goal is to estimate the unknown probability vector $\theta = (\theta_1, \ldots, \theta_K)$.

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- **The Multinomial Likelihood**
  Instead of the raw sequence, the likelihood is completely defined by the empirical counts $n_k = \sum_{i=1}^{n} \mathbf{1}_{\{x_i = k\}}$ for each category:

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- **The Frequentist Baseline (MLE)**
  For comparison, recall that the MLE is simply the empirical frequency:
  $$\hat{\theta}_k^{\mathrm{MLE}} = \frac{n_k}{n}$$

The Dirichlet Prior

- **The General Dirichlet Family**
  To act as our prior over the probability vector θ, we use the *Dirichlet distribution*, parameterized by $\alpha = (\alpha_1, \ldots, \alpha_K)$ where $\alpha_k > 0$:

$$\pi_\alpha(\theta) \propto \prod_{k=1}^{K} \theta_k^{\alpha_k - 1}$$

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  If we have no prior reason to favor one category over another, we set all elements of α to *identical* values.
- **The Concentration Parameter**
  Because all elements are identical, the symmetric Dirichlet is parameterized by a *single* scalar α, known as the *concentration parameter*.

The Effect of the Concentration Parameter (α)

- **The Uniform Case** ($\alpha = 1$)
  - The symmetric Dirichlet becomes a *flat* distribution over the $(K-1)$-simplex.
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- A nice visualization for $K = 3$ can be found in this blog

The Dirichlet Posterior Update

- **The Conjugacy Proof**
  To see that the Dirichlet family is *conjugate* to the multinomial likelihood, we multiply them and group the exponents:

$$\begin{aligned}
\pi_\alpha(\theta|x) &\propto_\theta p_\theta(x)\pi_\alpha(\theta) \\
&\propto_\theta \left(\prod_{k=1}^{K} \theta_k^{n_k}\right)\left(\prod_{k=1}^{K} \theta_k^{\alpha_k - 1}\right) \\
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- **The Dirichlet Mean**
  Recall that we need the *posterior mean* to find our MMSE Bayes estimator. For any $\Theta \sim \mathrm{Dir}(\alpha)$, the expected value is:

$$\mathbb{E}\alpha[\Theta_k] = \frac{\alpha_k}{\alpha_0}, \quad k \in [K],$$

  where $\alpha_0 = \sum_{k=1}^K \alpha_k$ is the sum of all concentration parameters.

The Bayes Estimator: Smoothing & Shrinkage

- **The Posterior Mean**
  Using the updated parameters $(n_k + \alpha_k)$, the Bayes estimator under squared-error loss is:

$$\mathbb{E}_\alpha[\Theta_k | X = x] = \frac{n_k + \alpha_k}{n + \alpha_0}, \quad \forall k \in [K]$$

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  When using a *symmetric* Dirichlet prior (e.g., $\alpha_k = 1$), this estimator is famously known in computer science and engineering as *Laplacian smoothing* (or additive smoothing).

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- **The Convex Combination**
  We can algebraically rewrite the posterior mean to reveal its true nature:

$$\mathbb{E}_\alpha[\Theta_k | X = x] = \underbrace{\left(\frac{n}{n + \alpha_0}\right)}_{\text{Weight on Data}} \underbrace{\left(\frac{n_k}{n}\right)}_{\text{MLE}} + \underbrace{\left(\frac{\alpha_0}{n + \alpha_0}\right)}_{\text{Weight on Prior}} \underbrace{\left(\frac{\alpha_k}{\alpha_0}\right)}_{\text{Prior Mean}}$$

- **The Shrinkage Estimator**
  - This is a *shrinkage estimator*: it "shrinks" the purely data-driven MLE toward our prior belief.
  - *Asymptotic Equivalence*: As $n \to \infty$, the data overwhelms the prior, and the Bayes estimator converges to the MLE.

Gaussian Conjugacy (Unknown Mean)

- **The Data Model: Gaussian Noise**
  Suppose we have n independent observations $X = (X_1, \ldots, X_n)$ drawn from a Gaussian distribution:

  $$X_i \overset{iid}{\sim} \mathcal{N}(\mu, \sigma^2),$$

  where the variance σ^2 is known (e.g., calibrated sensor noise), but the mean μ is the *unknown* parameter we want to estimate.

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- **The Likelihood & Frequentist Baseline**
  - The joint likelihood of our data is:

$$f_\mu(x) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2\right)$$

  - The Baseline (MLE): Recall that maximizing this likelihood yields the standard sample mean:

$$\hat{\mu}_{\text{MLE}} = \overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

- **The Gaussian Prior**
  To set up our Bayesian framework, we assign a Gaussian prior to the unknown mean μ:

$$\pi_{\mu_0,\sigma_0^2}(\mu) = \frac{1}{\sqrt{2\pi\sigma_0^2}} \exp\left(-\frac{(\mu-\mu_0)^2}{2\sigma_0^2}\right)$$

  parameterized by our prior mean μ_0 and prior variance σ_0^2.

Gaussian Conjugacy (The Posterior Update)

- **Multiplying the Exponentials**
  To find the posterior, we multiply the Gaussian prior and the Gaussian likelihood. Because both are exponentials, we simply add their exponents and ignore any constants that do not depend on μ:

$$\pi_{\mu_0,\sigma_0^2}(\mu|x) \propto_\mu \exp\left(-\frac{1}{2\sigma_0^2}(\mu^2 - 2\mu_0\mu)\right)\exp\left(-\frac{n}{2\sigma^2}(\mu^2 - 2\overline{x}\mu)\right)$$

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- **Grouping and Completing the Square**
  By grouping the μ^2 and μ terms, we construct a new quadratic in the exponent:

$$\propto_\mu \exp\left(-\underbrace{\left(\frac{1}{2\sigma_0^2} + \frac{n}{2\sigma^2}\right)}_{\text{New inverse-variance}}\mu^2 + \underbrace{\left(\frac{\mu_0}{\sigma_0^2} + \frac{n\overline{x}}{\sigma^2}\right)}_{\text{Shift term}}\mu\right)$$

  Completing the square on this expression returns it to the standard Gaussian form of $\exp(-a(\mu - b)^2)$.

- **The Posterior Mean Formula**
  By separating the terms of our updated mean μ_n, we can rewrite the Bayes estimator under squared-error loss:

$$\mu_n = \frac{\mu_0 \sigma^2 + n\sigma_0^2 \overline{x}}{\sigma^2 + n\sigma_0^2} = \frac{n\sigma_0^2}{\sigma^2 + n\sigma_0^2}\overline{x} + \frac{\sigma^2}{\sigma^2 + n\sigma_0^2}\mu_0$$

The Bayes Estimator: Gaussian Shrinkage

- **The Exact Gaussian Posterior**
  The algebra proves the posterior is a new Gaussian distribution, $\mathcal{N}(\mu_n, \sigma_n^2)$, with updated parameters:
  - Posterior Mean (μ_n): $\frac{\mu_0 \sigma^2 + n \sigma_0^2 \overline{x}}{\sigma^2 + n \sigma_0^2}$
  - Posterior Variance (σ_n^2): $\frac{\sigma_0^2 \sigma^2}{\sigma^2 + n \sigma_0^2}$

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- **The Convex Combination**
  Just like the Multinomial-Dirichlet case, this is a *shrinkage estimator*. It is a weighted average of the data ($\overline{x}$) and the prior (μ_0):

$$\mu_n = \underbrace{\left(\frac{n\sigma_0^2}{\sigma^2 + n\sigma_0^2}\right)}_{\text{Weight on MLE}} \overline{x} + \underbrace{\left(\frac{\sigma^2}{\sigma^2 + n\sigma_0^2}\right)}_{\text{Weight on Prior}} \mu_0$$

- **Engineering Intuition (The Limits)**
  - Infinite Data ($n \to \infty$): The weight on the MLE approaches 1. The data overpowers the prior: $\mu_n \approx \overline{x}$.
  - Terrible Sensor ($\sigma^2 \to \infty$): The weight on the prior approaches 1. We ignore the noisy data entirely: $\mu_n \approx \mu_0$.
  - Total Ignorance ($\sigma_0^2 \to \infty$): The prior is completely flat. We trust the data entirely: $\mu_n \approx \overline{x}$.

The biasedness of Bayes estimators

- **The Reality of Shrinkage**
  - In both the Dirichlet and Gaussian examples, the Bayes estimators are inherently *biased* for all finite sample sizes.
  - This is a *fundamental* feature, not a bug. They trade a little bias for a massive reduction in variance.

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  - In both the Dirichlet and Gaussian examples, the Bayes estimators are inherently *biased* for all finite sample sizes.
  - This is a *fundamental* feature, not a bug. They trade a little bias for a massive reduction in variance.
- **The Theorem: Unbiased = Degenerate**
  Under squared-error loss, a Bayes estimator $\delta_\pi(x)$ is only unbiased in the *degenerate* case where the Bayesian risk is exactly *zero* (i.e., perfect estimation, which is impossible in noisy environments).

- **The Proof (By Contradiction)**
  Assume the Bayes estimator is unbiased: $\mathbb{E}[\delta_\pi(X)|\Theta = \theta] = \theta$ for all θ. Let's evaluate the cross-term $\mathbb{E}[\delta_\pi(X)\Theta]$ in two different ways using the Law of Total Expectation:

  - Path A (Conditioning on Θ):

  $$\mathbb{E}[\delta_\pi(X)\Theta] = \mathbb{E}\big[\mathbb{E}[\delta_\pi(X)|\Theta]\Theta\big] = \mathbb{E}[\Theta \cdot \Theta] = \mathbb{E}[\Theta^2]$$

  - Path B (Conditioning on X):

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- **The Proof (By Contradiction)**
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- **The Conclusion (Zero Risk)**
  Expanding the Bayesian risk and substituting Path A and Path B:

$$\begin{aligned}
&\mathbb{E}_\pi[(\delta_\pi(X) - \Theta)^2] \\
&\quad = \mathbb{E}_\pi[\delta_\pi^2(X)] - 2\mathbb{E}_\pi[\delta_\pi(X)\Theta] + \mathbb{E}_\pi[\Theta^2] \\
&\quad = \underbrace{\Big(\mathbb{E}_\pi[\delta_\pi^2(X)] - \mathbb{E}_\pi[\delta_\pi(X)\Theta]\Big)}_{=0 \text{ (from Path B)}} + \underbrace{\Big(\mathbb{E}_\pi[\Theta^2] - \mathbb{E}_\pi[\delta_\pi(X)\Theta]\Big)}_{=0 \text{ (from Path A)}} = 0
\end{aligned}$$

Is the Sample Mean a Bayes Estimator?

- **The Diagnostic Test**
  We can use our previous result as a strict test: Under squared-error loss, if an estimator is *unbiased* but has *non-zero* risk, it cannot be a Bayes estimator.

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- **The Test Case: Gaussian Mean**
  Let's test the standard frequentist Maximum Likelihood Estimate (MLE). Can the sample mean $\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$ be the Bayes estimator for some prior π under squared-error loss?

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- **Running the Test**
  - Condition 1 (Unbiased): $\overline{X}$ is an unbiased estimator of μ.
  - Condition 2 (Non-zero Risk): Its Mean Squared Error (MSE) is strictly positive for any finite sample size n:

    $$\mathbb{E}_\mu[(\overline{X} - \mu)^2] = \frac{\sigma^2}{n} > 0, \quad \forall \mu$$

    Because the MSE is positive for all μ, the overall Bayesian risk will be non-zero for any prior π.

- **The Conclusion**
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- **A Critical Caveat**
  This mutual exclusivity between “unbiased” and “Bayes” is a special property of the *squared-error loss*. Under other loss functions (like absolute-error), it is mathematically possible to have an unbiased Bayes estimator with a non-zero risk.

Generalized Bayes estimators

- **The Loophole: Improper Priors**
  While $\overline{X}$ cannot be a standard Bayes estimator, it can be a *generalized* Bayes estimator if we allow the prior distribution to be *improper* (i.e., it does not integrate to 1):

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- **The Key Insight**
  Surprisingly, multiplying an improper prior by a highly concentrated likelihood can still yield a perfectly valid, *proper posterior*.
- **The Proof: The Flat Prior**
  Let's assign a completely flat, improper prior to our Gaussian mean: $\pi(\mu) = 1$ for all $\mu \in \mathbb{R}$. The posterior update becomes:
  $$\begin{aligned} \pi(\mu|x) &\propto_\mu \pi(\mu) f_\mu(x) \\ &\propto_\mu 1 \cdot \exp\left(-\frac{n}{2\sigma^2}(\mu^2 - 2\mu\overline{x})\right) \\ &\propto_\mu \exp\left(-\frac{1}{2(\sigma^2/n)}(\mu - \overline{x})^2\right) \end{aligned}$$

- **The Grand Conclusion**
  The resulting posterior is exactly $\mathcal{N}(\overline{x}, \sigma^2/n)$.
  - Under squared-error loss, we take the posterior mean.
  - Therefore, the sample mean $\overline{X}$ is exactly the *generalized* Bayes estimator under a *flat* improper prior.

Generalized Bayes & The Pitman Estimator

- **The Location Family Model**
  Let's generalize beyond the Gaussian. Consider n independent observations drawn from any shifted distribution:

$$X_i \overset{iid}{\sim} g(x_i - \theta)$$

  Here, g is a known base probability density, and $\theta \in \mathbb{R}$ is the unknown *location parameter* we want to estimate.

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- **The General Posterior Mean**
  Given an arbitrary prior $\pi(\theta)$, the posterior is proportional to the prior times the likelihood:

  $$\pi(\theta|x) \propto_\theta \pi(\theta) \prod_{i=1}^{n} g(x_i - \theta)$$

  Under squared-error loss, our Bayes estimator is the posterior mean:

  $$\mathbb{E}_\pi[\Theta|X = x] = \frac{\int_{-\infty}^{\infty} \theta\pi(\theta) \prod_{i=1}^{n} g(x_i - \theta)d\theta}{\int_{-\infty}^{\infty} \pi(\theta) \prod_{i=1}^{n} g(x_i - \theta)d\theta}$$

- **The Flat Prior & The Pitman Estimator**
  If we model total ignorance by choosing the *improper prior* $\pi(\theta) = 1$, the $\pi(\theta)$ terms vanish from the integrals.
  - The resulting formula is exactly the *Pitman estimator of location* as we discussed previously.
  - The Grand Connection: The Pitman estimator is famously known as the Uniformly Minimum Risk Equivariant Estimator (UMREE) under squared-error loss.

The Catch: Limits of Generalized Bayes

- **The Warning**
  While generalized Bayes estimators elegantly bridge the gap to classical frequentist methods (like the MLE and Pitman estimator), they do *not* inherit the strict theoretical safety nets of proper Bayes estimators.

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- **Loss of Guaranteed Admissibility**
  - Proper Bayes: If a prior is proper and strictly positive, the resulting Bayes estimator is mathematically guaranteed to be *admissible* (no other estimator can strictly beat it for all θ).
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- **The Risk Problem**
  Because an improper prior integrates to infinity ($\int_\Omega \pi(\theta)d\theta = \infty$), the overall integrated Bayesian risk is technically infinite. We cannot rely on standard risk-minimization theorems to prove optimality.

- **The Grand Summary: Which do we choose?**

Feature	Proper Bayes	Generalized Bayes
Prior $\pi(\theta)$	Integrates to 1	Integrates to ∞
Bias	Inherently Biased	Often Unbiased
Admissibility	Guaranteed Admissible	Can be Inadmissible
Equivalence	MAP / MMSE Filtering	MLE / UMREE

ECEN 662: Estimation & Detection Theory

Lecture 13: Variational Inference

Tie Liu

Texas A&M University

The reality check: Intractable posteriors

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$$\pi(\theta|x) = \frac{\pi(\theta) f_\theta(x)}{\int_\Omega \pi(\theta) f_\theta(x) d\theta}$$

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- When exact inference fails, we generally have two paths forward:
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- **The Core Idea of VI:** Instead of computing the highly complex true posterior $\pi(\theta|x)$, we define a family of simpler, well-understood distributions $q_\phi(\theta)$. We then optimize the parameters ϕ to make $q_\phi(\theta)$ as *close* to $\pi(\theta|x)$ as possible.

Example: The Gaussian Mixture Model (GMM)

- We wish to group data into K clusters. We face two sets of *unknown* parameters:
  - **The Centers:** $\mu = (\mu_1, \dots, \mu_K)$.
  - **The Assignments:** n unlabeled data points $X = (X_1, \dots, X_n)$, each originating from an unknown cluster C_i.

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$$\mu_k \sim \mathcal{N}(0, \sigma^2)$$

The Computational Trap: The Exact Posterior

- To solve the GMM, we target the joint posterior of the assignments c and centers μ:

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- However, because each of the n points can belong to K clusters, there are K^n terms in the outer sum.
- **Conclusion:** Computing the evidence scales *exponentially* in n. It is strictly intractable for real data, forcing us to use *approximate* methods like Variational Inference.

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- Algorithms like *Metropolis-Hastings* and the *Gibbs Sampler* are indispensable. Given infinite time, MCMC is asymptotically exact.
- **The Catch:** MCMC is inherently sequential and often *highly computationally expensive*. For massive modern datasets or complex models, MCMC is simply too slow. We need a faster approach.

Variational Inference (VI)

Rather than use sampling, the main idea behind *variational inference (VI)* is to use *optimization*:

- **Propose a Variational Family:** Posit a family $\mathcal{Q}$ of approximate densities $q(\theta)$ over the unknown parameters.

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- **Minimize Divergence:** Find the member of $\mathcal{Q}$ that is "closest" to the exact posterior using the *Kullback-Leibler (KL) divergence*:

$$q^* = \arg\min_{q \in \mathcal{Q}} D_{KL}\big(q(\theta) \,\|\, f(\theta|x)\big)$$

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- **Approximate:** Use the optimized $q^*(\theta)$ as the surrogate for the true posterior $f(\theta|x)$.

- *Note:* This allows us to use standard optimization tools like **Stochastic Gradient Descent**, making inference incredibly fast.

Comparison: VI vs. MCMC

Choosing between these paradigms is a trade-off between *precision* and *speed*:

- **MCMC:** The "Gold Standard."
  - **Pro:** Asymptotically exact samples from the target.
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- *Note:* The relative accuracy of VI vs. MCMC in finite-time scenarios remains an active area of research.

The Evidence Lower Bound (ELBO)

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- **The Decomposition:** Expanding the KL divergence using $f(\theta|x) = \frac{f(\theta,x)}{m(x)}$:

$$\begin{aligned} D_{KL}\big(q(\theta) \,\|\, f(\theta|x)\big) &= \mathbb{E}_q[\log q(\Theta)] - \mathbb{E}_q[\log f(\Theta|x)] \\ &= \underbrace{\mathbb{E}_q[\log q(\Theta)] - \mathbb{E}_q[\log f(\Theta, x)]}_{\text{Computable}} + \underbrace{\log m(x)}_{\text{Intractable Constant}} \end{aligned}$$

  where all expectations are over $\Theta \sim q(\theta)$.

- **The Strategy:** Since $\log m(x)$ is constant with respect to q, we can ignore it during optimization.

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- Since we cannot minimize the KL divergence directly, we maximize a computable surrogate objective:

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- **The Optimization Link:** Recall the identity from the previous slide:

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- **The "Lower Bound":** Because $D_{KL} \geq 0$, it follows that:

$$ELBO(q) \leq \log m(x)$$

  for any q. The ELBO provides a *rigorous lower bound* on the log-evidence. By "tightening" this bound, we improve our posterior approximation.

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- **The Trade-off:**
  - *More Complex $\mathcal{Q}$:* Better approximation, but much harder to optimize.
  - *Simpler $\mathcal{Q}$:* Easier optimization, but potentially less accurate.
- **The Mean-field Assumption:** We assume that all latent variables are mutually *independent*:

$$q(\theta) = \prod_{j=1}^{m} q_j(\theta_j)$$

- Each unknown θ_j is governed by its own factor q_j. During optimization, we update these factors iteratively to maximize the ELBO.
- *Key Insight:* This turns a high-dimensional global optimization into a sequence of simpler, low-dimensional coordinate updates.

Beyond Mean-Field: Increasing Fidelity

- While the mean-field family is our workhorse, we can expand $\mathcal{Q}$ to capture more complex posterior structures:
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  - **Mixture Models:** Uses a *mixture of densities* to approximate multi-modal posteriors.

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- **The Trade-off:**
  - **Pro:** Improved fidelity; the approximation can better "mimic" the true posterior.
  - **Con:** Leads to a much more *difficult optimization problem*. You lose the simplicity of independent coordinate updates.

Coordinate Ascent Variational Inference (CAVI)

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$$ELBO(q_j) = \mathbb{E}_j\left[\mathbb{E}_{-j}[\log f(\Theta_j, \Theta_{-j}, x)]\right] - \mathbb{E}_j[\log q_j(\Theta_j)] + \text{const}$$

where $\mathbb{E}_{-j}$ denotes the expectation over all factors $\ell \neq j$.

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  where $\mathbb{E}_{-j}$ denotes the expectation over all factors $\ell \neq j$.
- **The Goal:** Find the q_j that maximizes this expression. As we will see, for many models, this leads to a very clean, closed-form update rule.

The Optimal Coordinate Update

- What is the optimal shape for the factor q_j?
- **The Target Density:** Let $q_j^*(\theta_j)$ be defined as:

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- **The CAVI Update Rule:** To update the factor q_j, we set:

$$\boxed{q_j^*(\theta_j) \propto \exp\left\{\mathbb{E}_{-j}[\log f(\theta_j, \Theta_{-j}, x)]\right\}}$$

- *Intuition:* The optimal factor q_j is the *exponentiated average* of the log-joint distribution over all other variables.

CAVI for Gaussian Mixture Models

- We now apply the general update rule to our clustering example:
- **Update Assignments (C_i):** For each point i, the probability $\psi_{i,k}$ of belonging to cluster k is:

$$\psi_{i,k} \propto \exp\left(-\frac{\mathbb{E}_q[(x_i - \mu_k)^2]}{2}\right)$$

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- **Update Centers (μ_k):** The new variational factor $q(\mu_k)$ is a Gaussian:

$$q(\mu_k) = \mathcal{N}\left(\frac{\sum_i \psi_{i,k} x_i}{1/\sigma^2 + \sum_i \psi_{i,k}}, \frac{1}{1/\sigma^2 + \sum_i \psi_{i,k}}\right)$$

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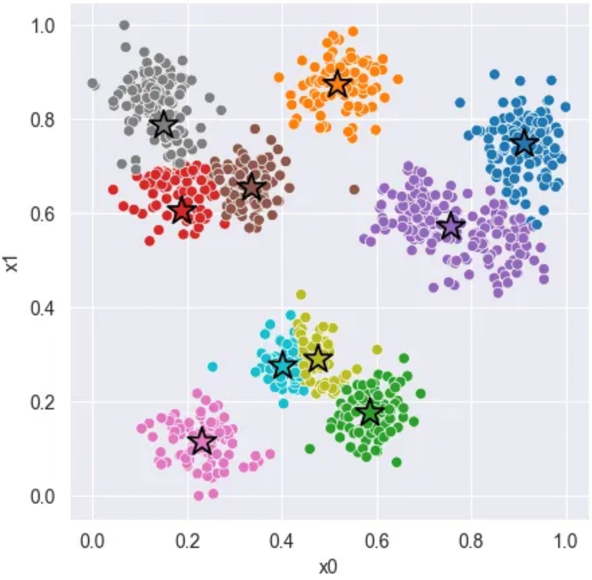
- **The Connection:** This iterative process—assigning points to centers and then re-calculating centers based on those assignments—is a *probabilistic* generalization of the *K-means* algorithm.
- *Key Benefit:* Unlike K-means, VI provides a full measure of *uncertainty* for both the assignments and the cluster locations.

K-means clustering



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- **General Parameterization:** For complex models, we parameterize the variational distribution $q_\lambda(\theta)$ by a vector of *variational parameters* λ.
- **Optimization Problem:** We maximize the ELBO with respect to λ:
$$\max_{\lambda} \left(\mathbb{E}_{q_\lambda}[\log f(\Theta, x)] - \mathbb{E}_{q_\lambda}[\log q_\lambda(\Theta)] \right)$$
- This can be solved using *Stochastic Gradient Ascent*, treating the model as a "black box" that only requires log-likelihoods and gradients.

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- This can be solved using *Stochastic Gradient Ascent*, treating the model as a "black box" that only requires log-likelihoods and gradients.
- *Key Challenge:* We must now compute the *gradient* of the ELBO with respect to λ, which involves differentiating an expectation.

Computing the Gradient of the ELBO

- To optimize the ELBO, we need its gradient with respect to λ:

$$ELBO(\lambda) = \int_{\Omega} q_{\lambda}(\theta)\left(\log f(\theta, x) - \log q_{\lambda}(\theta)\right) d\theta$$

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- **Implementation:** This true gradient is intractable. However, we can generate a noisy, *unbiased estimate* by sampling $\theta^{(s)} \sim q_\lambda$ and computing:

$$\nabla_\lambda ELBO(\lambda) \approx \frac{1}{S} \sum_{s=1}^{S} \left[\left(\nabla_\lambda \log q_\lambda(\theta^{(s)})\right)\left(\log f(\theta^{(s)}, x) - \log q_\lambda(\theta^{(s)})\right)\right]$$

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The Reparameterization Trick

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- **The Trick:** To reduce variance, express the random variable Θ as a deterministic, differentiable function of parameters λ and an auxiliary noise variable ϵ with a *fixed* distribution $p(\epsilon)$:

$$\Theta = g(\lambda, \epsilon) \quad \text{where} \quad \epsilon \sim p(\epsilon)$$

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- **Example (Gaussian):** If $\Theta \sim \mathcal{N}(\mu, \sigma^2)$, we set $\lambda = (\mu, \sigma)$ and let:

$$\Theta = \mu + \sigma \odot \epsilon \quad \text{where} \quad \epsilon \sim \mathcal{N}(0, I)$$

- **Benefit:** We can propagate gradients directly through $g(\lambda, \epsilon)$, leading to much *lower variance* estimators. This is the foundation of *Variational Autoencoders (VAEs)*.

Black Box VI Algorithm (Reparameterized)

- **Objective:** Maximize the ELBO with respect to variational parameters λ:

$$ELBO(\lambda) = \mathbb{E}_{\epsilon \sim p(\epsilon)} [\log f(g(\lambda, \epsilon), x) - \log q_\lambda(g(\lambda, \epsilon))]$$

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- **Stochastic Gradient Ascent Algorithm:**
  1. **Initialize** λ.
  2. **Loop** until convergence:
     - 2.1 Sample noise $\epsilon \sim p(\epsilon)$.
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     - 2.3 Compute gradient estimate: $\hat{g} = \nabla_\lambda \left[\log f(\theta, x) - \log q_\lambda(\theta) \right]$.
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- *Key Advantage:* We can compute $\hat{g}$ efficiently using automatic differentiation libraries (e.g., PyTorch, TensorFlow).

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- **The Consequence:** If the true posterior is multi-modal, a simpler q will "choose" a single mode and fit tightly inside it, rather than spanning across low-probability valleys.
- *Takeaway:* VI gives you a fast, localized approximation, but it is often *overconfident* about its predictions.

Summary: Variational Inference

- **The Core Problem:** For most interesting models, the posterior $f(\theta|x)$ is *intractable* because computing the evidence $m(x)$ requires high-dimensional integration.
- **The VI Paradigm:** We transform *inference* into *optimization*. We find the closest approximation q^* within a family $\mathcal{Q}$ by maximizing the *ELBO* (Evidence Lower Bound).
- **Methods:**
  - **CAVI:** Provides *closed-form* updates for conjugate models (e.g., K-means).
  - **Black Box VI:** Uses *gradients* to optimize general families (e.g., Neural Networks).
- **The Reparameterization Trick:** Allows direct gradients through stochastic nodes, crucial for low-variance gradients in deep generative models.
- **Trade-off:** VI is *faster and more scalable* than MCMC, but it is an *approximation* and does not guarantee exactness.

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- **Bayesian Neural Networks (BNNs):**
  - Uses *Black Box VI* to place distributions over network weights. Provides crucial *uncertainty quantification* for safety-critical systems like autonomous driving and medical AI.

ECEN 662: Estimation & Detection Theory

Lecture 14: Minimax Estimators and Worst-Case Optimality (Part I)

Tie Liu

Texas A&M University

Setting

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- The derivation of *minimax* estimators is often less prescriptive and more problem-specific
- In this lecture, we will develop a few tools which, when they apply, will help to identify minimax estimators

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- The problem of finding a least favorable prior can be thought of as the *dual* to the problem of finding a minimax estimator, in that the objective value of *any* feasible solution to the dual is a *lower bound* on the objective value of *any* feasible solution to the primal problem

- To see why, simply note that for *any* given prior π and *any* given estimator δ, we have

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- In particular, let δ_π be a *Bayes* estimator that corresponds to the prior distribution π. If the average risk of δ_π *coincides* with its own maximum risk, then the Bayes estimator δ_π is also a *minimax* estimator

Example: Bernoulli with unknown mean

- Consider $X = (X_1, \dots, X_n)$, where

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- The MSE of $\delta_{\pi_{a,b}}$ can be calculated as

$$\begin{aligned} R_{\delta_{\pi_{a,b}}}(\theta) &= \mathbb{E}_\theta\left[\left(\frac{n\overline{X} + a}{n + a + b} - \theta\right)^2\right] \\ &= \frac{n\theta(1-\theta) + (a(\theta - 1) + b\theta)^2}{(n + a + b)^2} \end{aligned}$$

- Note that if we can choose $a, b > 0$ such that the MSE $R_{\delta_{\pi_{a,b}}}(\theta)$ is *constant* for $\theta \in [0, 1]$, then the corresponding Bayes estimator $\delta_{\pi_{a,b}}$ is also a minimax estimator

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- Towards this purpose, let us write the numerator as

$$((a+b)^2 - n)\theta^2 + (n - 2a(a+b))\theta + a^2$$

and set

$$(a+b)^2 - n = 0 \quad \text{and} \quad n - 2a(a+b) = 0$$

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$$(a+b)^2 - n = 0 \quad \text{and} \quad n - 2a(a+b) = 0$$

- Solving the above equations gives $a = b = \sqrt{n}/2$. We thus conclude that the Bayes estimator

$$\delta_{\pi_{\sqrt{n}/2, \sqrt{n}/2}} = \frac{\overline{X} + 1/(2\sqrt{n})}{1 + 1/\sqrt{n}}$$

is a minimax estimator with a *constant* risk

$$R_{\delta_{\pi_{\sqrt{n}/2, \sqrt{n}/2}}}(\theta) = \frac{1}{4(\sqrt{n}+1)^2}$$

- By comparison, the risk function of the standard sample mean estimate $\overline{X}$ is given by

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- Therefore, for *Bernoulli* measurements, the standard sample mean estimator is *not* minimax

Least favorable sequence of priors

- Previously, we mentioned that the *minimaxity* of an estimator can be *certified* using a least favorable prior

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- Sometimes, however, the least favorable prior may *not* exist. Instead, if there exists a *sequence* of priors $(\pi_m : m \in \mathcal{N})$ such that

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- The above sequence of priors is usually known as a *least favorable sequence of priors*

Example: Gaussian with unknown mean

- Consider $X = (X_1, \ldots, X_n)$, where

$$X_i \overset{iid}{\sim} \mathcal{N}(\theta, \sigma^2)$$

Here, $\theta \in \mathcal{R}$ is the unknown parameter, and σ^2 is known

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- We thus *suspect* that the sample mean estimator $\overline{X}$ is, in fact, minimax under the squared-error loss
- To prove this, let us consider the family of priors
$$\pi_m = \mathcal{N}(0, m^2)$$
- By the *conjugacy* between the Gaussian priors and Gaussian likelihood with known variance, the posterior distribution of Θ given X is again Gaussian and is given by
$$\mathcal{N}\left(\frac{\overline{X}}{1 + \frac{\sigma^2}{nm^2}}, \frac{\frac{\sigma^2}{n}}{1 + \frac{\sigma^2}{nm^2}}\right)$$

- Note that the posterior variance of Θ given X is *not* a function of X. Therefore, the Bayesian risk of the posterior mean estimator is simply given by the the posterior variance of Θ given X:

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- We may thus conclude that for Gaussian measurements with known variance, the sample mean estimator is, in fact, minimax

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- This technique is mainly based on the following simple observation. Suppose that δ is a minimax estimator for the sub-model $\theta \in \Omega_0 \subseteq \Omega$. Now if

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- To see this. simply note that for any estimator δ', we have

$$\sup_{\theta \in \Omega} R_{\delta'}(\theta) \geq \sup_{\theta \in \Omega_0} R_{\delta'}(\theta) \geq \sup_{\theta \in \Omega_0} R_\delta(\theta) = \sup_{\theta \in \Omega} R_\delta(\theta)$$

  where the second inequality follows from the *minimaxity* of δ over the sub-model $\theta \in \Omega_0$

- This technique is particularly useful when there are *multiple* unknown parameters, and a minimax estimator is much easier to find when some of the parameters are assumed to be *fixed*

Example: Gaussian with unknown mean and variance

- Consider $X = (X_1, \ldots, X_n)$, where

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- Therefore, we restrict attention to the family parameterized by
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where B is a known constant
- To see that in this case, the sample mean $\overline{X}$ is, once again, a minimax estimator for θ, let us first restrict the value of σ^2 to be B

- When $\sigma^2 = B$, as we have shown previously, the sample mean $\overline{X}$ is minimax, with the maximum risk given by B/n

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  so the maximum risk over Ω remains to be B/n
- We many thus conclude that the sample mean $\overline{X}$ remains to be minimax even when σ^2 is unknown but has a known upper bound

ECEN 662: Estimation & Detection Theory

Lecture 15: Minimax Estimators and Worst-Case Optimality (Part II)

Tie Liu

Texas A&M University

Estimating the mean of an unknown distribution

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Here, the density f is assumed to be *unknown*, so our setting here is *non-parametric*

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- To see this, let us first *restrict* f to be *Gaussian*. Under this restriction, previously we have shown that the sample mean $\overline{X}$ is minimax with a maximum risk B/n

- Further note that for any f, the risk of the sample mean $\overline{X}$ is

$$R_{\overline{X}}(f) = \mathbb{E}_f[(\overline{X} - \theta_f)^2] = \sigma_f^2/n$$

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- What if the constraint is on the *support* of f instead, say the support of f is $[0, 1]$?

- Previously, we have shown that if f is *Bernoulli*, then the sample mean $\overline{X}$ is *not* minimax. Instead, a minimax estimator is given by

$$\delta(X) = \frac{\overline{X} + 1/(2\sqrt{n})}{1 + 1/\sqrt{n}}$$

with the maximum risk given by

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- To show that the above estimator is, in fact, minimax for *all* f supported in $[0,1]$, all we need to do is to show that the maximum risk of δ remains to be

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- To see this, simply note that

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R_\delta(f) &= \mathbb{E}_f[(\delta(X) - \theta_f)^2] \\
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- We now use the fact that $X_1 \in [0,1]$ so we have $X_1^2 \leq X_1$ with probability 1 and hence

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- We may thus conclude that

$$R_\delta(f) \leq \frac{1}{4(\sqrt{n}+1)^2}$$

for any f supported on $[0,1]$, where the equality holds when f is *Bernoulli*

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- One situation for which the *uniqueness* of the minimax estimator can be readily recognized is as follows: If δ_π is a *unique* Bayes estimator such that

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then δ_π is also a *unique* minimax estimator
- To see this, let δ' be another estimator that is *different* from δ_π. We have
$$R_{\delta'}^{(M)} \geq R_{\delta',\pi} > R_\pi^{(B)} = R_{\delta_\pi}^{(M)}$$
where the *strict* inequality follows from the *uniqueness* of δ_π as a Bayes estimator

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- To show that that $\overline{X}$ is admissible, assume the opposite so there exists another estimator δ that *strictly* dominates $\overline{X}$
- Note that the risk function of $\overline{X}$ is given by $R_{\overline{X}}(\theta) = \sigma^2/n$. By the fact that the risk function for any estimator is *continuous*, there must exists an interval (θ_0, θ_1) such that

$$R_\delta(\theta) \leq \sigma^2/n - \epsilon, \quad \forall \theta \in (\theta_0, \theta_1)$$

for some $\epsilon > 0$

- To find a contradiction, let us consider the family of priors $\mathcal{N}(0, \delta_0^2)$ and denote the corresponding Bayes estimators as $\delta_{\pi_{\sigma_0^2}}$

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$$R_{\delta_{\pi_{\sigma_0^2}}, \pi_{\sigma_0^2}} = \frac{\sigma^2}{\sigma^2/\sigma_0^2 + n}$$

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- The Bayesian risk of δ can be bounded from above as:

$$
\begin{aligned}
R_{\delta,\pi_{\sigma_0^2}} &= \frac{1}{\sqrt{2\pi\sigma_0^2}} \int_{-\infty}^{\infty} R_\delta(\theta) \exp\left(-\frac{\theta^2}{2\sigma_0^2}\right) d\theta \\
&= \frac{1}{\sqrt{2\pi\sigma_0^2}} \left(\int_{-\infty}^{\theta_0} + \int_{\theta_0}^{\theta_1} + \int_{\theta_1}^{\infty} \right) R_\delta(\theta) \exp\left(-\frac{\theta^2}{2\sigma_0^2}\right) d\theta \\
&\leq \frac{1}{\sqrt{2\pi\sigma_0^2}} \left(\left(\int_{-\infty}^{\theta_0} + \int_{\theta_1}^{\infty} \right) \left(\frac{\sigma^2}{n}\right) \exp\left(-\frac{\theta^2}{2\sigma_0^2}\right) d\theta + \right. \\
&\qquad \left. \int_{\theta_0}^{\theta_1} \left(\frac{\sigma^2}{n} - \epsilon\right) \exp\left(-\frac{\theta^2}{2\sigma_0^2}\right) d\theta \right) \\
&= \frac{1}{\sqrt{2\pi\sigma_0^2}} \int_{-\infty}^{\infty} \left(\frac{\sigma^2}{n}\right) \exp\left(-\frac{\theta^2}{2\sigma_0^2}\right) d\theta - \\
&\qquad \frac{1}{\sqrt{2\pi\sigma_0^2}} \int_{\theta_0}^{\theta_1} \epsilon \exp\left(-\frac{\theta^2}{2\sigma_0^2}\right) d\theta \\
&= \frac{\sigma^2}{n} - \frac{\epsilon}{\sqrt{2\pi\sigma_0^2}} \int_{\theta_0}^{\theta_1} \exp\left(-\frac{\theta^2}{2\sigma_0^2}\right) d\theta
\end{aligned}
$$

- Note that when $\sigma_0^2 \to \infty$, both the upper bound of $R_{\delta,\pi_{\sigma_0^2}}$ and $R_{\delta_{\pi_{\sigma_0^2}},\pi_{\sigma_0^2}}$ approach σ^2/n, but with *different* convergence rates

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- In particular, we have

$$\frac{\sigma^2/n - R_{\delta,\pi_{\sigma_0^2}}}{\sigma^2/n - R_{\delta_{\pi_{\sigma_0^2}},\pi_{\sigma_0^2}}} \geq \frac{\frac{\epsilon}{\sqrt{2\pi\sigma_0^2}} \int_{\theta_0}^{\theta_1} \exp\left(-\frac{\theta^2}{2\sigma_0^2}\right) d\theta}{\frac{\sigma^4}{n(\sigma^2+n\sigma_0^2)}}$$

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- Therefore, for sufficiently large σ_0^2, we will have

$$\frac{\sigma^2/n - R_{\delta,\pi_{\sigma_0^2}}}{\sigma^2/n - R_{\delta_{\pi_{\sigma_0^2}},\pi_{\sigma_0^2}}} > 1$$

and hence

$$R_{\delta,\pi_{\sigma_0^2}} < R_{\delta_{\pi_{\sigma_0^2}},\pi_{\sigma_0^2}}$$

contradicting the fact that $\delta_{\pi_{\sigma_0^2}}$ is the *Bayes* estimator for $\pi_{\sigma_0^2}$

Randomized minimax estimators

- *Randomized* estimators played little role in our exploration of *average* risk optimality, since non-randomized estimators of equal or better average risk are always available. However, they turn out to play a role when we consider the *minimax criterion*

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- Notice that when working with *convex* losses, we can dispense with randomized estimators, because we can find a deterministic estimator with the same or better performance. However, there are *non-convex* losses for which no deterministic minimax estimator exists, as the following example demonstrates

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- Let $X \sim Binomial(n, \theta)$, where $\theta \in [0, 1]$ is the unknown parameter. Consider estimation of θ under the *0-1* loss:

$$L_\alpha(\theta, d) = 1_{\{|\theta - d| \geq \alpha\}}$$

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- Claim: If $\alpha < \frac{1}{2(n+2)}$, the maximum risk of *any* deterministic estimator is 1

- On the other hand, consider the *completely* random guess $\delta = U$ where $U \sim \mathcal{U} \in [0,1]$ and is *independent* of X

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- Hence, in this setting, there can be *no* deterministic minimax estimator

ECEN 662: Estimation & Detection Theory

Lecture 16: Empirical Bayes Estimation

Tie Liu

Texas A&M University

Empirical Bayes

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- When the learning of the prior distribution follows a *frequentist* approach, the overall approach for building such estimators is called *empirical Bayes*
- Next, we shall illustrate how this approach works with an example that stunned the entire statistics community in 1961

Gaussian with unknown means

- Let $X = (X_1, \ldots, X_n)$, where

$$X_i \overset{ind}{\sim} \mathcal{N}(\theta_i, 1)$$

Here, $\theta = (\theta_1, \ldots, \theta_n)$ are the unknown parameters

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- Now suppose we want to *simultaneously* estimate θ_i for all $i \in [n]$, and our loss function is the (completely natural) *total* squared-error loss

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- It is only natural to think that $X = (X_1, \ldots, X_n)$ is (at least) admissible. A stunning result of James and Stein in 1961, however, showed that this is actually *not* the case when $n \geq 3$

James-Stein estimator

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- To compare the risks of the natural estimator X and the James-Stein estimator $\delta(X)$, note that

$$\begin{aligned}
\|\delta(X) - \theta\|^2 &= \|(\delta(X) - X) + (X - \theta)\|^2 \\
&= \|\delta(X) - X\|^2 + \|X - \theta\|^2 + 2(\delta(X) - X)^t (X - \theta) \\
&= \frac{(n-2)^2}{\|X\|^2} + \|X - \theta\|^2 - 2(n-2)\left(\frac{X}{\|X\|^2}\right)^t (X - \theta)
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- Taking expectations on both sides gives

$$\begin{aligned}
R_\delta(\theta) = R_X(\theta) + (n-2)^2 \mathbb{E}_\theta \left[\frac{1}{\|X\|^2}\right] - \\
2(n-2)\mathbb{E}_\theta \left[\left(\frac{X}{\|X\|^2}\right)^t (X - \theta)\right]
\end{aligned}$$

- (*Stein's lemma*) Let $Z = (Z_1, \ldots, Z_n)$ be a *standard* Gaussian vector, and let $g : \mathcal{R}^n \to \mathcal{R}^n$ be a differential function such that

$$\mathbb{E}\left[\sqrt{\sum_{i=1}^{n}\sum_{j=1}^{n}\left(\frac{\partial}{\partial Z_j} g_i(Z)\right)^2}\right] < \infty$$

Then

$$\mathbb{E}[g(Z)^t Z] = \sum_{i=1}^{n} \mathbb{E}\left[\frac{\partial}{\partial Z_i} g_i(Z)\right]$$

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$$\mathbb{E}_\theta\left[\left(\frac{X}{\|X\|^2}\right)^t (X-\theta)\right] = (n-2)\mathbb{E}_\theta\left[\frac{1}{\|X\|^2}\right]$$

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- It thus follows that

$$R_\delta(\theta) = R_X(\theta) - (n-2)^2 \mathbb{E}_\theta\left[\frac{1}{\|X\|^2}\right] < R_X(\theta), \quad \forall n \geq 3$$

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- The main intuition for the James-Stein estimator is that

$$R_X(\theta) = \mathbb{E}_\theta[\|X\|^2] - \|\theta\|^2$$

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- It is thus natural to consider *shrinking* the natural estimator X to make up the gap between $\mathbb{E}_\theta[\|X\|^2]$ and $\|\theta\|^2$
- While *shrinkage* is a common characteristic of *Bayes* estimators, here that the shrinkage factor $1 - (n-2)/\|X\|^2$ depends on the observation X naturally reminds us of *empirical* Bayes

James-Stein as an empirical Bayes estimator

- To see how the James-Stein estimator can be derived via *empirical* Bayes estimation, consider the family of priors

$$\Theta_i \overset{iid}{\sim} \mathcal{N}(0, A)$$

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- To find an estimate of $\frac{1}{A+1}$ based on the observation A, note that

$$X_i \overset{iid}{\sim} \mathcal{N}(0, A+1)$$

- One possible choice is to find the *UMVUE* for $\frac{1}{A+1}$

- To do so, simply note that $\|X\|^2$ is a complete sufficient statistic for A, and

$$\mathbb{E}_A\left[\frac{1}{\|X\|^2}\right] = \frac{1}{(n-2)(A+1)}$$

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- Substituting

$$\frac{1}{A+1} \approx \frac{n-2}{\|X\|^2}$$

into the Bayes estimator

$$\delta_{\pi_A} = \left(1 - \frac{1}{A+1}\right) X$$

immediately gives rise to the James-Stein estimator

Shrinkage to the mean

- Assume that

$$X_i \overset{iid}{\sim} \mathcal{N}(\theta_i, \sigma_0^2)$$

where σ_0^2 is known. For any $i \in [n]$, let

$$\hat{\theta}_i = \overline{X} + \left(1 - \frac{(n-3)\sigma_0^2}{S}\right)(X_i - \overline{X})$$

where

$$\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

$$S = \sum_{i=1}^{n}(X_i - \overline{X})^2$$

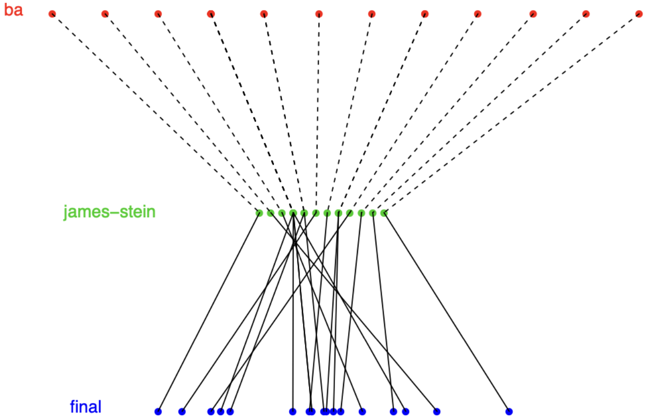
For any $n \geq 4$, we have

$$\mathbb{E}_\theta[\|\hat{\theta} - \theta\|^2] < n\sigma_0^2$$

James-Stein estimator in the wild

Table 6: Batting averages of 18 baseball players. *Column 1*, observed batting average after 45 attempts. *Column 2*, James–Stein estimate (3.4), $\sigma_0^2 = 0.066$. *Column 3*, average for remainder of the 1970 season. *Column 4*, number of attempts in remainder of the season.

	MLE	JS	final avg	# final AB
Clemente	.400	.294	.346	367
F. Robinson	.378	.289	.298	426
F. Howard	.356	.284	.276	521
Johnstone	.333	.279	.222	275
Berry	.311	.275	.273	418
Spencer	.311	.275	.270	466
Kessinger	.289	.270	.263	586
L. Alvarado	.267	.265	.210	138
Santo	.244	.261	.269	510
Swoboda	.244	.261	.230	200
Unser	.222	.256	.264	277
Williams	.222	.256	.256	270
Scott	.222	.256	.303	435
Petrocelli	.222	.256	.264	538
E. Rodriguez	.222	.256	.226	186
Campaneris	.200	.251	.285	558
Munson	.178	.246	.316	408
Alvis	.156	.242	.200	70



ba
james–stein
final

- The prediction error for the MLE and the James-Stein estimator are:

$$\begin{aligned} \|\hat{\theta}_{MLE}(x_{current}) - x_{final}\|^2 &= 0.0745 \\ \|\hat{\theta}_{JS}(x_{current}) - x_{final}\|^2 &= 0.0213 \end{aligned}$$

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- More generally, empirical Bayes methods share data across individuals, for the benefit of overall performance but at possible peril to unusual individuals

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- Let $m_\pi(x)$ be the marginal density of X. Then

$$m_\pi(x) = \int_\Omega \varphi(x-\theta)\pi(\theta)d\theta$$

where the φ is the density function of a standard Gaussian variable

- (*Tweedie's formula*) The posterior mean estimator

$$\delta_\pi(x) = \mathbb{E}[\Theta | X = x]$$

can be conveniently expressed in terms of the marginal density $m_\pi(x)$ as follows:

$$\delta_\pi(x) = x + \frac{d}{dx} \log m_\pi(x)$$

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- Note that in the above procedure, a model is placed on the marginal density $m_\pi(x)$, instead of the prior distribution $\pi(\theta)$

The missing species problem

- *Corbet*, a leading naturalist, had been trapping butterflies in Malaysia (then Malaya) for two years:

Table 2: Butterfly data; number y_x of species seen x times each in two years of trapping; 118 species trapped just once, 74 trapped twice each, etc.

x	1	2	3	4	5	6	7	8	9	10	11	12
y_x	118	74	44	24	29	22	20	19	20	15	12	14

x	13	14	15	16	17	18	19	20	21	22	23	24
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- Corbet asked *Fisher* a seemingly unanswerable missing-species question: If he continued trapping for one additional year, how many *new* species could he expect to see?

A Poisson process model

- Let X_i be the number of times species $i \in [S]$ is trapped in the original one unit of time (the unit being two years for Corbet), and let $X_i(t)$ the number of times trapped in a hypothetical future period of t units

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$$X_i \overset{ind}{\sim} Poisson(\theta_i)$$
$$X_i(t) \overset{ind}{\sim} Poisson(\theta_i t)$$

where $(X_i : i \in [S])$ and $(X_i(t) : i \in [S])$ are independent of each other

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- Let X_i be the number of times species $i \in [S]$ is trapped in the original one unit of time (the unit being two years for Corbet), and let $X_i(t)$ the number of times trapped in a hypothetical future period of t units

- Under the *Poisson process* assumption, we have

$$X_i \stackrel{ind}{\sim} Poisson(\theta_i)$$
$$X_i(t) \stackrel{ind}{\sim} Poisson(\theta_i t)$$

  where $(X_i : i \in [S])$ and $(X_i(t) : i \in [S])$ are independent of each other

- Under this model, the probability that species i is not seen in the initial trapping period but is seen in the future period is given by

$$\mathbb{P}_{\theta_i}(X_i = 0, X_i(t) > 0) = e^{-\theta_i}(1 - e^{-\theta_i t})$$

Constructing an empirical Bayes estimator

- Assuming that $\Theta_i \overset{iid}{\sim} \pi(\theta)$, the expected number of Corbet's hoped-for "new" species N is given by

$$\mathbb{E}[N] = S \int_0^\infty e^{-\theta}(1 - e^{-\theta t})\pi(\theta) d\theta$$

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- It follows immediately that

$$\mathbb{E}[N] = \mathbb{E}[Y_1]t - \mathbb{E}[Y_2]t^2 + \mathbb{E}[Y_3]t^3 - \cdots$$

- Replacing $\mathbb{E}[Y_x]$ by the unbiased estimate y_x leads to *Good* and *Toulmin*'s intriguing estimator:

$$\widehat{\mathbb{E}[N]} = y_1 t - y_2 t^2 + y_3 t^3 - \cdots$$

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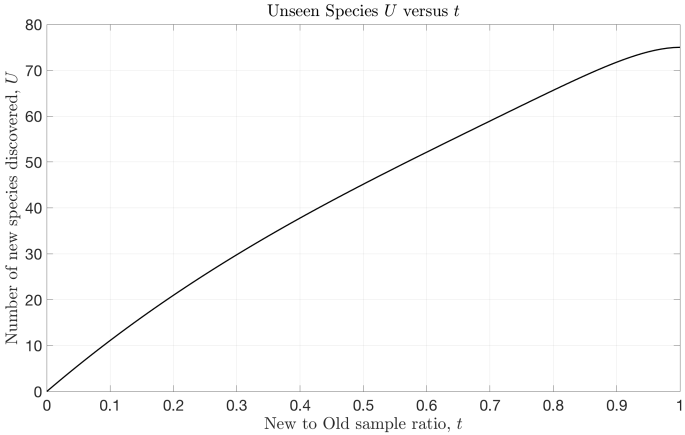
- There are interesting recent work on how to improve on the Good–Toulmin estimator for $t > 1$

Species discovery curve



ECEN 662: Estimation & Detection Theory

Lecture 17: Transform and Rejection Sampling

Tie Liu

Texas A&M University

Sampling/random variable generation/simulation

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- Here we shall focus on two of the most basic sampling methods: *transformation* and *rejection*
- Our main criterion for evaluating different sampling algorithms are the *required knowledge* of the distribution and *algorithm efficiency*

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- Thus, in order to sample from a CDF F on $\mathbb{R}$, it suffices to generate a random variable $U \sim \mathcal{U}([0,1])$ and make the transformation $x = F^{-1}(u)$
- For example, if $X \sim Exp(1)$, then $F(x) = 1 - e^{-x}$ and $F^{-1}(u) = -\log(1-u)$. Therefore, if $U \sim \mathcal{U}([0,1])$, then $X = -\log(U) \sim Exp(1)$ (as U and $1-U$ are both uniform between 0 and 1)

Proof that inverse transform works

- First note that CDF is *right-continuous*. Therefore, for any $u \in [0, 1]$, the generalized inverse satisfies

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- It follows immediately that

$$\mathbb{P}(F^{-1}(U) \leq x) = \mathbb{P}(U \leq F(x)) = F(x)$$

where the last equality follows from the fact that $U \sim \mathcal{U}([0,1])$ and $F(x) \in [0,1]$

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- Most noticeably, the CDF of the standard Gaussian distribution *cannot* be expressed explicitly, since

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} \exp(-z^2/2) dz$$

- Thus, to use the inverse transform to generate a standard Gaussian random variable, one may have to use an *approximate* expression for the generalized inverse Φ^{-1}

- One known such approximation is

$$\Phi^{-1}(u) \approx t - \frac{2.30753 + 0.27061t}{1 + 0.99299t + 0.04481t^2}$$

where

$$t = \sqrt{-2\log u}$$

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- If no other fast sampling method was available, this approximation could be used in settings which do *not* require much precision in the *tails* of $\mathcal{N}(0, 1)$

General transformation method

- While the inverse transform offers a *generic* link between an arbitrary random variable on $\mathbb{R}$ and the uniform random variable, sometimes there is a *special* link between a specific random variable to some random variables that are easy to simulate. This special link can often be exploited to sample from the targeted distribution

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- For example, let X_1 and X_2 be two independent standard Gaussian random variables. It is known that the polar coordinates (R, Θ) of (X_1, X_2) are distributed as:

$$R \sim Exp(1/2) \quad \text{and} \quad \Theta \sim \mathcal{U}([0, 2\pi])$$

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  where R and Θ are independent of each other

- Therefore, if U_1 and U_2 are i.i.d. $\mathcal{U}([0, 1])$

$$X_1 = \sqrt{-2 \log U_1} \cos(2\pi U_2), \quad X_2 = \sqrt{-2 \log U_1} \sin(2\pi U_2)$$

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- Alternatively, if U and V are i.i.d. $\mathcal{U}([0,1])$, the distribution of

$$\frac{U^{1/\alpha}}{U^{1/\alpha} + V^{1/\beta}}$$

conditioned on

$$U^{1/\alpha} + V^{1/\beta} \leq 1$$

is $Beta(\alpha, \beta)$

- However, given the constraint on $U^{1/\alpha} + V^{1/\beta}$, this result does *not* provide a good algorithm to generate $Beta(\alpha, \beta)$ random variables for *large* values of α and β

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- The basic idea is to find an alternative probability distribution G, with density function g, from which we already have an efficient algorithm for sampling from (e.g., inverse transform method or whatever), but also such that g is "close" to the target density function f
- In particular, we assume that the ratio $f(x)/g(x)$ is *bounded* by a constant $M > 0$:
$$\sup_{x \in \mathcal{X}} \frac{f(x)}{g(x)} \leq M$$

- Here is the algorithm for generating a random variable X distributed as F:

  1) Generate a random variable Y according to the distribution G
  2) Generate a random variable U that is uniform between 0 and 1 and independent of Y
  3) If
  $$U \leq \frac{f(Y)}{Mg(Y)}$$
  then set $X = Y$ ("accept"); otherwise, go back to Step 1 ("reject")

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- Next we provide a few remarks in regard to the algorithm before proving that it actually works

Remarks

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- The success probability p can be calculated as follows:

$$\mathbb{P}\left(U \leq \frac{f(Y)}{Mg(Y)} \middle| Y = y\right) = \frac{f(y)}{Mg(y)}$$

and hence

$$p = \int_{\mathcal{X}} \frac{f(y)}{Mg(y)} g(y)dy = \frac{1}{M} \int_{\mathcal{X}} f(y)dy = \frac{1}{M}$$

- Thus, on average the number of iterations required is given by

$$p^{-1} = M$$

and we can now indeed see that it is desirable to choose our alternative density g so as to minimize this constant

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- In the end, we obtain our X as having the *conditional* distribution of Y given that the event $U \leq \frac{f(Y)}{Mg(Y)}$ occurs

Proof that the algorithm works

- Let $B \subseteq \mathcal{X}$. Note that

$$
\begin{aligned}
\mathbb{P}\left(U \le \frac{f(Y)}{Mg(Y)}, Y \in B\right) &= \int_B \mathbb{P}\left(U \le \frac{f(y)}{Mg(y)}\right) g(y)dy \\
&= \int_B \frac{f(y)}{Mg(y)} g(y)dy \\
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- It follows that

$$\mathbb{P}\left(Y \in B | U \le \frac{f(Y)}{Mg(Y)}\right) = \frac{\mathbb{P}\left(U \le \frac{f(Y)}{Mg(Y)}, Y \in B\right)}{\mathbb{P}\left(U \le \frac{f(Y)}{Mg(Y)}\right)}$$
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$$= \frac{\frac{1}{M}\int_B f(y)dy}{\frac{1}{M}} = \int_B f(y)dy$$

- That is, the conditional distribution of Y given $U \leq \frac{f(Y)}{Mg(Y)}$ is indeed F

Example: Standard Gaussian distribution

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- One possible choice for the proposal distribution is the *two-sided exponential* distribution

$$g(x) = \frac{1}{2}\exp(-|x|)$$

  which we already know how to sample from using the *inverse transform* method

- The density ratio

$$\frac{f(x)}{g(x)} = 2A\exp\left(|x| - \frac{x^2}{2}\right)$$

is *maximized* at $x = \pm 1$, giving

$$M = \sup_x \frac{f(x)}{g(x)} = 2\exp(1/2)A$$

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- The *acceptance* rule is thus given by

$$U \leq \frac{f(Y)}{Mg(Y)} = \exp\left(|Y| - \frac{Y^2+1}{2}\right) = \exp\left(-\frac{(|Y|-1)^2}{2}\right)$$

or equivalently

$$-\log U \geq \frac{(|Y|-1)^2}{2}$$

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- Note that $-\log(U) \sim Exp(1)$, so in each iteration of the algorithm, we cam simply use two independent *exponential* random variables to perform simulation

- Last but not the least, we see from this example that rejection sampling *only* requires the density function of the target distribution up to a *multiplicative* constant: The same constant *cancels out* in $f(Y)$ and M!

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- This makes makes rejection sampling potentially *suitable* for sampling from *posterior* distributions in Bayesian statistics

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- However, as the dimensions of the problem get larger, the ratio of the embedded volume to the "corners" of the embedding volume tends towards zero, thus a lot of rejections can take place before a useful sample is generated, thus making the algorithm inefficient and impractical
- In high dimensions, it is necessary to use a different approach, typically a *Markov chain Monte Carlo* method such as Metropolis sampling or Gibbs sampling

ECEN 662: Estimation & Detection Theory

Lecture 18: Monte Carlo Integration and Importance Sampling

Tie Liu

Texas A&M University

Classical Monte Carlo integration

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- Consider the generic problem of evaluating the expectation/integral

$$\mathbb{E}_f[h(X)] = \int_{\mathcal{X}} h(x) f(x) dx$$

- For cases where the above expectation/integral *cannot* be evaluated analytically, it is natural to propose using multiple *independent* samples $(X_1, \ldots, X_m)$ drawn from the density f to approximate it:

$$\hat{h}_m = \frac{1}{m} \sum_{i=1}^{m} h(X_i)$$

- This approach is often referred to as the *Monte Carlo* method

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- Second, by the *weak law of large numbers*,

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  in the limit as $m \to \infty$, which is the main justification of the Monte Carlo method
- In practice, however, the sample size m is always *finite*. In order for the method to work, the *variance* of the estimate

$$\mathrm{Var}_f[\hat{h}_m] = \frac{\mathrm{Var}_f[h(X)]}{m}$$

  has to be *sufficiently small*

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so
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$$\mathbb{E}_f[h(X)] = p_B \quad \text{and} \quad \mathrm{Var}_f[\hat{h}_m] = \frac{p_B - p_B^2}{m}$$

- If we require the standard deviation of $\hat{h}_m$ to be in the *same order* as p_B, we need

$$\sqrt{\frac{p_B - p_B^2}{m}} \approx p_B \quad \Longrightarrow \quad m \approx \frac{1}{p_B} - 1$$

which can be exceedingly *large* for a very *rare* event B

Importance sampling

- The method of *importance sampling* is an evaluation of the expectation/integral

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- The method is based on the following alternative representation of the expectation/integral:

$$\mathbb{E}_f[h(X)] = \int_{\mathcal{X}} h(x)\omega(x)g(x)dx = \mathbb{E}_g\left[\omega(X)h(X)\right]$$

where $\omega(x) = f(x)/g(x)$ is the density ratio

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- The above representation immediately leads to the following approximation based on samples $(X_1, \ldots, X_m)$ drawn from g:

$$\hat{h}_m = \frac{1}{m}\sum_{i=1}^{m} \omega(X_i)h(X_i)$$

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- The aforementioned importance sampling method is a manifest to the fact that a given integral is *not* intrinsically associated with a given distribution
- Importance sampling is thus of considerable interest because it puts very *little* restriction on the choice of the proposal distribution, which can be chosen from distributions that are easy to sample from
- Moreover, the same samples from g can be used *repeatedly* for different functions h but also for different densities f

Variance control

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- Among proposal distributions leading to finite variances for the importance sampling estimator, it is possible to identify *optimal* proposal distribution corresponding to a given function h and a fixed distribution f

- More specifically, the choice of g that minimizes the *variance* of the importance sampling estimator is

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- From a practical point of view, the above result suggests looking for proposal distributions g for which $\omega(x)|h(x)|$ is almost *constant*

Self-normalizing importance sampling

- An alternative to the vanilla importance sampling estimator

$$\hat{h}_m = \frac{1}{m}\sum_{i=1}^{m}\omega(X_i)h(X_i)$$

is the following *self-normalizing importance sampling* estimator

$$\hat{h}'_m = \frac{\sum_{i=1}^{m}\omega(X_i)h(X_i)}{\sum_{i=1}^{m}\omega(X_i)}$$

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- Note that as $m \to \infty$,

$$\frac{1}{m}\sum_{i=1}^{m}\omega(X_i) \overset{i.p.}{\sim} \mathbb{E}_g[\omega(X)] = \int_{\mathcal{X}} \frac{f(x)}{g(x)}g(x)dx = \int_{\mathcal{X}} f(x)dx = 1$$

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- Therefore, just like the vanilla importance sampling estimator, the *self-normalizing* importance sampling estimator is also a *consistent* estimator of the expectation $\mathbb{E}_f[h(X)]$

- In addition, the *self-normalizing* importance sampling estimator also enjoys a couple of important additional benefits

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- In addition, the *self-normalizing* importance sampling estimator also enjoys a couple of important additional benefits

- First, unlike the vanilla importance sampling estimator, the *self-normalizing* importance sampling estimator is slightly *biased* and thus may enjoy the benefit of having a *smaller* variance

- Second, unlike the vanilla importance sampling estimator, which requires precise knowledge of the density f, the *self-normalizing* importance sampling estimator can be implemented based on an *un-normalized* density $\tilde{f}$ with an *unknown* scaling factor, which makes it particularly appealing when f is a *posterior* distribution in Bayesian statistics

Monte Carlo marginalization

- Consider the problem of computing the marginal density $f(x)$ from the joint density $f(x, z)$ via *importance sampling*:

$$f(x) = \int_{\mathcal{Z}} f(x, z) dz = \int_{\mathcal{Z}} \frac{f(x, z)}{g(z)} g(z) dz$$

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- Based on the previous analysis, the *optimal* proposal distribution that minimizes the variance must ensure that the ratio $f(x, z)/g(z)$ is *constant* with respect to z:

$$g(z) \propto_z f(x, z)$$

  and hence

$$g(z) = f(z|x)$$

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- However, knowing the joint density $f(x,z)$, computing the posterior distribution $f(z|x)$ is the *same* as computing the marginal distribution $f(x)$

- Therefore, to perform *marginalization* using importance sampling, one may wish to find an *easy-to-sample-from* and *easy-to-evaluate* proposal distribution that also *approximates* the posterior distribution

- Therefore, to perform *marginalization* using importance sampling, one may wish to find an *easy-to-sample-from* and *easy-to-evaluate* proposal distribution that also *approximates* the posterior distribution
- This is the essential idea of *variational auto-encoding (VAE)*

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- The conditional distribution $p_\theta(\boldsymbol{x}|\boldsymbol{z})$ must also be *easy* to draw from
- The generative model $p_\theta(\boldsymbol{x})$ is *implicitly* defined as the *marginal* distribution of $\boldsymbol{x}$:

$$p_\theta(\boldsymbol{x}) = \int p(\boldsymbol{z}) p_\theta(\boldsymbol{x}|\boldsymbol{z}) d\boldsymbol{z} = \mathbb{E}_{\boldsymbol{Z} \sim p(\boldsymbol{z})}[p_\theta(\boldsymbol{x}|\boldsymbol{Z})]$$

Model training

- The goal of learning is thus to identify a model parameter θ such that the generative model

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- In statistics, one often use the so-called *K-L divergence* to measure the "distance" between two probability distributions p and q:

$$D_{KL}(p\|q) := \int p(x) \log \frac{p(x)}{q(x)} dx = \mathbb{E}_{X \sim p(x)} \left[\log \frac{p(X)}{q(X)} \right]$$

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- One possibility is to learn a parameter θ by minimizing the K-L divergence

$$\begin{aligned} D_{KL}(p \| p_\theta) &= \mathbb{E}_{\boldsymbol{X} \sim p(\boldsymbol{x})} \left[\log \frac{p(\boldsymbol{X})}{p_\theta(\boldsymbol{X})} \right] \\ &= \mathbb{E}_{\boldsymbol{X} \sim p(\boldsymbol{x})} [\log p(\boldsymbol{X})] - \mathbb{E}_{\boldsymbol{X} \sim p(\boldsymbol{x})} [\log p_\theta(\boldsymbol{X})] \end{aligned}$$

Maximum-likelihood (ML) estimate

- Note that minimizing the K-L divergence $D_{KL}(p \| p_\theta)$ with respect to θ is equivalent to maximizing

$$\mathbb{E}_{\boldsymbol{X} \sim p(\boldsymbol{x})} [\log p_\theta(\boldsymbol{X})] \approx \frac{1}{m} \sum_{i=1}^{m} \log p_\theta(\boldsymbol{x}_i)$$

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- To compute the maximum-likelihood estimate, we need to evaluate the marginal distribution

$$p_\theta(\boldsymbol{x}) = \int p(\boldsymbol{z})p_\theta(\boldsymbol{x}|\boldsymbol{z})d\boldsymbol{z} = \mathbb{E}_{\boldsymbol{Z}\sim p(\boldsymbol{z})}[p_\theta(\boldsymbol{x}|\boldsymbol{Z})]$$

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  for any given $\boldsymbol{x}$

- Note, however, that the integral for evaluating $p_\theta(\boldsymbol{x})$ *cannot* be evaluated analytically

Monte-Carlo (MC) integration

- In practice, one may *estimate* $p_\theta(\boldsymbol{x})$ through *Monte Carlo integration*:

$$p_\theta(\boldsymbol{x}) \approx \frac{1}{n}\sum_{j=1}^{n} p_\theta(\boldsymbol{x}|\boldsymbol{z}_j)$$

  where $(\boldsymbol{z}_1, \ldots, \boldsymbol{z}_n)$ are independent samples of the random source $\boldsymbol{z}$

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- However, the conditional probability $p_\theta(\boldsymbol{x}|\boldsymbol{z})$ usually *fluctuates* significantly for different values of $\boldsymbol{z}$. As a result, the above estimate only becomes reliable when the number of samples drawn from the random source $\boldsymbol{z}$ becomes very large

Importance sampling to the rescue

- In statistics, there is a clever approach for solving this problem known as *importance sampling*

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- In statistics, there is a clever approach for solving this problem known as *importance sampling*
- The main idea is to introduce a new *sampling distribution* $q(\boldsymbol{z})$ to rewrite the expression for $p_\theta(\boldsymbol{x})$:

$$\begin{aligned} p_\theta(\boldsymbol{x}) &= \int p(\boldsymbol{z}) p_\theta(\boldsymbol{x}|\boldsymbol{z}) d\boldsymbol{z} \\ &= \int \frac{p(\boldsymbol{z}) p_\theta(\boldsymbol{x}|\boldsymbol{z})}{q(\boldsymbol{z})} q(\boldsymbol{z}) d\boldsymbol{z} \\ &= \mathbb{E}_{\boldsymbol{Z} \sim q(\boldsymbol{z})} \left[\frac{p(\boldsymbol{Z}) p_\theta(\boldsymbol{x}|\boldsymbol{Z})}{q(\boldsymbol{Z})} \right] \\ &\approx \frac{1}{n} \sum_{j=1}^{n} \frac{p(\boldsymbol{z}_j) p_\theta(\boldsymbol{x}|\boldsymbol{z}_j)}{q(\boldsymbol{z}_j)} \end{aligned}$$

where $(\boldsymbol{z}_1, \ldots, \boldsymbol{z}_n)$ are independent samples drawn according to $q(\boldsymbol{z})$

Ideal sampling distribution

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- Letting

$$\frac{p(\boldsymbol{z})p_\theta(\boldsymbol{x}|\boldsymbol{z})}{q(\boldsymbol{z})} = c$$

gives

$$q(\boldsymbol{z}) = \frac{p(\boldsymbol{z})p_\theta(\boldsymbol{x}|\boldsymbol{z})}{c}$$

where the constant c needs to be chosen such that

$$\int q(\boldsymbol{z})d\boldsymbol{z} = 1$$

- This gives

$$c = \int p(\boldsymbol{z}) p_\theta(\boldsymbol{x}|\boldsymbol{z}) d\boldsymbol{z} = p_\theta(\boldsymbol{x})$$

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- That is, the *ideal* sampling distribution is, in fact, given by the *posterior* distribution of $\boldsymbol{z}$ given $\boldsymbol{x}$

- This gives

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- In practice, one may introduce a second parameterized model $q_\phi(\boldsymbol{z}|\boldsymbol{x})$ to *approximate* the posterior distribution $p_\theta(\boldsymbol{z}|\boldsymbol{x})$

The encoder and decoder

- To summarize, even though our original goal was simply to train a generator $p_\theta(\boldsymbol{x}|\boldsymbol{z})$ via maximum-likelihood estimate, estimating the marginal likelihood $p_\theta(\boldsymbol{x})$ requires training an additional sampling distribution $q_\phi(\boldsymbol{z}|\boldsymbol{x}) \approx p_\theta(\boldsymbol{z}|\boldsymbol{x})$

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- Further note that while the generator $p_\theta(\boldsymbol{x}|\boldsymbol{z})$ *generates* data from random source, the goal of the sampling distribution $q_\phi(\boldsymbol{z}|\boldsymbol{x}) \approx p_\theta(\boldsymbol{z}|\boldsymbol{x})$ is to *recover* the random source from the data

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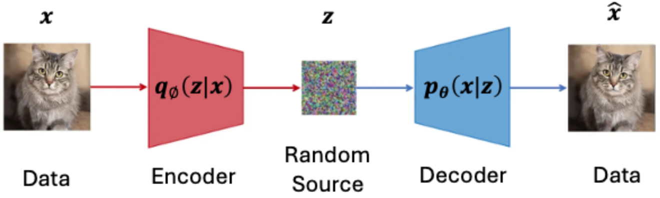
- We can think of the role of generator and sampling distribution as a process of first transforming data into a random source using $q_\phi(\boldsymbol{z}|\boldsymbol{x})$ and then recovering the data using the generator $p_\theta(\boldsymbol{x}|\boldsymbol{z})$



- This process of *data self-recovery* is usually known as *auto-encoding*. Under the context of *auto-encoding*, we call the sampling distribution $q_\phi(\boldsymbol{z}|\boldsymbol{x})$ the *encoder* and the generator $p_\theta(\boldsymbol{x}|\boldsymbol{z})$ the *decoder*

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- It is important to note that the aforementioned auto-encoding is *stochastic* in nature in that the recovered data is only equivalent to the original data *statistically* and in general is *not* the same as the original data

Joint training of encoder and decoder

- To jointly learn the encoder and decoder, let us consider the following *variational* representation of $\log p_\theta(\boldsymbol{x})$:

$$\log p_\theta(\boldsymbol{x}) = \max_{q(\boldsymbol{z}|\boldsymbol{x})} \left[\log p_\theta(\boldsymbol{x}) - D_{KL}(q(\boldsymbol{z}|\boldsymbol{x}) \| p_\theta(\boldsymbol{z}|\boldsymbol{x}))\right]$$

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- The above representation holds due to the fact that the K-L divergence

$$D_{KL}(q(\boldsymbol{z}|\boldsymbol{x}) \| p_\theta(\boldsymbol{z}|\boldsymbol{x})) \geq 0$$

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- Note that for any given $\boldsymbol{x}$,

A variational characterization of the log likelihood

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\begin{aligned}
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Modeling the sampling distribution

- To use *Monte Carlo* sampling to estimate the objective function

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- We thus introduce a second parameterized *model* $q_\phi(\boldsymbol{z}|\boldsymbol{x})$ and use it to give an *approximate* expression for $\log p_\theta(\boldsymbol{x})$:

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- Note that in order for the above approximation to be accurate, the model class needs to be sufficiently large so that one can always find a sampling distribution $q_\phi(\boldsymbol{z}|\boldsymbol{x})$ such that

$$q_\phi(\cdot|\boldsymbol{x}) \approx p_\theta(\cdot|\boldsymbol{x})$$

Importance sampling at work

- The key point here is that when $q_\phi(\cdot|\boldsymbol{x}) \approx p_\theta(\cdot|\boldsymbol{x})$, it is *easier* to use Monte Carlo integration to estimate the objective function

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because the expectant

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which is *constant* with respect to the random source $\boldsymbol{Z}$. This is exactly *importance sampling* at work!

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- Conclusion: We can *jointly* learn an encoder and a decoder by maximizing

$$\frac{1}{m}\sum_{i=1}^{m}\mathbb{E}_{\boldsymbol{Z}\sim q_\phi(\boldsymbol{z}|\boldsymbol{x}_i)}\left[\log \frac{p(\boldsymbol{Z})p_\theta(\boldsymbol{x}_i|\boldsymbol{Z})}{q_\phi(\boldsymbol{Z}|\boldsymbol{x}_i)}\right]$$

  over both θ and ϕ, where the expectation over $\boldsymbol{Z}$ can be estimated using Monte Carlo integration

Rejection sampling *v.s.* importance sampling

- Finally, let us make an informal comparison between rejection sampling and importance sampling in terms of computing the expectation/integration

$$\mathbb{E}_f[h(X)] = \int_{\mathcal{X}} h(x)f(x)dx$$

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- Finally, let us make an informal comparison between rejection sampling and importance sampling in terms of computing the expectation/integration

$$\mathbb{E}_f[h(X)] = \int_{\mathcal{X}} h(x) f(x) dx$$

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- As discussed before, the *self-normalizing* importance sampling estimator is given by

$$\hat{h}'_m = \frac{\sum_{i=1}^{m} \omega(X_i) h(X_i)}{\sum_{i=1}^{m} \omega(X_i)}$$

  where $(X_1, \ldots, X_m)$ are independent samples drawn from the proposal distribution g

- Alternatively, one may select a *subset* of samples $(X_{i_1}, \dots, X_{i_k})$ from $(X_1, \dots, X_m)$ using the *acceptance* rule

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$$\hat{h}''_m = \frac{1}{k} \sum_{j=1}^{k} h(X_{i_j}) = \frac{\sum_{i=1}^{m} \tilde{\omega}(X_i, U_i) h(X_i)}{\sum_{i=1}^{m} \tilde{\omega}(X_i, U_i)}$$

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- Note the *similarity* between the above two estimators

ECEN 662: Estimation & Detection Theory

Lecture 19: Markov Chain Monte Carlo Sampling

Tie Liu

Texas A&M University

Markov Chain Monte Carlo (MCMC) Sampling

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  - The samples *converge* in distribution to the target distribution.
  - We can construct a *consistent* estimator for expectations, even though the samples are *not* independent.

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- **Continuous Case:** If P has a *transition density* $p(x, y)$, π is stationary if it satisfies the integral equation:

$$\int_{\mathcal{X}} \pi(x) p(x, y) dx = \pi(y)$$

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- *Note:* We assume these technical conditions hold for the algorithms discussed herein.

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- **Detailed Balance Condition:** If P has a *density* $p(x,y)$, reversibility is equivalent to the *detailed balance condition*:

$$\pi(x)p(x,y) = \pi(y)p(y,x) \quad \forall x, y \in \mathcal{X}$$

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- Since $\mathbb{P}(X_1 \in \mathcal{X}) = 1$ and $\mathbb{P}(X_0 \in \mathcal{X}) = 1$, this simplifies to:

$$\mathbb{P}(X_0 \in A) = \mathbb{P}(X_1 \in A) \quad \forall A$$

Reversibility Implies Stationarity

- Assume $X_0 \sim \pi$ and P is *reversible* with respect to π.
- By the definition of reversibility, for *any* measurable sets A, B:

$$\mathbb{P}(X_0 \in A, X_1 \in B) = \mathbb{P}(X_0 \in B, X_1 \in A)$$

- **Proof of Stationarity:** Setting $B = \mathcal{X}$ (the entire state space) in the equation above gives:

$$\mathbb{P}(X_0 \in A, X_1 \in \mathcal{X}) = \mathbb{P}(X_0 \in \mathcal{X}, X_1 \in A)$$

- Since $\mathbb{P}(X_1 \in \mathcal{X}) = 1$ and $\mathbb{P}(X_0 \in \mathcal{X}) = 1$, this simplifies to:

$$\mathbb{P}(X_0 \in A) = \mathbb{P}(X_1 \in A) \quad \forall A$$

- **Conclusion:** $X_0 \sim \pi \implies X_1 \sim \pi$. Reversibility is a sufficient condition for stationarity.

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  - *Continuous Space* ($\mathcal{X} = \mathbb{R}^d$): $Q(x, x') = \mathcal{N}(x, \sigma^2 I)$.
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- **Requirement:** For each x, we must be able to efficiently *sample* from $Q(x, \cdot)$.

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- **Goal:** Choose $\alpha(x, y)$ such that the resulting Markov chain is *reversible* with respect to π.

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$$P(x, B) = \int_B \alpha(x, y)q(x, y)dy + \mathbf{1}_{\{x \in B\}} \int_{\mathcal{X}} (1 - \alpha(x, z))q(x, z)dz$$

and the corresponding transition density is:

$$p(x, y) = \alpha(x, y)q(x, y) + \delta_x(y) \int_{\mathcal{X}} (1 - \alpha(x, z))q(x, z)dz$$

Choosing Acceptance Probability α

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- **Key Insight:** The second term in $p(x, y)$ is only non-zero when $x = y$. For $x = y$, detailed balance trivially holds. Therefore, for $x \neq y$, it suffices to pick $\alpha(x, y)$ to satisfy:

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- **Constraint:** Because α is an acceptance probability, we must ensure that $\alpha(x, y) \in [0, 1]$ for all x, y.

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- The subscript B in the second choice stands for Barker. While both yield the correct stationary distribution, the *Metropolis-Hastings* choice (α^*) is the standard in practice.

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  - If the proposal is symmetric, i.e., $q(x, y) = q(y, x)$ for all x, y (e.g., Gaussian $\mathcal{N}(x, \sigma^2 I)$).
  - The proposal terms cancel out perfectly:

$$\alpha(x, y) = \min\left(1, \frac{\pi(y)}{\pi(x)}\right)$$

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  - *Action:* Always accept moves to higher probability states; probabilistically accept moves to lower ones.
- **Independent Proposal:**
  - If the proposal does not depend on the current state, i.e., $q(x, y) = q(y)$.
  - The acceptance probability becomes a ratio of importance weights:

$$\alpha(x, y) = \min\left(1, \frac{\pi(y)q(x)}{\pi(x)q(y)}\right) = \min\left(1, \frac{w(y)}{w(x)}\right)$$

    where $w(\cdot) = \pi(\cdot)/q(\cdot)$.

Gibbs Sampling: A Powerful Special Case

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- **The Algorithm:** At iteration $t+1$, we sweep through all d coordinates sequentially:

$$\begin{aligned}
x_1^{(t+1)} &\sim \pi(x_1 \mid x_2^{(t)}, x_3^{(t)}, \ldots, x_d^{(t)}) \\
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- **Key Property:** Gibbs sampling is equivalent to MH using the full conditionals as the proposal distributions. Remarkably, this guarantees the acceptance probability is *exactly 1* for every step.

Asymptotic Variance

- Let $(X_t)_{t \geq 0}$ be a Markov chain with stationary distribution π.

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- **Definition:** We define the *asymptotic variance* of $\hat{h}_T$ as:

$$\sigma_h^2(P) := \lim_{T\to\infty}\left(T\mathrm{Var}(\hat{h}_T)\right)$$

  which is independent of the distribution of X_0 and is the standard metric used to compare different Markov chains (kernels P) that have the *same* stationary distribution.

Peskun ordering of Markov chains

- (*Peskun-Tierney theorem*) Let P_1 and P_2 be two transition kernels, both *reversible* with respect to π. We have $\sigma_h^2(P_1) \leq \sigma_h^2(P_2)$ if

$$P_1(x, B \setminus \{x\}) \geq P_2(x, B \setminus \{x\})$$

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Optimality of the Metropolis-Hastings Ratio

- **Claim:** For any acceptance probability $\alpha(x,y)$ satisfying the *detailed balance condition*:

  $$\pi(x)\alpha(x,y)q(x,y) = \pi(y)\alpha(y,x)q(y,x)$$

  we have:

  $$P_{\alpha^*}(x, B \setminus \{x\}) \geq P_{\alpha}(x, B \setminus \{x\})$$

  for any $x \in \mathcal{X}$ and $B \in \mathcal{B}(\mathcal{X})$, where α^* is the standard Metropolis-Hastings choice:

  $$\alpha^*(x,y) = \min\left(1, \frac{\pi(y)q(y,x)}{\pi(x)q(x,y)}\right)$$

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- **Implication:** The standard Metropolis-Hastings acceptance probability α^* guarantees the highest probability of moving to a new state among *all* reversible algorithms with the same proposal Q. By Peskun's Theorem, it therefore minimizes the asymptotic variance!

Proof of Optimality: Setup

- To prove $P_{\alpha^*}(x, B \setminus \{x\}) \geq P_\alpha(x, B \setminus \{x\})$, it suffices to show that the continuous part of the transition density satisfies:

$$p_{\alpha^*}(x, y) \geq p_\alpha(x, y) \quad \text{for any } y \neq x$$

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- Recall that for any valid acceptance probability α and any $y \neq x$:

$$p_\alpha(x, y) = \alpha(x, y) q(x, y)$$

  Since $q(x, y)$ is the same for both, we just need to prove $\alpha^*(x, y) \geq \alpha(x, y)$.

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- Because $\alpha^*(x, y) = \min\left(1, \frac{\pi(y)q(y,x)}{\pi(x)q(x,y)}\right)$, the natural way to proceed is to split the proof into two cases based on the ratio:

$$\frac{\pi(y)q(y, x)}{\pi(x)q(x, y)} \geq 1 \quad \text{and} \quad \frac{\pi(y)q(y, x)}{\pi(x)q(x, y)} < 1.$$

Proof of Optimality: Analyzing the Cases

- **Case 1:** $\frac{\pi(y)q(y,x)}{\pi(x)q(x,y)} \geq 1$. In this case, we have

  $$\alpha^*(x, y) = 1 \geq \alpha(x, y)$$

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- **Case 2:** $\frac{\pi(y)q(y,x)}{\pi(x)q(x,y)} < 1$. In this case, we have

$$\alpha^*(x,y) = \frac{\pi(y)q(y,x)}{\pi(x)q(x,y)}$$

and hence,

$$p_{\alpha^*}(x,y) = \alpha^*(x,y)q(x,y) = \frac{\pi(y)q(y,x)}{\pi(x)}$$

- To compare this to $p_\alpha(x, y)$, we use the *detailed balance condition* for α:
$$p_\alpha(x, y) = \frac{\pi(y)\alpha(y, x)q(y, x)}{\pi(x)}$$
Since $\alpha(y, x) \leq 1$, it immediately follows that:
$$p_\alpha(x, y) \leq \frac{\pi(y)q(y, x)}{\pi(x)} = p_{\alpha^*}(x, y)$$
completing the proof!

Autocorrelation and Effective Sample Size

- To understand why the asymptotic variance $\sigma_h^2(P)$ differs from the i.i.d. case, we expand the variance of our estimator:

$$\mathrm{Var}(\hat{h}_T) = \frac{1}{T^2}\sum_{t=0}^{T-1}\mathrm{Var}(h(X_t)) + \frac{2}{T^2}\sum_{0\leq s<t\leq T-1}\mathrm{Cov}(h(X_s), h(X_t))$$

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- The covariance terms exist because MCMC samples are *correlated*. We quantify this using the *integrated autocorrelation time* (IACT), denoted as τ:

$$\tau = 1 + 2\sum_{k=1}^{\infty}\rho(k)$$

where $\rho(k)$ is the autocorrelation at lag k.

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- **Effective Sample Size (ESS):** We define $\mathrm{ESS} = T/\tau$. This represents the equivalent number of *independent* samples that would yield the same estimation variance as our T correlated MCMC samples.

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- **Visual Diagnostics:** We often use *trace plots* (plotting X_t versus t) to assess both burn-in (transient phase) and mixing (stationarity and exploration).

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- **Practical Evaluation:** Unlike i.i.d. sampling, MCMC estimators suffer from initialization bias and correlation. We must discard early transient samples (*burn-in*) and monitor the *Effective Sample Size (ESS)* to evaluate chain mixing.

ECEN 662: Estimation & Detection Theory

Lecture 20: Diffusion Monte Carlo Sampling

Tie Liu

Texas A&M University

Brownian Motion and Diffusion

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- *Note:* In $\mathbb{R}^d$, B_t consists of d independent 1D Brownian motions.
- **General Diffusion Process:** Defined by the Stochastic Differential Equation (SDE):

$$\begin{aligned} X_0 &\sim p_0 \\ dX_t &= \underbrace{b(X_t, t)dt}_{\text{Drift}} + \underbrace{a(X_t, t)dB_t}_{\text{Diffusion}} \end{aligned}$$

where $b : \mathbb{R}^d \times [0, \infty) \to \mathbb{R}^d$ and $a : \mathbb{R}^d \times [0, \infty) \to \mathbb{R}^{d \times d}$.

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- Therefore, the deterministic *drift* b describes the infinitesimal change in the mean, and the *volatility* function a describes the infinitesimal standard deviation.

The Fokker-Planck Equation

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$$
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- **Key Significance:** The change of the density p_t over *time* t can be expressed entirely using spatial derivatives over the *coordinates* x.

- **Crucial Consequence:** Because the evolution of the density is deterministic, the marginal distributions of *every* SDE can also be achieved by simulating a deterministic *ordinary differential equation (ODE)*.

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- **The SDE Formulation:** Consider the *SDE* given by:

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- **The Probability Flow ODE:** The answer is (surprisingly) yes! The corresponding vector field F is given by:

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- **The Score Function:** The effect of a stochastic SDE can be achieved by running a deterministic ODE. However, to actually compute this, we must know the *score function* $\nabla_x \log p_t(x)$.

Langevin Diffusion

- **The Langevin SDE:** In *Langevin diffusion*, the governing stochastic differential equation is:

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- **The Practical Advantage:** To compute the score function $s(x)$, we only need to know the target density $\pi(x)$ up to an unknown normalizing constant!

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- **The Learning Goal:** One way to learn a score-based model $s_\theta(x)$ is by *matching* it to the true score function $\nabla_x \log \pi(x)$ on a *per-sample* basis.

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- **Hyvärinen's Trick:** While this seems *untenable* because π is *unknown*, a remarkable result by Hyvärinen (2005) uses integration by parts to show this is equivalent to minimizing:

$$\mathbb{E}_{X\sim\pi}\left[\|s_\theta(X)\|^2 + 2\sum_{j=1}^{d}\frac{\partial s_{\theta,j}(X)}{\partial X_j}\right]$$
$$\approx \frac{1}{m}\sum_{i=1}^{m}\left[\|s_\theta(x[i])\|^2 + 2\sum_{j=1}^{d}\frac{\partial s_{\theta,j}(x[i])}{\partial x_j}\right]$$

The Computational Bottleneck

- **The Trace Term:** Recall the critical term in the Hyvärinen objective:

$$\mathrm{Tr}(\nabla_x s_\theta(x)) = \sum_{j=1}^{d} \frac{\partial s_{\theta,j}(x)}{\partial x_j}$$

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- **The Path Forward:** To make Score Matching scalable, we must avoid the explicit computation of the Jacobian. This leads to two main families of methods: *Sliced Score Matching* and *Denoising Score Matching*.

Sliced Score Matching (SSM)

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$$J_{ssm}(\theta) := \mathbb{E}_{p_v}\mathbb{E}_\pi\left[\frac{1}{2}\|v^\top s_\theta(X) - v^\top \nabla_X \log \pi(X)\|^2\right]$$

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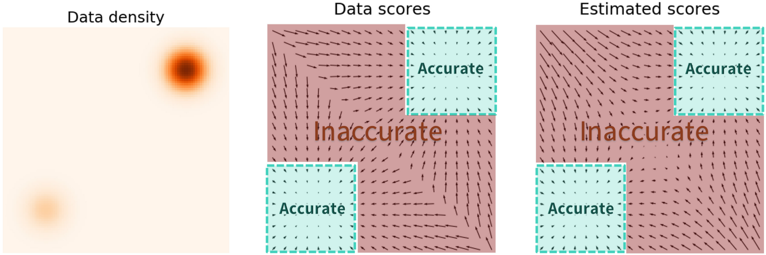
  - This can be computed efficiently using forward-mode or reverse-mode AD in *one or two passes*.
- **Efficiency:** We have reduced the complexity from $O(d)$ backprops to $O(1)$ (independent of the dimension d).

Pitfalls of Score Matching

- **The Data Sparsity Problem:** In practice, the aforementioned approaches have *limited* success. The main issue is that estimated score functions are inaccurate in *low-density* regions.
  - Score matching relies on samples to "see" the gradient.
  - Where there are *few* data points, the objective provides no signal to the model.



- **The Sampling Failure:** When sampling with Langevin dynamics, our initial sample (usually pure noise) is almost certain to start in a *low-density* region.
  - An inaccurate score model derails the dynamics from the *very beginning*.
  - The "gravity" of the score function points in random, wrong directions, preventing the model from ever reaching the high-quality data manifold.

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  where we dropped the dependency of the velocity function a on X_t for simplicity.
- **Reverse-time SDE:** Consider the following *reverse-time* SDE (moving from T to 0):

$$\begin{aligned} \overline{X}_T &\sim p_T \quad \text{(Prior distribution)} \\ d\overline{X}_t &= \left(b(\overline{X}_t, t) - a^2(t)\nabla_{\overline{X}_t} \log p_t(\overline{X}_t)\right) dt + a(t)d\overline{W}_t \end{aligned}$$

  where p_t is the marginal distribution of X_t, and $\overline{W}_t$ is the *reverse-time* Brownian motion.

Anderson's Reverser-time SDE Theorem

- **Theorem:** It was shown by Anderson (1982) that the reverse-time SDE generates the same marginal distributions as the forward process:

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- **Generation Process:** The actual sampling is done by reversing the process:
  1. Sample $x_T \sim p_T$.
  2. Simulate the *reverse-time* SDE from $t = T$ down to $t = 0$.

Annealed Langevin Diffusion (1/2)

- **Forward Process:** Let the forward process be the *Ornstein–Uhlenbeck (OU)* process:

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\begin{aligned}
X_0 &\sim \pi \quad \text{(Target data distribution)} \\
dX_t &= -\beta X_t dt + \sigma dW_t
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- **Prior Distribution:** Sampling from $\mathcal{N}\left(0, \frac{\sigma^2}{2\beta} I\right)$ is *easy* (Gaussian noise).

Annealed Langevin Diffusion (2/2)

- **Reverse-time SDE:** Using Anderson's theorem, the generative process is:

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- **Learning the Score:** These can be learned using *Implicit Score Matching* or *Denoising Score Matching* (Vincent, 2010).

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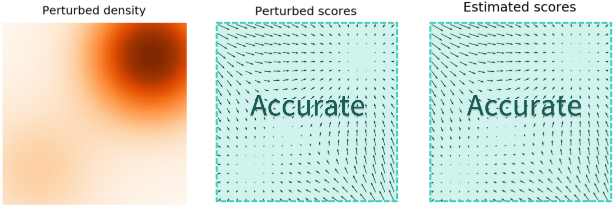

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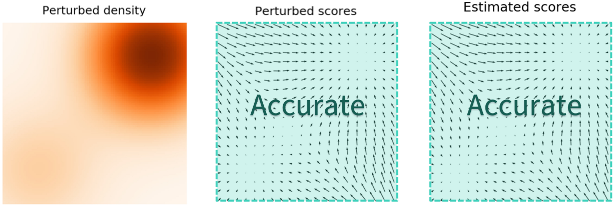

- **Key Question:** How do we choose an appropriate noise *scale* for the perturbation process? If it's too high, we lose the data; if it's too low, we're back to square one.

Multi-scale Noise Perturbation

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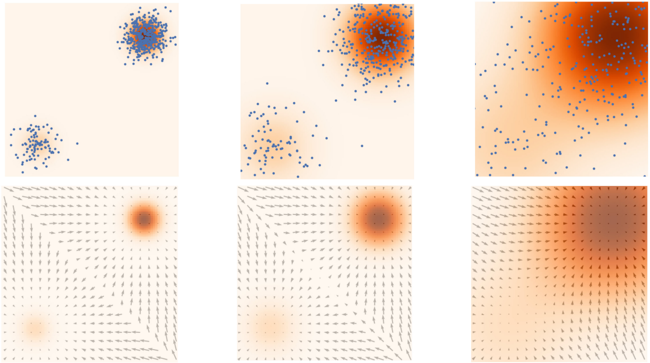
- **The Synthesis:** To achieve the best of both worlds, we use *multiple scales* of noise perturbations simultaneously. This allows the model to learn:
  - **Global structure** from high noise.
  - **Local details** from low noise.

$\sigma_1 < \sigma_2 < \sigma_3$



Diffusion GAN: The Density Ratio Perspective

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- **Variational Characterization:** We can learn this ratio through an adversarial objective. The optimal discriminator $d^*(x)$ satisfies:

$$d^*(x) = \frac{\pi(x)}{\pi(x) + p_t(x)}$$

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- **Discriminator-Guided Steps:** We train a *discriminator* $d_\theta(x, t)$ using samples from π and the current model p_t, then use its gradients to simulate *one step* of the ODE.